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Electrical Eng. Dept. at KSU
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King Saud University
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EE 312

Laboratory Manual



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OBJECTIVES

- Analyze the fundamental operational characteristics, parameters, and primary applications of core semiconductor devices.
- Utilize standard laboratory instrumentation, prototyping circuit boards, and modern electronic design automation (EDA) tools for experimental verification.
- Design and evaluate fundamental semiconductor circuits, with a specific focus on biasing configurations and application-specific topologies.
- Interpret manufacturer datasheets to evaluate device specifications, operational constraints, and practical design limitations.
- Develop professional technical writing skills through the documentation of engineering reports and experimental data.

EVALUATION AND GRADING POLICY

Reports	30%
Midterm Exam	20%
Design and Simulation Quiz	10%
Final Exam	40%

EE312 Website**:** http://fac.ksu.edu.sa/talmadhi/course/157758

Course Schedule

Week No.	Activity		To be submitted
	Wednesday Sections	*Other* Sections	
1	Experiment 1	Experiment 1	
2	Experiment 2	Experiment 2	Report for E1
3	Experiment 3	Experiment 3	Report for E2
4	Experiment 4	Experiment 4	Report for E3
5	اليوم الوطني National Day	Experiment 5	Report for E4
6	Experiment 5	Revision	
7	*Midterm Exam*	*Midterm Exam*	Report for E5
8	Experiment 6	Experiment 6	
9	Experiment 7	Experiment 7	Report for E6
10	Experiment 8	Experiment 8	Report for E7
11	Experiment 9	Experiment 9	Report for E8
12	Experiment 10	Experiment 10	Report for E9
13	Design and Simulation Quiz	Design and Simulation Quiz	Report for E10
14	إجازة الخريف	إجازة الخريف	
15	Revision	Revision	
15	*Final Exam*	*Final Exam*	

Guidelines for Writing EE 312 Reports

Your reports should be computer-typed and spell-checked. Paragraphs should be written in Times New Roman, size 12 points. Headings should be in upper case, bolded and size 14 points. Use formal English, direct language, and simple terms.

Your reports/assignments should be uploaded via LMS: https://lms.ksu.edu.sa. The report consists of three parts.

PART I [8 points]

1- **COVERSHEET**

It should show the name of the course, title and number of the experiment, your name and student KSUID number, section number and date of submission.

2- **OBJECTIVES**

3- **CIRCUIT DIAGRAMS**

4- **EQUIPMENT**

PART II [12 points]

5- **SUMMARY OF EXPERIMENTAL STEPS**

A descriptive **list** of the **major steps** and the corresponding measurements and equipment used to obtain them.

Example:

- Started out with assembling the circuit shown in Fig 1.
- Applied a 14-$V_{p\text{-}p}$ sinusoid to the input port of that circuit. The corresponding $V_O(t)$ was obtained using an oscilloscope, and it is shown in Fig 2.

6- **RECORDS OF MEASUREMENTS**

It should include screen shots of waveforms, table of measured quantities, and post-lab plots if applicable. *You should place a figure number **below the figure**, whereas a table number should be placed **above the table**.*

PART III [10 points]

7- **POST-LAB QUESTIONS (PLQs)**

8- **CONCLUSION**

- Briefly restate the purpose of the experiment and the principles being tested.
- Compare experimental results with accepted theoretical values.
- Explain what the results mean in the context of the relevant engineering principles (e.g., "The linear relationship confirms Ohm's Law...").
- Describe any trends or patterns observed (e.g., linear, exponential, inverse).
- Discuss the main sources of measurement uncertainty (e.g., equipment precision, environmental factors).
- Discuss the implications of the findings for engineering or industrial applications.

9- **REFERENCES** (if applicable**)**

Course Learning Outcomes

Upon successful completion of EE 312, the student is expected to:

CLO 1: Be able to conduct experiments related to the applications, characterization and testing of electronic devices using basic laboratory test equipment.

CLO 2: Understand experimentally obtained characteristic curves, be able to extract basic circuit models and device parameters from them, and use those models to interpret experimental data.

CLO 3: Understand the role of basic electronic devices in the conducted application-related experiments, and look at some of their possible limitations.

CLO 4: Apply knowledge obtained in the course to design basic electronic circuits, including the design of necessary bias schemes.

CLO 5: Identify the basic analysis types of SPICE, and be able to apply them to explore and verify prototype designs of electronic circuits.

CLO 6: Communicate efficiently by applying technical writing skills to prepare clear, concise, and observation-based lab reports/presentations.

EXPERIMENT

1

Exploring and Testing Junction Diodes

OBJECTIVES:

- To identify and test common types of junction diodes.
- To explore the relatively constant-voltage-drop and the open-circuit behaviors of junction diode.
- To build and test a simple voltage reference.

MATERIALS:

- Laboratory setup, including plug-in panel or a proto-board
- 1 rectifier diodes (e.g., the 1N4001 rectifier diode)
- 1 germanium (Ge) diode
- 1 light emitting diode (LED)
- 1 Zener diode (e.g., 1N4733A)
- **Two** resistors (100 Ω)
- Several wires and bridging plugs

INTRODUCTION

A p-n junction diode is a two-terminal electronic device, which has important applications that we shall explore in this lab. The electrode attached to the p-side of the junction is called the ***anode*** while the one attached to the n-side is called the ***cathode*** (Fig. 1.1). The cathode of a diode is usually marked by a band or a dot. It also can be identified based on a simple test using a digital multimeter (DMM).

A diode conducts current only in the ***forward direction***–from anode to cathode. However, the forward current of a real diode starts to be significant only for positive anode-to-cathode voltage (forward voltage) values which exceeds a minimum value called the ***cut-in*** *voltage*. On the other hand, a diode conducts ***negligibly small current*** (assumed to be zero) in the reverse direction as along as the magnitude of the reverse voltage does not exceed the breakdown voltage of the diode.

In this experiment, four types of diodes –the rectifier diode,[1] the germanium diode, the Zener diode and the light emitting diode (LED)– are introduced (Fig. 1.2). A Zener diode is a special type of diode which has small values of breakdown voltage, and can be operated safely in that mode of operation. An LED is a special type of diode that emits light while conducting a forward current.

An important feature of a forward-biased diode is its ability to maintain its ***terminal voltage*** approximately ***constant*** *over* ***a wide range of currents***. Therefore, a ***forward-biased*** diode can be approximately ***modeled*** by a ***constant voltage*** that depends on its type and characteristics. On the other hand, a ***reverse-biased*** diode can be modeled by an ***open circuit*** , as long as the reverse bias is less than the breakdown voltage of that diode.

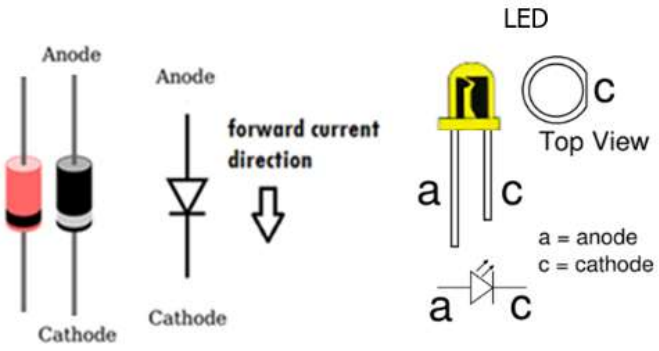


Figure 1.1 Visually identifying the cathode of a diode.

[1] Diodes for high-power applications, which draw lots of current or rectify high voltages, are called *rectifier diodes*. On the other hand, diodes that are designed to have high switching speeds go by names such as *fast recovery* or *switching diodes* [9].

PROCEDURE A: Testing Common Types of Diodes

1. Set your digital multimeter (DMM) to the diode-test mode.
2. Observe the ***open-circuit reading*** of the DMM and record it in Table 1.1.
3. Connect any type of diode (Fig. 1.2) between the red and black input jacks of the DMM (labeled V. Ω. and **COM,** respectively) in such a way to have the DMM continue to display the same ***open-circuit reading***. In this case, the **cathode** lead of the diode will be the one connected to the higher-potential (red) input jack of the DMM.
4. Reverse the connection of the diode. The DMM should display the ***forward voltage drop***. Record the forward voltage drop in Table 1.1. Make sure that this value is consistent with the reference range given in the table.
5. Repeat step 3 and 4 for the remaining types of diodes.

- If a diode does not show this behavior (a voltage drop in one direction and an open-circuit in the other), that implies that the diode is most probably defective.



FIGURE 1.2 Common types of diodes. From right: LED, Ge diode, Zener diode, and rectifier diode.

PROCEDURE B: Exploring the Diode Behavior

1. Connect the circuit shown in Fig. 1.3a, initially with only a digital multimeter (DMM) across the open circuit (no diode).
2. Make sure that your DMM is on the **dc voltage** measurement mode.
3. Turn on your dc supply using the ~ POWER switch. Use the CURRENT knob to set the current limit at **0.05** A. Use the COARSE and FINE knobs to set the output voltage at 5 V. Turn ON the dc switch ⎓.
4. Record the open-circuit voltage V_{OC} reading displayed by the DMM in Table 1.2.
5. Connect the rectifier diode backwards (as shown in Fig. 1.5a), and then record the resulting reverse voltage drop V_{rev} across the diode.
6. Turn off the dc switch of the supply. Flip the diode upside down (Fig. 1.3b).
7. Turn on the dc switch of the supply. Adjust the dc supply output voltage until the voltage across the resistor V_{RS} reads **0.1 V**. The current in this case will be 1 mA. Record the value of the diode forward drop V_D in Table 1.2.
8. Increase the dc supply output voltage until V_{RS} reaches **1 V**. The current in this case will be 10 mA. Record the new value of V_D in Table 1.2.
9. Increase the dc supply output voltage further until V_{RS} reaches **2 V**. The current in this case will be 20 mA. Record the new value of V_D in Table 1.2.
10. Connect a 100-Ω resistor (R_L) in parallel with the diode. Record V_D' in Table 1.2
11. Turn off the dc supply, disconnect all parts, and tidy up your table.

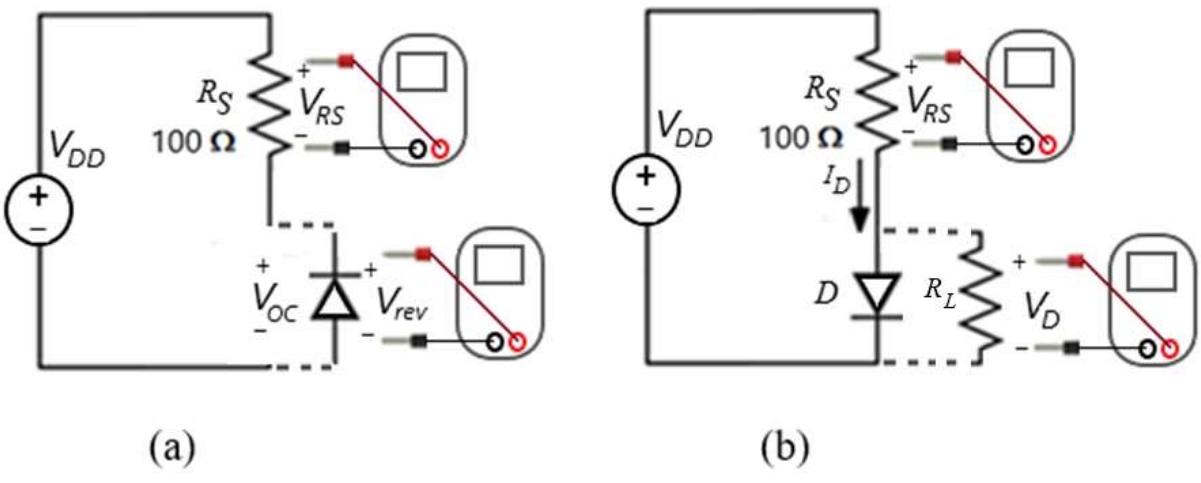


FIGURE 1.3 Exploring the essence of diode operation.

RESULTS

Table 1.1 Procedure A Testing Junction Diodes

DMM Test-Position Reading in Case of an Open Circuit:		
DMM Test-Position Reading in Case of a Reverse-Biased Diode:		
Type of diode	*Expected* Forward Voltage Drop	Measured *Forward Voltage* Drop *(Round off to 3 Decimal Places)*
Rectifier	~ 0.5 – 0.8 V (made of Si)	
Zener		
LED	~ 1.5 – 3.5 V	
Germanium	~ 0.2 – 0.4 V	

Table 1.2 Procedure B Measured Data *(Rounded off to 3 Decimal Places)*

Reverse Bias			
Step			
4 & 5	Supply Voltage V_{DD} = 5 V		
	The corresponding open-circuit voltage V_{OC} =		
	The diode's reverse voltage drop V_{rev} =		
Forward Bias			
Step	Arrangement	I_D (mA)	V_D (V)
7	Adjust V_{DD} so that V_{RS} = 0.1 V	1	
8	Adjust V_{DD} so that V_{RS} = 1 V	10	
9	Adjust V_{DD} so that V_{RS} = 2 V	20	
7→8	ΔV_D associated with increasing I_D ***10 times*** =		
8→9	ΔV_D associated with **doubling** I_D =		
Loading Effect (Forward Biased Diode)			
10	V_D *(no load across the diode)* =		
	V_D' *(100-Ω resistor across the diode)* =		
	$\Delta V_D = V_D' - V_D$ =		
	% Change = $(\Delta V_D / V_D) \times 100$ =		

POST LAB QUESTIONS (PLQs)

1) *Procedure A*: Arrange the values of the forward voltage drop of a rectifier diode, germanium diode and a light emitting diode in an ascending order.
2) *Procedure B, step 4 and 5*: For the circuit shown in Figure 1.3a, what is the relationship between the supply voltage V_{DD}, the open circuit voltage drop V_{OC}, and the reverse voltage drop across the diode V_{rev}.
3) *Procedure B, steps 7→ 9*: Calculate the expected increment of diode terminal voltage (ΔV_D) associated with increasing its current from 10 mA to 100 mA, and from 100 mA to 200 mA. Based on that, calculate the expected V_D at 200 mA.
   Hint: $V_{D|_{200\ mA}} = V_{D|_{10\ mA}} + \Delta V_{D|_{10\ mA \rightarrow 100\ mA}} + \Delta V_{D|_{100\ mA \rightarrow 200\ mA}}$
4) *Procedure B, step 10*: What is you observation on the percentage change in V_D between the no-load case and the R_L = 100 Ω case?
5) A certain load operates at 2 V and 10 mA. Use standard 1.5 batteries and the voltage reference circuit shown in Fig. 1.4 to power this load.
   i. How many series-connected diodes do you need to connect in parallel with the load to get V_{LOAD} = 2 V?
   ii. Calculate the required value of R_S.
   iii. Sketch the final circuit and label it with relevant information.

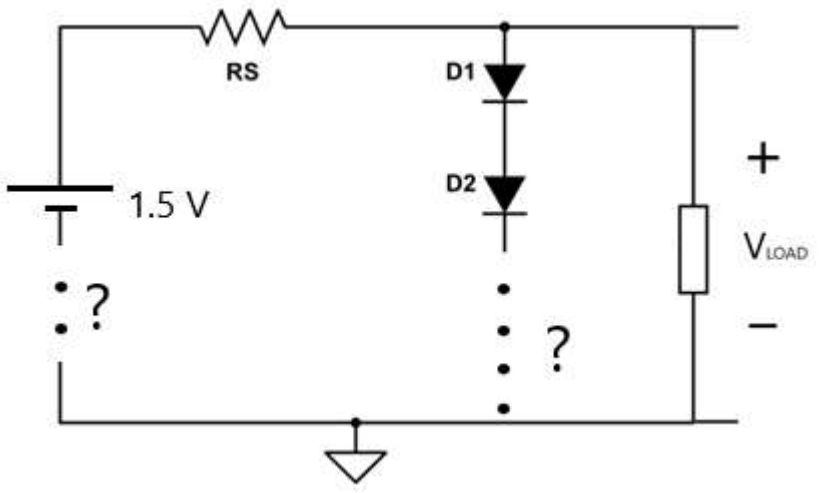


FIGURE 1.4 Use identical diodes that have 0.67 V@10 mA. The load operates at 2 V and 10 mA.

EXPERIMENT

2

Terminal Characteristics Of Junction Diodes

OBJECTIVES:

- To experimentally obtain the *i-v* characteristic curves for a rectifier diode and a Zener diode.
- To use experimental data to calculate important device parameters.
- To be able to extract a piecewise-linear model for a Zener diode.
- To understand the mechanism of reverse breakdown in Zener diodes and its applications

MATERIALS:

- Laboratory setup, including a plug-in panel or a proto-board
- 1 rectifier diode (e.g., the 1N4001 rectifier diode)
- 1 Zener diode (e.g., 1N4733A)
- 1 resistor (1 kΩ)
- Several wires and bridging plugs

INTRODUCTION

Figure 2.1 shows the $i - v$ relationship of a junction diode. This relationship consists of three distinct *regions of operation*: The forward-bias region ($v > 0$), the reverse-bias region ($v < 0$), and the breakdown region ($v < -V_{ZK}$). In the forward region the $i - v$ relationship is closely approximated by

$$i = I_S\left(e^{\frac{v}{nV_T}} - 1\right) \qquad (2.1)$$

where I_S is the **saturation current**, V_T is the **thermal voltage** ≅ 26 mV at T = 300 K, n is the **emission coefficient** whose value $1 \le n \le 2$.
If one has two pairs of $i - v$ measurements in the forward bias region ($V_D \ge 0.5$ V), n for a given diode can be found using the following equation:

$$n = \frac{V_{D2} - V_{D1}}{2.3 \times V_T \times \log\left(\frac{I_{D2}}{I_{D1}}\right)} \qquad (2.2)$$

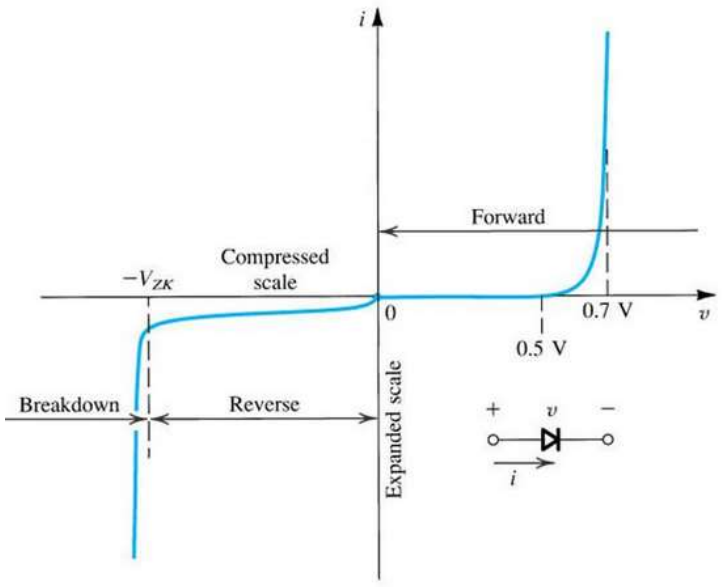


Figure 2.1 The current-voltage characteristic curve of a junction diode [1].

Recall that diodes specifically manufactured to operate in reverse breakdown region at low breakdown voltages are called **Zener diodes**. On the other hand, the applications of rectifier diodes require unipolar conduction. Thus, rectifier diodes are manufactured with large breakdown voltages.

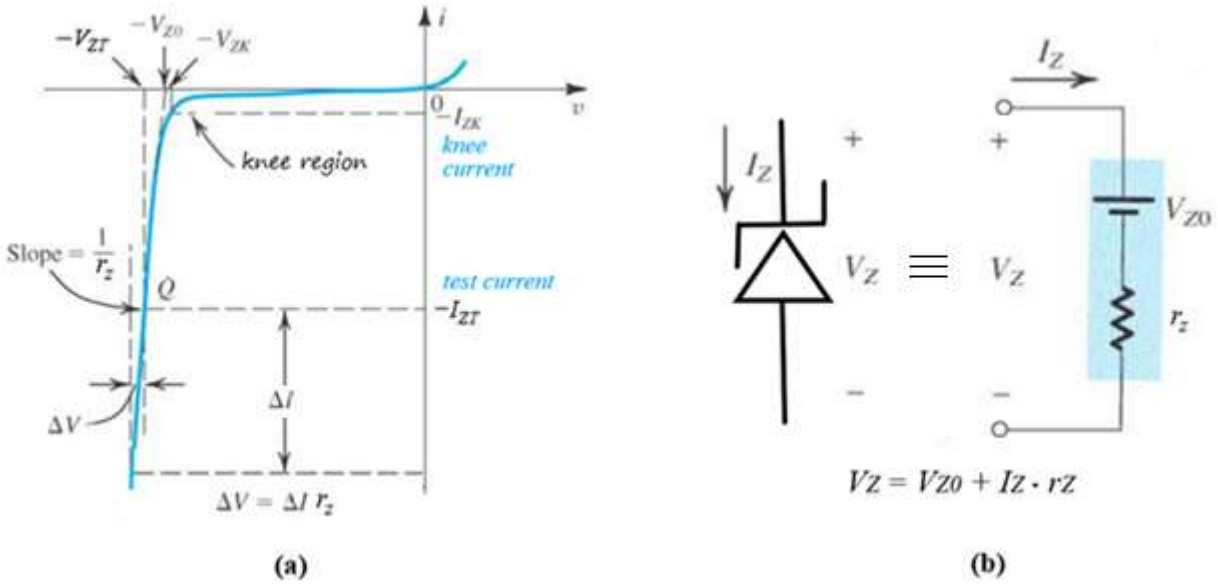


Figure 2.2 (a) Breakdown characteritic of a Zener diode and **(b)** its linearized model.

The breakdown region (Fig. 2.2a) is characterized by a near-vertical line (constant drop behavior), which is desirable if the diode is to be used in **voltage regulation**. An **equivalent circuit** based on a piecewise-linear approximation of a Zener diode is shown in Fig. 2.2b. The inverse of the slope in the breakdown segment determines the Zener **incremental resistance** r_Z.

*Smaller r_Z indicates steeper characteristic and consequently smaller ΔV_Z for large ΔI_Z, which suggests that we can use a Zener diode in that region as a **voltage reference** or as a **voltage regulator**.*

A piece-wise linear model for the Zener diode in the reverse breakdown region is shown in (Fig. 2.2b). If we have two operating points on the current-voltage characteristic in the breakdown region, we can calculate the parameters of the linear model. First we calculate r_Z as the inverse of the slope, i.e., $r_Z = \Delta V_Z/\Delta I_Z$. V_{Z0} can then be obtained as the difference between V_Z and ($I_Z \times r_Z$).

To insure satisfactory voltage regulation a Zener diode, the current of the Zener diode must be sufficiently high under the maximum loading condition, to avoid operating near the **knee region**, i.e., I_Z must be greater than I_{ZK} by a couple of milliamperes.

In their datasheets manufacturers of Zener diodes usually provide information related to the **nominal** voltage V_Z and the maximum incremental resistance r_Z (denoted by $\boldsymbol{Z_{ZT}}$ by many manufacturers) at a given reverse test current I_{ZT}. The maximum reverse current I_{ZM} is given as well (refer to Table 2.1).

When a Zener diode operates at a current I_{Z2} which is different than I_{ZT}, the ***change*** in its voltage from the nominal value can be obtained by multiplying r_Z by the ***change*** in the current:

$$\Delta V_Z = r_Z \times \Delta I_Z \qquad (2.3)$$

Table 2.1 The 1N7xxx Zener diode series specifications. *Source: futurlec.com*.

Type Number	Nominal Zener Voltage V_Z @ I_{ZT}[3] (Volts)	Test Current I_{ZT} (mA)	Maximum Zener Impedance Z_{ZT} @ I_{ZT}[1] (Ω)	Maximum Regulator Current I_{ZM}[2] (mA)
1N746A	3.3	20	28	110
1N747A	3.6	20	24	100
1N748A	3.9	20	23	95
1N749A	4.3	20	22	85
1N750A	4.7	20	19	75
1N751A	5.1	20	17	70
1N752A	5.6	20	11	65
1N753A	6.2	20	7	60
1N754A	6.8	20	5	55
1N755A	7.5	20	6	50
1N756A	8.2	20	8	45
1N757A	9.1	20	10	40
1N758A	10	20	17	35
1N759A	12	20	30	30

PROCEDURE A: Reverse Biasing

1. Connect the circuit shown in Fig. 2.3a, initially with a rectifier diode.
2. Turn on your dc supply using the ~ POWER switch. Use the CURRENT knob to set the current limit at **0.05** A. Use the COARSE and FINE knobs to set the output voltage V_{BB} at 25 V. Turn ON the dc switch ⎓.
3. Using a DMM, measure and record the values of V_D and V_{RS} (*rounded off to 3 decimal places*) in Table 2.2. *Note that* V_{RS} (in volts) = I_D (in milliamperes).[2]
4. Replace the rectifier diode with a Zener diode and record V_D and V_{RS} in the table.
5. Repeat the above steps for the remaining values of V_{BB} in the table.
6. Turn off the dc switch of the power supply.

PROCEDURE B: Forward Biasing

1. Modify *only* the supply connection to implement the circuit shown in Fig. 2.3b.
2. Use the dc power supply VOLTAGE knobs to set the output voltage at 0.3 V.
3. Turn on the dc switch of your supply. Record the multimeter voltage readings for the resistor and the diode in Table 2.2.
4. Replace the rectifier diode with a Zener diode and measure and record V_{RS} and V_D. Repeat the above steps for the remaining values of V_{FF} in the table.
5. Switch off your dc supply, disconnect all parts, and tidy up your table.

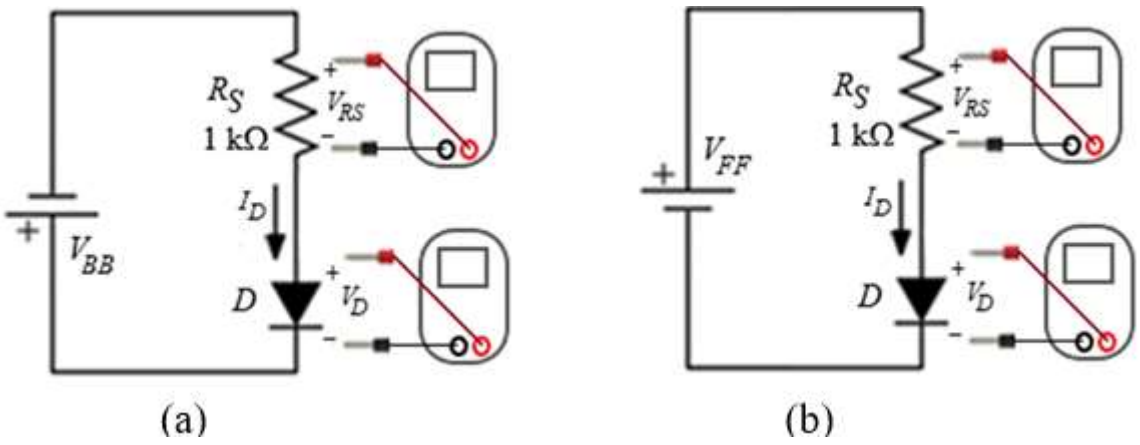


Figure 2.3 Obtaining the current-voltage characteristics. **(a)** Reverse bias. **(b)** Forward bias.

[2] $I_D = I_{RS} = V_{RS} / 1000 = V_{RS} \times 10^{-3} \Rightarrow$ $\boldsymbol{I_D}$ (expressed in **milliamperes**) = $\boldsymbol{V_{RS}}$ (expressed in **volts**)

RESULTS

Table 2.2 Measured Data for plotting the *i-v* curve **(Rounded Off to 3 Decimal Places).**

Bias Condition	dc Supply Voltage	Rectifier Diode		Zener Diode	
Reverse Biasing Case **(Fig 2.3a)**	V_{BB}(V)	V_D (V)	I_D (mA) = V_{RS} (V)	V_D (V)	I_D (mA) = V_{RS} (V)
	25				
	20				
	15				
	10				
	7				
	5				
	4				
	3				
	2	-2	0.00	-2	0.00
	0	0	0	0	0
Forward Biasing Case **(Fig 2.3b)**	V_{FF}(V)	V_D (V)	I_D (mA)= V_{RS} (V)	V_D (V)	I_D (mA) = V_{RS} (V)
	0.3				
	0.4				
	0.5				
	0.55				
	0.60				
	0.65				
	0.75				
	1.7				
	10.7				
	20.7				

POST LAB QUESTIONS (PLQs)

1) **Plot I_D vs. V_D** for both diodes using MATLAB (see Appendix B). Label the three regions of operation on the graph. Find and label the **cut-in** voltage for the rectifier diode. *Include this plot as well as a copy of the code after entering your measured data in your report.*
2) Use two pairs of $i - v$ forward biasing measurements (obtained at V_{FF} = 10.7 V and 1.7 V) and Equation (2.2) to calculate *n* for your rectifier diode; also, using Equation (2.1), calculate I_S for that diode.
3) Calculate the incremental resistance r_Z parameter (Fig. 2.2b) for the Zener diode. Use the ***absolute*** values of the terminal voltage and current of the Zener diode measured at V_{BB} = 15 V and V_{BB} = 20 V. Using the above calculated value of r_Z and the absolute values of the voltage and current at V_{BB} = 20 V, calculate the V_{Z0} parameter.
4) Redraw the circuit in **Fig. 2.3a**, replacing the Zener diode with its linearized circuit model consisting of r_Z and V_{Z0}. For the V_{BB} = 25 V case, calculate I_Z, and then use it to calculate $Vz = V_{Z0} + I_Z \times r_Z$, as shown in Fig. 2.2b.
5) The load line equation of the circuit shown in Fig. 2.3b is given by:

$$V_{FF} = I_D \times R_S + V_D \qquad (2.4)$$

   i. Plot the load line on the i-v characteristic curve of the rectifier diode for the V_{FF} = 10.7 V case.
   ii. Find the coordinates of the **dc operating point** (V_{DQ}, I_{DQ}) at the intersection of the load line with the i-v characteristic curve.
   iii. Compare V_{DQ} and I_{DQ} with your measurements for the V_{FF} = 10.7 case.

EXPERIMENT

3

Limiting Circuits

OBJECTIVES:

- To explore the basic principles of some limiting (clipping) circuits and be able to predict their responses.
- To experimentally obtain and examine the output waveforms and the voltage transfer characteristics of some typical limiting circuits.
- To be able to design and implement a circuit to satisfy a given limiting transfer characteristic.

MATERIALS:

- Laboratory setup, including a plug-in panel or a proto-board.
- 2 ordinary silicon diodes (e.g., the 1N4001 rectifier diode)
- 2 Zener diode (e.g., 1N4733A)
- 1 resistor (1 kΩ)
- Several wires and bridging plugs

INTRODUCTION

Limiting or clipping is a function performed by a circuit that has the ability to limit or clip off portions of a signal that are above or below a specified level [5]. A junction diode is a suitable candidate for this type of circuits because, as we have learned, once it conducts enough current it tends to have little variations in its voltage, voltage reference behavior.

Fig. 3.1 shows the general voltage transfer characteristic (VTC) for a **double limiter**. The circuit acts linearly for inputs larger than the **lower threshold** and less than the **upper threshold**, $(V_O^-/A_V) < v_I < (V_O^+/A_V)$. If, however, $|v_I|$ exceeds the magnitude of the threshold levels, the output voltage is limited or clamped to the **limiting levels**. If there is no load connected across its output port, a limiter will have a unity slope A_V in the linear region of its VTC, and consequently the threshold levels will be roughly equal to the limiting levels.

The limiting levels of a given limiting circuit can be designed to meet a given specification. Toward that end, the diode forward drop (0.6–0.8 V for a silicon diode) and/or the nominal Zener voltage V_Z (a list of which is given in Appendix E) can be utilized to obtain specified limiting levels of a desired transfer characteristic.

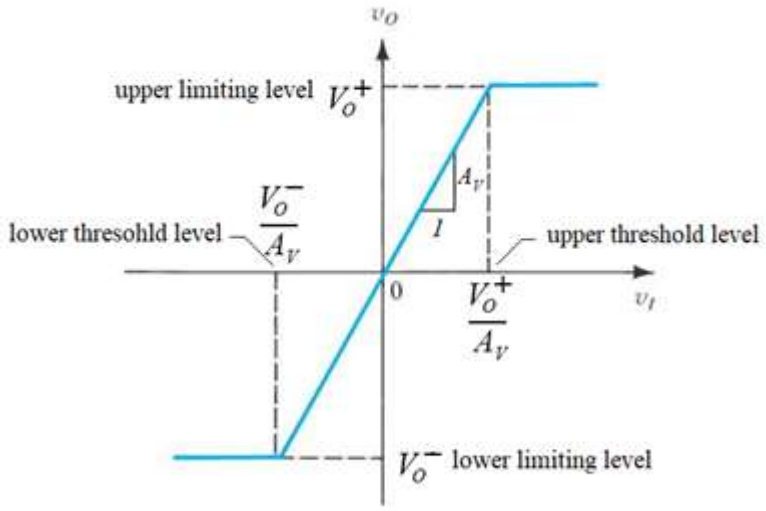


Figure 3.1 General transfer characteristic for a double limiter circuit.

In order to measure the limiting levels experimentally, we will use an extremely useful lab instrument for the first time in this course: the oscilloscope (Fig. G.3). Oscilloscopes are used to observe time-varying waveforms and measure their characteristics. They come in different shapes and brands. Essentially an oscilloscope can be used to measure voltage, but it can be used indirectly to measure current by probing the voltage across a one-ohm precision resistor.

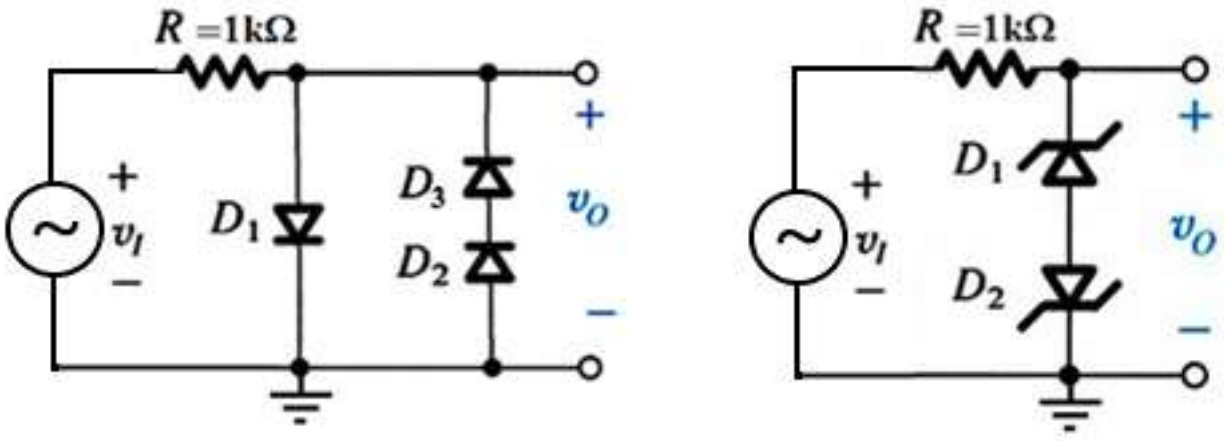
Another valuable tool to any electronics lab is the waveform (function) generator (Fig. G.4). It can be used to obtain several waveform types to test our circuits. It has the capabilities of controlling the frequency, amplitude, and the offset of its output waveform. We are going to use a waveform generator in this experiment to obtain input signal required to test and characterize limiting circuits.

PROCEDURE A: Limiter Using Ordinary Diodes

1. Assemble the circuit shown in Fig. 3.4a.
2. Use a coaxial cable with banana plugs to connect the input of your circuit to the output of channel one (**CH1)** of a waveform generator.
3. Turn your generator on. Set the input signal type to sinusoidal. Then set the frequency at **200 Hz** and the amplitude at **500 mV_{p-p}** (peak to peak).

   To set the frequency: Press **F1** → Enter 200 → Press **F3** to select **Hz**.

   To set the amplitude: Press **F2** → Enter 500→ Press **F1** to select **mVpp**.
4. Press the **CH1** key of your waveform generator **to activate it**.
5. Use a coaxial cable to connect the input of your circuit to channel one (**CH1**) of an **oscilloscope**, and another cable to connect the output of the circuit to channel two (**CH2)** of the scope. The black ground leads of the two coaxial cables should be connected to the reference node of the circuit (i.e., to ground).
6. Turn on your scope; press the **Default** key, then press the **Autoset** key.
7. If not already activated press the **CH2** key of the scope to activate it.
8. Press on the **POSITION** knobs of both channels of the scope to align the zero reference for the input and output waveforms at the centerline of the grid.



(a) (b)

Figure 3.4 **(a)** Antiparallel-diode double limiter. **(b)** Double-anode Zener limiter.

9. Use the **SCALE** knobs of **CH1** and **CH2** to set the vertical sensitivity of each channel at **200 mV/div**.
10. To obtain a convenient number of cycles, use the **SCALE** knob from the **Horizontal Controls** to control the timebase (time/div).
11. Press the **Measure** key→ Select **Add Measurement** from the bottom-side menu → Activate the **Source 1** submenu → Use the **VARIABLE** knob to highlight **CH2** from the list →Press **Select** → Activate the **V/I** submenu → Use the **VARIABLE** knob to highlight **Max** → Press **Select** → Use the **VARIABLE** knob to highlight **Min** → Press **Select**. Press the **Menu OFF** key twice.
12. Use the knob on your **waveform generator** to increase the amplitude of the input waveform gradually until you reach **1.5 $V_{p\text{-}p}$.**
13. Record the **Max.** value of the output, i.e., the ***upper limiting level*** V_O^+.

14. Use the knob on your **waveform generator** to increase the amplitude of the input waveform gradually until you reach **4 $V_{p\text{-}p}$.**
15. Use the **SCALE** knobs of **CH1** and **CH2** to obtain a scale of **500 mV/div** for both of them.
16. Record the **Min.** value of the output, i.e., the ***lower limiting level*** V_O^-.

17. Record v_I(t) and v_O(t) waveforms on Fig. 3.6a.
18. To obtain the voltage transfer characteristic for the circuit, we need to put the scope on the **XY mode**. Press the **Acquire** function key, select **XY** from the bottom side menu, then select **Triggered XY** from the right side menu. Press the **POSITION** keys of **CH1** and **CH2** to align the characteristic at point (0, 0).
19. Record the displayed voltage transfer characteristic (VTC) on Fig. 3.6b.
20. Turn the XY mode off by selecting **OFF(YT)** from the right menu.

PROCEDURE B: Double Anode Limiter Using Zener Diodes

1. Turn off your waveform generator.
2. Assemble the circuit shown in Fig. 3.4b.
3. Use the **SCALE** knobs of **CH1** and **CH2** to obtain a scale of **2 V/div** for each one.
4. Turn on your waveform generator.
5. Gradually increase the amplitude of the input until you reach **15 $V_{p\text{-}p}$**.
6. Record v_I(t) and v_O(t) waveforms on Fig. 3.7a.
7. Record the **Max.** value of the output, i.e., the ***upper limiting level*** V_O^+.

8. Record the **Min.** value of the output, i.e., the ***lower limiting level*** V_O^-.

9. Repeat step 18 of Procedure A to obtain the voltage transfer characteristic for this circuit.
10. Record the displayed voltage transfer characteristic (VTC) on Fig. 3.7b.
11. Turn the XY mode off by selecting **OFF(YT)** from the right menu.
12. Turn off your waveform generator, and then turn off your scope. Disconnect your circuit and put all components and cables back in place.

POST LAB QUESTIONS (PLQs)

1) *Procedure A: For the 500 m$V_{p\text{-}p}$ input case:* Is there any difference between the waveforms of v_I(t) and v_O(t)? Why?
2) *Procedure A*: Draw an equivalent circuit for circuit **(a)** of Fig 3.4 to justify your measured upper limiting level. In that circuit, use an appropriate constant drop model for each diode contributing to the limiting level.
3) *Procedure A*: Draw an equivalent circuit for circuit **(a)** of Fig 3.4 to justify your measured lower limiting level. In that circuit, use an appropriate constant drop model for each diode contributing to the limiting level.
4) *Procedure B: For the 4 $V_{p\text{-}p}$ input case:* Is there any difference between the waveforms of v_I(t) and v_O(t)? Why?
5) *Procedure B*: Draw an equivalent circuit for circuit **(b)** of Fig 3.4 to justify your measured upper limiting level. In that circuit, use an appropriate constant drop model for each diode contributing to the limiting level.
6) *Procedure B*: Draw an equivalent circuit for circuit **(b)** of Fig 3.4 to justify your measured lower limiting level. In that circuit, use an appropriate constant drop model for each diode contributing to the limiting level.

RESULTS

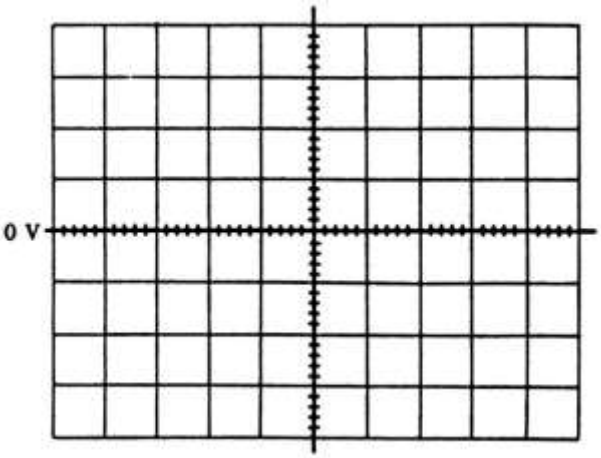


Figure 3.6a

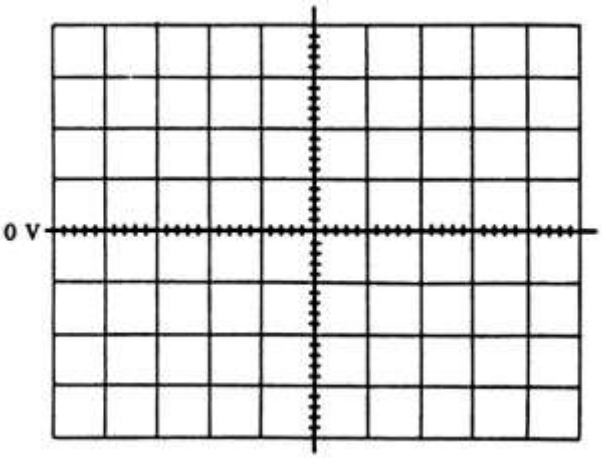


Figure 3.6b

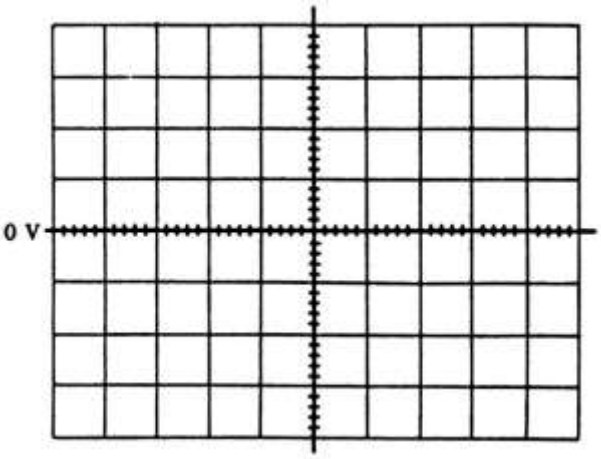


Figure 3.7a

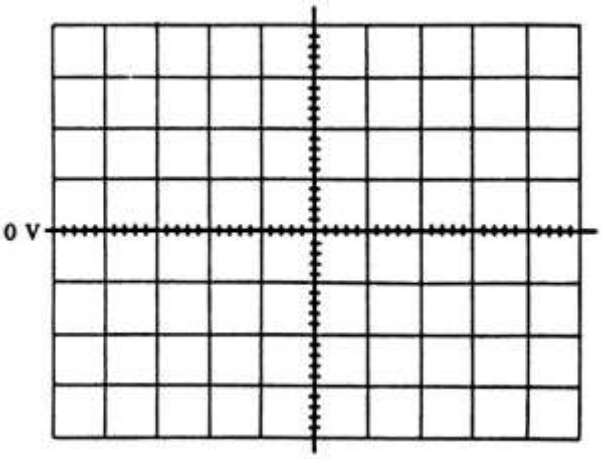


Figure 3.7b

EXPERIMENT

4

Rectifiers and Voltage Regulation (ac to dc Conversion)

OBJECTIVES:

- To experimentally explore an important application of diodes, namely rectification and voltage regulation.
- To understand the importance of filtering and voltage regulation in the process of ac to dc conversion.
- To learn and apply some practical design tips for designing simple ac to dc conversion system.

MATERIALS:

- Laboratory setup, including plug-in panel or a proto-board
- 1 center-tapped transformer
- 2 rectifier diodes (e.g., the 1N4001 rectifier diode)
- 1 Zener diode
- 2 electrolytic capacitors (10 μF, 470 μF)
- 2 resistors (4.7 kΩ, 270 Ω)
- Several wires and bridging plugs

INTRODUCTION

Electronic equipment needs dc power supply to operate. A block diagram of such a system is shown in Fig. 4.1 [1]. This system involves rectification, filtering and voltage regulation.

A **power transformer** is used to step down the input ac voltage and provide electrical isolation, which is important for safety. A **diode rectifier** uses the unidirectional-current property of diodes to convert an input sinusoid to a unipolar but pulsating output. Acting as a simple low-pass filter.

A **filter capacitor** is used to reduce the **ripple** (pulsation) of the resulting output waveform. The output voltage peak-to-peak ripple $V_{r(pp)}$ is inversely proportional to the capacitance C of the filter capacitor

$V_{r(pp)}$ is inversely proportional to R_L, or equivalently directly proportional to the dc load current I_L, as well. Therefore, a **shunt voltage regulator** (dotted box in Fig. 4.3) is needed to keep the dc output voltage fixed *despite possible variations in the load current and/or the ac input ac voltage*. A Zener diode of suitable nominal breakdown voltage is used based on the desired output dc level. Using a regulator enables the designer to use a capacitor of lower capacitance as well.

There are two main implementations of a full-wave rectifier: the one that utilizes a **center-tapped** transformer and requires only two diodes (see Fig. 4.2) and the **bridge rectifier** that does not require a center-tapped transformer but requires four diodes. Please refer to the textbook by Sedra and Smith for further details [1].

Two important ratings are needed to be specified when an engineer needs to select diodes for a given rectifier design: the **p**eak **i**nverse **v**oltage (**PIV**) and current-handling capability. The PIV is the maximum reverse voltage that the diode ever experiences in a given circuit [4].

A manufacturer's **datasheet** gives detailed information on a device so that it can be used properly in a given application. Appendix C shows the datasheet for general-purpose rectifier diodes (1N4001 – 1N4007).

In those datasheets, manufacturers list a number of important maximum ratings: the **R**epetitive **R**everse voltage **M**aximum that can be applied across the diode, $\mathbf{V_{RRM}}$, the maximum average rectified forward current at T_A=75°C, $\mathbf{I_{F(avg)}}$, and the maximum not repetitive forward **surge** current the diode can sustain, $\mathbf{I_{FSM}}$.

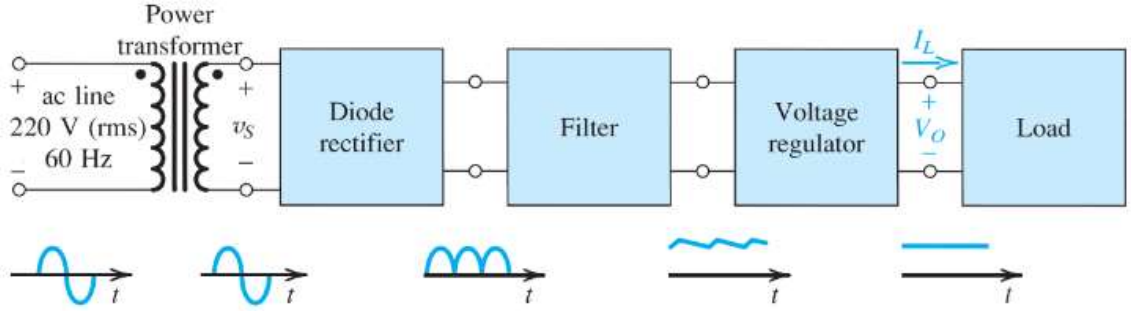


Figure 4.1 A block diagram of a dc power supply [1].

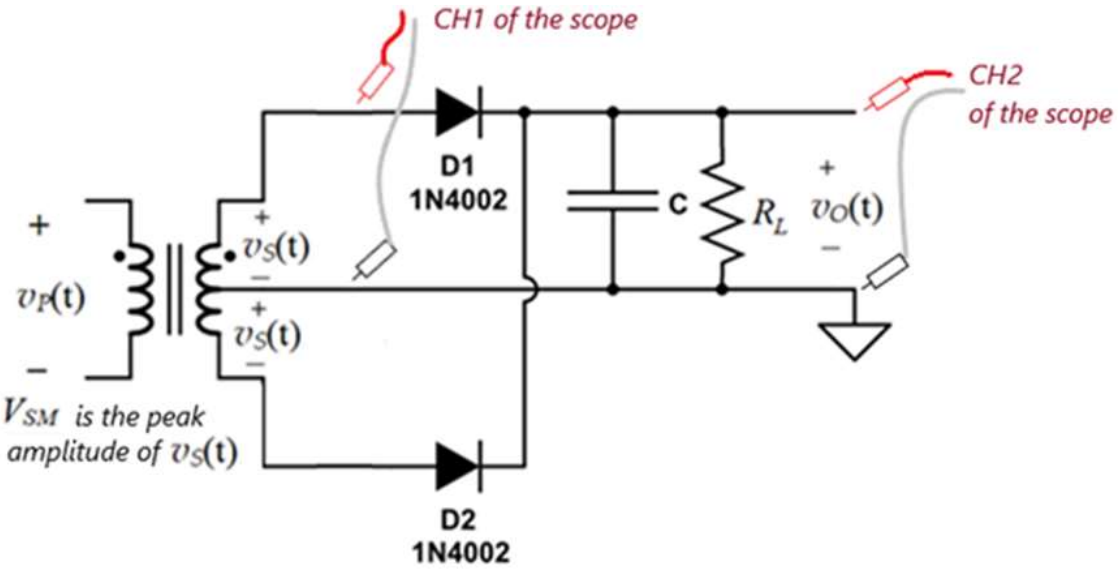


Figure 4.2 Rectifier circuit which uses a center-tapped transformer, with a filter capacitor.

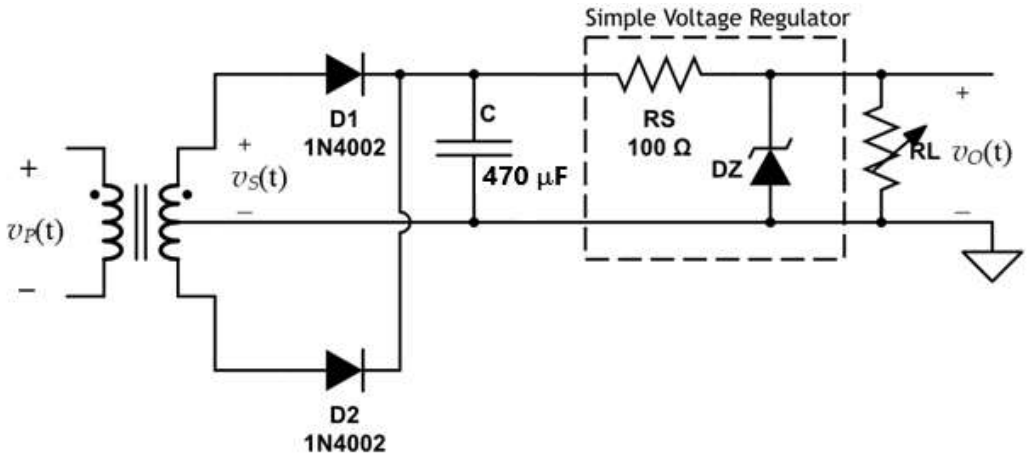


Figure 4.3 Simple Regulated dc Power Supply.

PROCEDURE A: Unregulated dc Power Supply

Assemble the circuit shown in Fig. 4.2, with R_L = 4.7 kΩ, and initially ***without*** a capacitor.



*Caution: Do not connect the transformer to the ac mains before the instructor inspect your circuit***.**

1. Use a coaxial cable to connect the input of your circuit, *as shown on the circuit diagram*, to **CH1** of an oscilloscope, and another cable to connect the output of your circuit to **CH2** of the scope.
2. Turn on your scope; wait for few seconds, press the **Default** key, and then the **Autoset** key.
3. Connect the transformer to an appropriate ac power outlet.
4. If not already activated, press the **CH2** key of the scope to activate it.
5. Press the **POSITION** knobs of both channels of the scope to align the zero reference for the input and output waveforms at the centerline of the grid.
6. Use the **SCALE** knobs of **CH1** and **CH2** to obtain a scale of **5 V/div** for each.
7. If necessary, use the **SCALE** knob from the ***Horizontal* Controls** to obtain an appropriate timebase (time/div) to display convenient number of cycles.
8. Press the **Measure** key→ Select **Add Measurement** from the bottom-side menu → Activate the **V/I** submenu → Use the **VARIABLE** knob to highlight the appropriate item (**Pk-Pk, Max.**) then press **Select** key to add anyone of them.
9. Activate the **Source 1** submenu → Use the **VARIABLE** knob to highlight **CH2** then press **Select** → Activate the **V/I** submenu → Use the **VARIABLE** knob to highlight the appropriate item (**Pk-Pk, Max.**) then press **Select** key to add anyone of them. Press the **Menu OFF** key twice.
10. Record the waveforms of v_S(t) and v_O(t) on Fig. 4.4.
11. Observe what happens when you disconnect a bridging plug between D_2 and the transformer. Reconnect the bridging plug to have D_2 back in the circuit.

12. To obtain the voltage transfer characteristic for the circuit, we need to put the scope on the **XY mode**. Press the **Acquire** function key, select **XY** from the bottom side menu, then select **Triggered XY** from the right side menu. Press the **POSITION** keys of **CH1** and **CH2** to align the characteristic at point (0, 0).
13. Record the displayed voltage transfer characteristic (VTC) on Fig. 4.5.
14. Turn the XY mode off by selecting **OFF(YT)** from the right menu.
15. Connect a 10 µF electrolytic capacitor in parallel with R_L; **the capacitor lead marked with negative signs should be connected to ground**. Record the waveforms of v_S(t) and v_O(t) on Fig. 4.6.
16. Replace the 10-µF capacitor with another one that has a capacitance of 470-µF. Record the waveforms of v_S(t) and v_O(t) waveforms on Fig. 4.7.
17. Press on the **CH2** key of the scope. Change the **Coupling** (using the leftmost bottom-menu button) to **AC**. Press the **Autoset** key. Record the output voltage peak to peak ripple $V_{r(pp)}$ in Table 4.1.
18. Disconnect the 4.7-kΩ resistor then connect a 270-Ω resistor in its place. Record the peak to peak ripple of the output voltage $V_{r(pp)}$ in Table 4.1.
19. Calculate the resulting ***change*** in the peak to peak ripple, i.e. $\Delta V_{r(pp)}$, of the output and record that in Table 4.1.

PROCEDURE B: The Effect of Adding a Voltage Regulator

1. Unplug the transformer from the ac power outlet.
2. Append a voltage regulator to the original circuit, relocating R_L to its new position, in parallel with D_Z (see Fig. 4.3). *Initially use the 4.7-kΩ for R_L.*
3. Relocate the two leads of the coaxial cable connected to **CH2** of the scope to put them across R_L in its new position.
4. Reconnect the transformer to an ac power outlet.
5. Change the **Coupling** of **CH2** back to **DC**. Press the **Autoset** key, and then press the **POSITION** key of **CH2**.
6. Record the waveforms of v_S(t) and v_O(t) waveforms on Fig. 4.8.
7. Change the **Coupling** of **CH2** to **AC**. Press the **Autoset** key.
8. Record the peak to peak ripple of the output voltage $V_{r(pp)}$ in Table 4.1.
9. Disconnect the 4.7-kΩ resistor then connect the 270-Ω resistor in its place. Record the peak to peak ripple of the output voltage $V_{r(pp)}$ in Table 4.1.
10. Calculate the ***change*** in the peak to peak ripple, i.e. $\Delta V_{r(pp)}$, of the output and record that in Table 4.1.
11. Compare this change to the change you calculated in the last step of Procedure A.
12. **Unplug the transformer from the ac outlet**; turn off your scope, and then disconnect all components and cables and put them back in place.

RESULTS

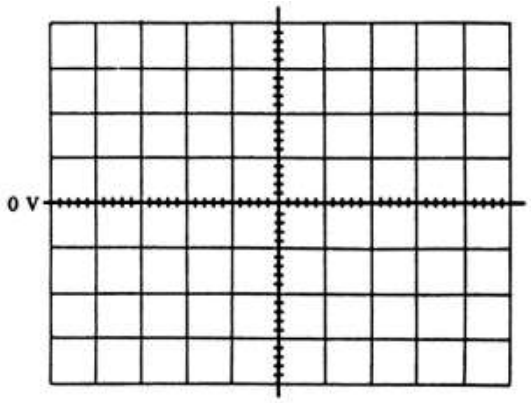


Figure 4.4

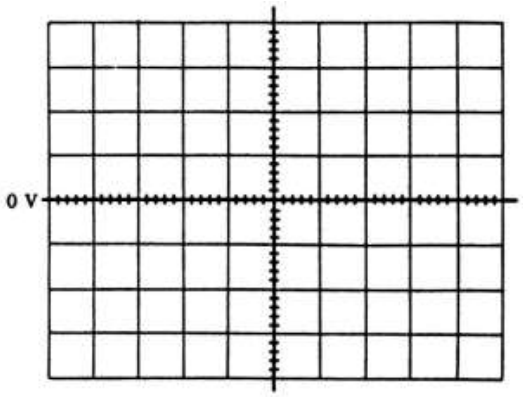


Figure 4.5

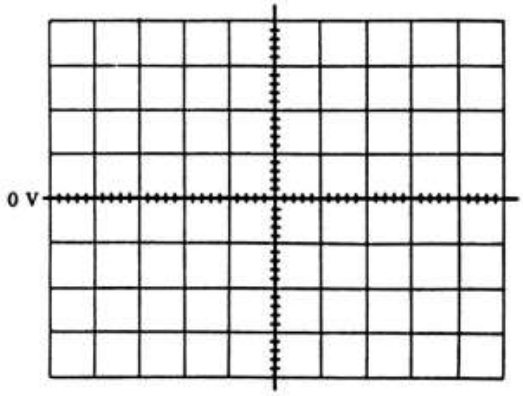


Figure 4.6

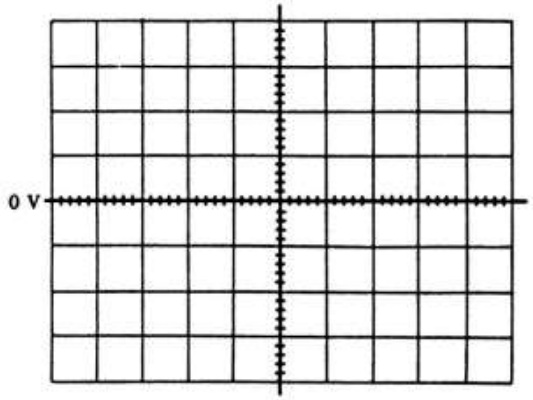


Figure 4.7

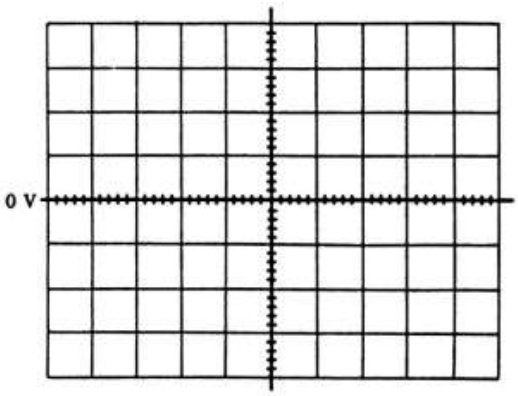


Figure 4.8

Table 4.1

Step	Case		Output Voltage Ripple $V_{r(pp)}$
17	No Voltage Regulator Procedure **A**	C = 470 μF, R_L = 4.7 kΩ	
18		C = 470 μF, R_L = 270 Ω	
19		$\Delta V_{r(pp)}$ =	
8	With Voltage Regulator Procedure **B**	C = 470 μF, R_L = 4.7 kΩ	
9		C = 470 μF, R_L = 270 Ω	
10		$\Delta V_{r(pp)}$ =	

POST LAB QUESTIONS (PLQs)

1) *Procedure A*: What is your observation on the effect of removing D_2 from the circuit? What is the name of the new circuit?
2) *Procedure A*: How does the output peak to peak ripple change if:
    i. The capacitance of the filter capacitor is increased.
    ii. R_L is reduced.
3) *Procedure B:* From your peak to peak ripple measurements $V_{r(pp)}$, comment on the ability of the voltage regulator to reduce the ripple for a load resistance $R_L = 270\ \Omega$ compared to the unregulated case for the same value of R_L.
4) *Procedure B:* From your $\Delta V_{r(pp)}$ calculations, comment on the ability of the voltage regulator to reduce the change in ripple resulting from varying R_L.

5) ***Using the design tips given in section H.2 of Appendix H***, design a regulated power supply (Fig. 4.2) to supply a dc load of $I_L = 20\times$TN mA, where TN is your table number in the lab. The center-tapped transformer available to you has a terminal voltage relative to the center of $V_S = 15$ V (rms).
    a. Calculate V_{SM}, the peak amplitude of V_S.
    b. Calculate the capacitance of the required capacitor based on a peak-to-peak ripple of 0.075 V_{SM}.
    c. Calculate $\mathbf{V_{RRM}}$ for the needed type of diode.
    d. Calculate $\mathbf{I_{F(avg)}}$ for the type of diode you are going to need.
    e. Look for a rectifier diode that can be used for your design using the table given in Appendix D. The selected diode should have maximum reverse voltage and average current ratings slightly above the ones calculated in part **c** and **d**.

EXPERIMENT

5

The *Op*erational *Am*plifier

As an amplifier and a comparator

OBJECTIVES:

- To be acquainted with the essence of operation of the operational amplifier.
- To observe the difference between ideal and real-life op amp.
- To build and test a non-inverting amplifier, measure its closed loop voltage gain and bandwidth, and see the output voltage swing limitation.
- To build and test a voltage comparator employing limiter at its output.

MATERIALS:

- Laboratory setup, including a plug-in panel or a proto-board
- One 741-type operational amplifier (op amp)
- 3 rectifier diodes
- 1 Zener diode
- 2 Resistors (10 kΩ, 100 kΩ)
- Several wires and bridging plugs

INTRODUCTION

The operational amplifier is a versatile *integrated circuit* (IC) that performs various types of analog signal processing tasks. The most popular op amp is the μA 741 (Fig. 5.1), which was introduced by Fairchild in 1968 to become the industry standard.

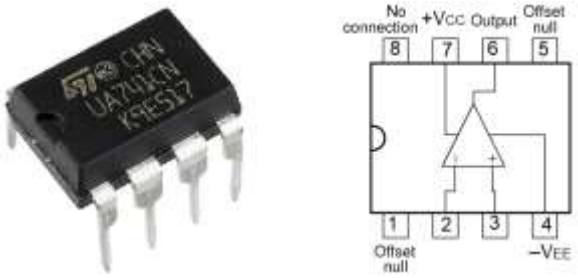
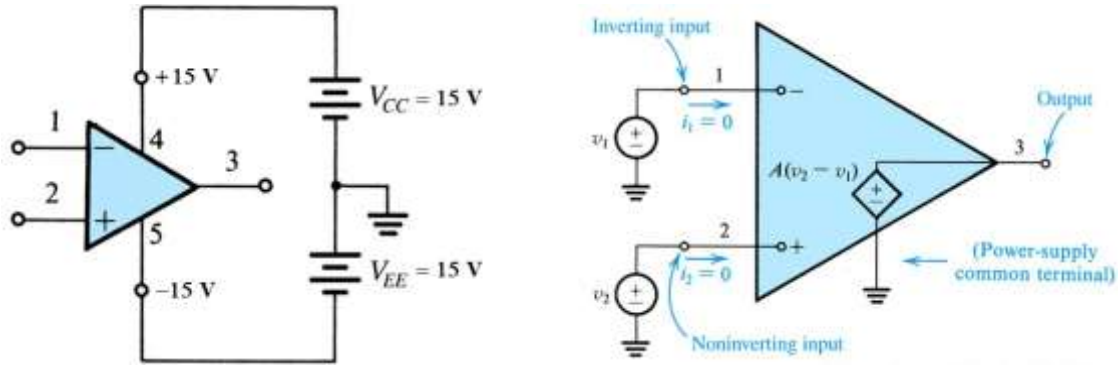


Figure 5.1 The popular μA 741 op amp IC package and its pin diagram.

Figure 5.2a shows the circuit symbol of the op amp, along with its power connection scheme. Powering the op amp with bipolar dual power supplies is the common practice, but a single power supply powering may be used in special applications.

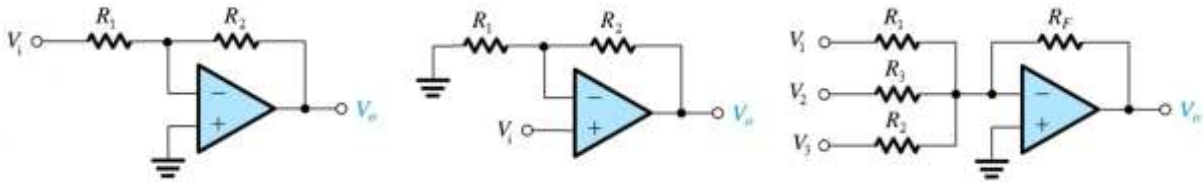
The power supplies bias the internal transistors of the op amp, and provide power to the output load and the feedback network. However, these **power rails** put limits on the maximum and minimum swings of the output voltage. As long as an op amp output is not saturated at its power rails limits, it operates linearly with an equivalent circuit like the one shown in Fig. 5.2b.

As suggested by its idealized equivalent circuit, an op amp **amplifies the difference** between two input voltages and produces a **single-ended output**. It is characterized by a very large **open-loop gain** *A*, a very large input impedance, and a very small output impedance. Very large input impedance implies that the two input terminals draw negligible current. A predictable **closed-loop gain** *G* can be obtained using **negative feedback**. The effect of negative feedback is to make the closed-loop gain almost entirely dependent on external components (e.g., R_1 and R_2). Presuming negative feedback with G << A, and v_o not saturated at the power rails limits, the two input terminals can be assumed to have the same potential (**virtual short**).



(a) (b)

Figure 5.2 (a) Circuit symbol and **(b)** equivalent circuit of the ideal op amp [1].



(a) (b) (c)

Figure 5.3 **(a)** Inverting amplifier. **(b)** Non-inverting amplifier. **(c)** Weighted summer.

Figure 5.3 shows a variety of op amp circuits: (a) an **inverting amplifier**, (b) a **noninverting amplifier**, and (c) a **weighted summer**. Please refer to the textbook by Sedra and Smith for further details [1].

In the first part of this experiment, we shall assemble and test a noninverting amplifier (Fig. 5.4a). In that circuit, since the inverting terminal of the op amp draws negligible current (ideally zero), R_1 and R_2 are effectively in series. Thus, after invoking the virtual short circuit property, v_i and v_O are related by a simple voltage divider rule,

$$v_I = v_O \frac{R_1}{R_1 + R_2}$$

$$\rightarrow G \equiv \frac{v_o}{v_I} = \frac{R_1 + R_2}{R_1} = 1 + \frac{R2}{R1} \qquad (5.1)$$

If **no** negative feedback is employed, an op amp is said to operate in the **open-loop** mode. In that mode, an op amp basically works as a comparator:

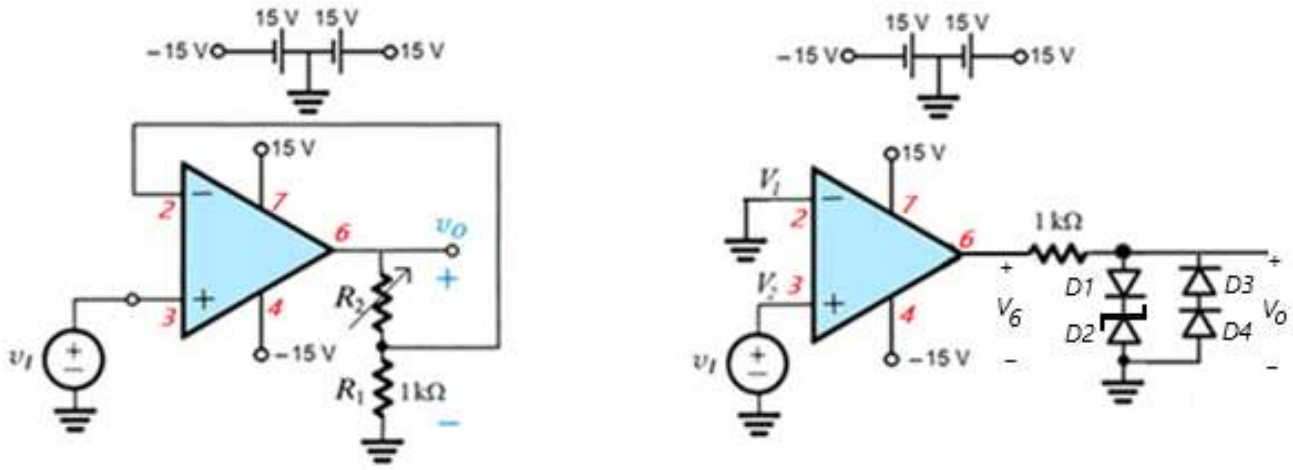
- If $v_2 > v_1$ by more than few tens of microvolts $\rightarrow v_O = V_O^+ \approx V_{CC}$
- If $v_1 > v_2$ by more than few tens of microvolts $\rightarrow v_O = V_O^- \approx -V_{EE}$

In the second part of this experiment, we shall assemble and test a comparator (Fig. 5.4b) as an example of the second main application of an op amp.

In that circuit, if $V_2 > V_1$ by more than few tens of microvolts (which occurs for positive half cycles) $\Rightarrow v_O \approx V_{CC} \Rightarrow D_1$ and D_2 conduct, and the output voltage gets approximately limited to **(0.7 + V_{Z2}) V**. The op amp **sources** a current in this case.

On the other hand, if $V_1 > V_2$ by more than few tens of microvolts (which occurs for negative half cycles) $\Rightarrow V_O \approx -V_{EE} \Rightarrow D_3$ and D_4 conduct, and the output voltage gets approximately limited to **– (0.7 + 0.7)** V. The op amp **sinks** a current in this case.

PROCEDURE A: Non-inverting Amplifier



(a) (b)

Figure 5.4 **(a)** The noninverting amplifier. **(b)** Comparator with limited output range.

1. Construct the circuit shown in Fig. 5.4a; start off with a value of 10 kΩ for R_2.
2. Using a waveform generator, apply a 1-kHz, 400 $mV_{p\text{-}p}$ sinusoid to the noninverting terminal of your op amp.
3. Turn on your scope; wait for few seconds, press the **Default** key, and then the **Autoset** key. Display $v_I(t)$ and $v_O(t)$ on **CH1** and **CH2** respectively.
4. Use the **SCALE** knob of **CH1** to set its vertical sensitivity at **100 mV/div**.
5. Use the **SCALE** knob of **CH2** to set its vertical sensitivity at **500 mV/div**.
6. Press the **Measure** key on your scope → **Add Measurement** → Activate the **Source 1** submenu → Use the **VARIABLE** knob to highlight **CH2** then press **Select** → Activate the **V/I** submenu → Highlight the appropriate item (**Pk-Pk, Max.** and **Min.**), then press the **Select** key to add anyone of them.
7. Switch the dc power supply, then the waveform generator on.
8. Record $v_I(t)$ and $v_O(t)$ on Fig. 5.5.
9. Record the peak-to-peak amplitude for the input and output voltage waveforms, then calculate the voltage gain *G*:

$$G = \frac{V_{o(\text{pp})}}{V_{i(\text{pp})}} \qquad (5.2)$$

10. Increase the frequency of the input signal until the peak amplitude of the output falls to 0.7 of its value at 1 KHz. Take note of that frequency from the **waveform generator reading**. Record that frequency (the bandwidth BW) in Fig. 5.5.
11. Reduce the frequency of the input signal down to 1 kHz.
12. Increase the scale of **CH2** to **5 V/div**. Replace the 10-kΩ resistor with a 100-kΩ one, then sketch the resulting waveforms of $v_I(t)$ and $v_O(t)$ on Fig. 5.6.
13. Obtain the *upper limiting level* V_O^+ and the *lower limiting level* V_O^- for $v_O(t)$ from the scope Max. and Min. measurements. Record those values in Fig. 5.6.
14. Swich the waveform generator then the the power supply off.

PROCEDURE B: Implementing a Comparator Using an OP-AMP

1. Assemble the circuit shown in Fig. 5.4b.
2. Use the same input signal that you used for part A.
3. Connect the non-inverting input of the op amp to display to **CH1** of the scope, the op-amp output to **CH2**, and the overall circuit output to **CH3**.
4. Add a **Max.** and **Min.** measurement for **CH3** of the scope.
5. Set the the sacle of **CH3** at 2 V/Div.
6. Swich the power supply, and then the waveform generator on.
7. Record $v_I(t)$, $v_6(t)$, and $v_O(t)$ on Fig. 5.7.
8. Obtain the upper and lower limiting levels for $v_6(t)$, and $v_O(t)$ from the oscilloscop Max. and Min. values, and then record them in lower part of Fig. 5.7.
9. Swich off the waveform generator then the the power supply off.

RESULTS

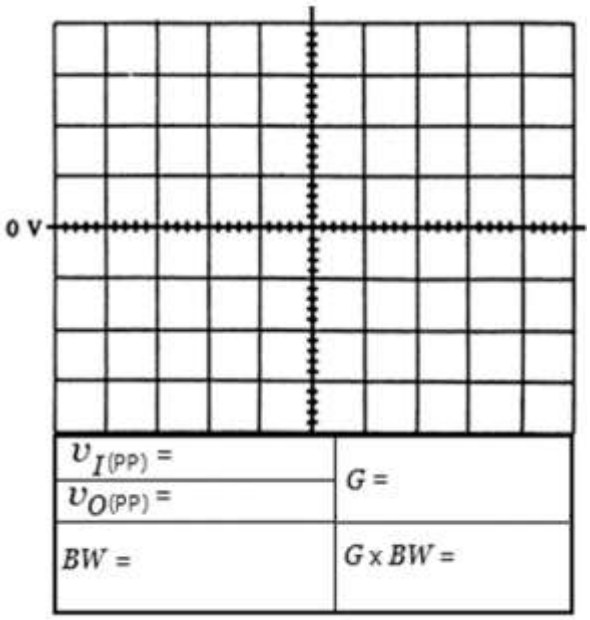


Figure 5.5

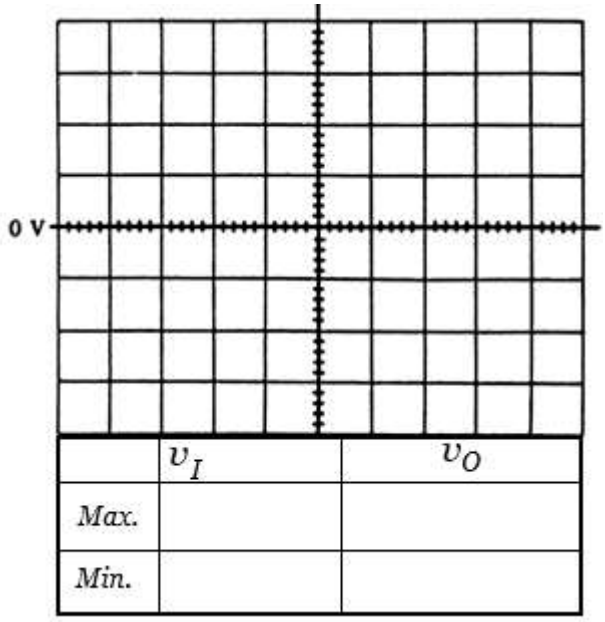


Figure 5.6

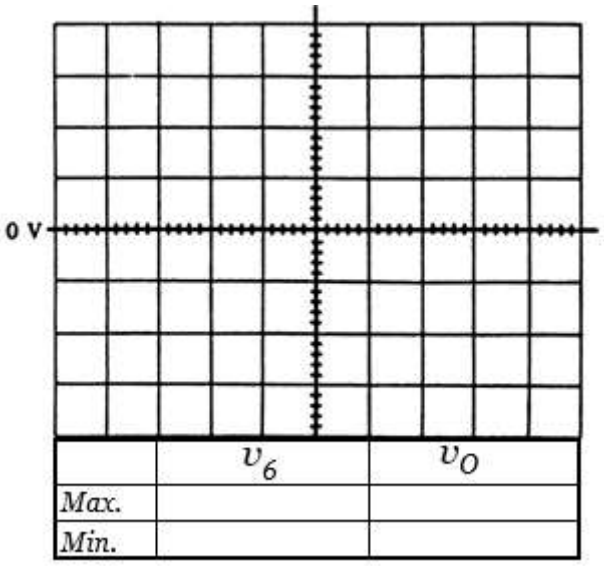


Figure 5.7

POST LAB QUESTIONS (PLQs)

1) *Procedure A*: Calculate the closed loop gain for the R_2 = 10 kΩ case using equation 5.2 and compare it to theoretical gain given by equation 5.1. Comment on the cause of the difference.
2) *Procedure A*: Calculate the gain bandwidth product for the R_2 = 10 kΩ case.
3) *Procedure A*:
    - State the cause of clipping (limiting) in the output voltage waveform resulting from increasing R_2 to 100 kΩ.
    - How are the limiting levels of $v_O(t)$ related to the power rails ± 15 V?
4) *Procedure A*: Compare the op amp you have used in the lab with an ideal op amp model in terms of bandwidth and output-voltage swing range.
5) *Procedure B*: Justify the value of the upper limiting levels of $v_6(t)$ and $v_O(t)$.
6) *Procedure B*: Justify the value of the lower limiting levels of $v_6(t)$ and $v_O(t)$.

EXPERIMENT

6

SPICE Simulation I

OBJECTIVES:

- To learn how to simulate electronic circuits using freely-available online design tools, the LTspice® software as an example.
- To see why a voltage regulator is needed in ac to dc conversion systems with the aid of simulation as a study tool.

MATERIALS:

- A computer with LTspice® installed. You can download LTspice from here:
- https://www.analog.com/en/design-center/design-tools-and-calculators/ltspice-simulator.html#

INTRODUCTION

In this experiment, we shall explore the importance of using a **voltage regulator** in the process of ac to dc conversion with the aid of computer simulation. We shall implement a simple voltage regulator using a Zener diode. If operated in the reverse breakdown, a Zener diode effectively behaves like a voltage reference owing to the semi-vertical shape of its characteristic curve in that region, which implies that the Zener diode will maintain a constant voltage, V_Z, across its terminals for a wide range of reverse breakdown currents.

LTspice® is a high performance SPICE simulation software, schematic capture and waveform viewer with enhancements and models for easing the simulation of analog circuits. Included in the download of LTspice are macromodels for a majority of Analog Devices switching regulators, amplifiers, as well as a library of devices for general circuit simulation.[3]

To place a new component on your schematic, you can either press F2 or go to Edit >> Component. To place a new resistor on your schematic, you can either press 'R' on the keyboard or go to Edit >> Resistor. To place a new capacitor on your schematic, you can either press 'C' or go to Edit >> Capacitor. To place a new inductor on your schematic, you can either press 'L' or go to Edit >> Inductor. To place a new diode on your schematic, you can either press 'D' or go to Edit >> Diode. To change the model of the diode, right click on it, and then click on Pick New Diode. To place a new voltage source on your schematic, you can either press 'v' or press F2, and then select voltage, then click OK. To place a ground (GND) on your schematic, you can either press 'G' or go to Edit >> Place GND. To wire the components, you can either press F3 or go to Edit >> Draw wire.

When you wish to move, mirror, rotate, drag or delete objects, first select the move, drag or delete command form the top Toolbar. Then you can select an object by clicking on it. You can select multiple objects by dragging a box about them. The program will stay in the move, drag, or delete mode until the right mouse button is clicked or the Esc key is pressed.

Simulating a circuit by LTspice involves the following basic steps:

1. Opening a new schematic (File >> New Schematic), placing required components, wiring them, and adding a ground.
2. Setting an appropriate SPICE analysis (Edit >> SPICE Analysis). The basic analysis types are: DC operating point (DC op pnt), DC sweep, AC analysis, and transient.
3. Running the simulation, and then selecting the desired response.
4. Saving the circuit before closing the program.

[3] https://www.analog.com/en/design-center/design-tools-and-calculators/ltspice-simulator.html#

PROCEDURE A: Unregulated dc Supply

1. Double click the LTspice icon on the desktop. Go to File >> New Schematic.
2. First, you have to add all components required to build the circuit (Fig. 6.1) on your schematic, except for the capacitor.[4] To add a coupling coefficient between the two inductors, go to Edit >> SPICE directive >> type: K1 L1 L2 1.
3. Add a circuit ground (GND), wire all components, and label the **S1**, **S2**, and the **out** nodes.
4. To adjust a component value, right click on it then type the required value. In SPICE, meg stands for 1e6 (mega), k stands for 1e3 (kilo), m stands for 1e-3 (milli), u stands for 1e-6 (micro), n stands for 1e-9 (nano), and p stands for 1e-12 (pico).
5. To adjust the voltage source parameters, right click on it, and then fill in the required fields, which are: (DC offset = 0, Amplitude = 311, Freq = 60, Tdelay = 0, Theta = 0, Phi = 0, Ncycles = 12).
6. Go to Edit >> SPICE analysis, and then fill in the required fields, which are: (Stop time = 0.2 , Time to start saving data = 0, Maximum Timestep = 0.2m).

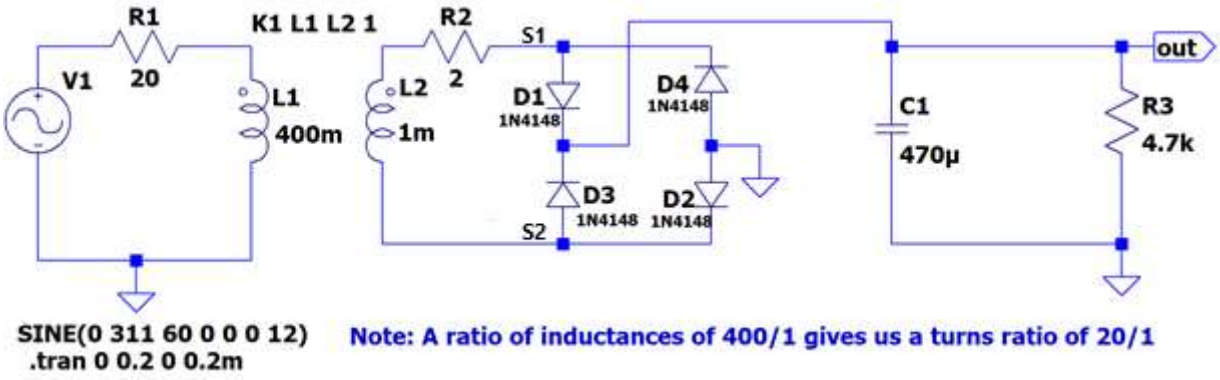


Figure 6.1 Bridge rectifier circuit.

7. To run your simulation, go to **Simulate** >> **run**.
8. Click on the **out** node on the circuit schematic to display the waveform of the output voltage.

[4] See the introduction on the previous page, or watch:
https://lms.ksu.edu.sa/bbcswebdav/users/talmadhi/EE312/LTSpice_OP.mp4

9. To display the secondary winding voltage waveform, put the mouse pointer on node S1, then left click and hold while dragging the mouse pointer to node S2, then release the left button of the mouse. What is the peak amplitude of this waveform, i.e., V_{SM}?
10. Similarly obtain and display the voltage difference across D1, V(S1, out). What is the peak inverse voltage **PIV** for D1? Note that it is equal in magnitude to the peak amplitude across the transformer secondary winding. The **PIV** must be less than the **VRRM** of the diode by at least 25% (See Fig. 6.3 for the absolute maximum ratings for the N4148).
11. Connect the capacitor as shown in Fig 6.1, and rerun the simulation to see how that affects the ripple in the output voltage waveform. Experiment with lower values of capacitance (10 μF for example) to see the effect on the output voltage waveform ripple.
12. Put the mouse pointer on diode D1 (note how pointer changes to a clamp meter shape) then left click on the diode. The resulting waveform is the diode's current. Note how the current peaks to a surge value at the beginning of the simulation.[5] The diode must have an $\mathbf{I_{FSM}}$ rating 1.5 times that surge current.
13. On the waveforms window, click and drag your mouse diagonally to show only steady state cycles (without surge). Press and hold the Ctrl key then click on I(D1). You should see a small window displaying the average value of the current waveform. That average current should not be greater than 0.7 $\mathbf{I_{F(AV)}}$ rating of the diode.
14. Press Ctrl+E to return to the original view.
15. Delete the I(D1) and V(S1,out) traces.
16. Decease the value of R_1 to 270 Ω and rerun to see how that affects the ripple in the output.

[5] This surge is caused by the capacitor initially drawing a very high charging current. If, the surge current is larger than $\mathbf{I_{FSM}}$ of the diode, a properly sized surge resistor must be used.

PROCEDURE B: Adding a Simple Voltage Regulator

1. Add a voltage regulator to your dc supply circuit as shown in Fig. 6.2. Change R_1 back to 4.7 kΩ. Rerun the simulation. Read off and record the dc value of $v_{out}(t)$.

Vout (dc) =

2. Reduce R_1 to 270 Ω and simulate the circuit again to see how that affects the ripple in the output. Read off and record the dc value of $v_{out}(t)$.

Vout (dc) =

3. Save the circuit schematic before you close the program.[6]

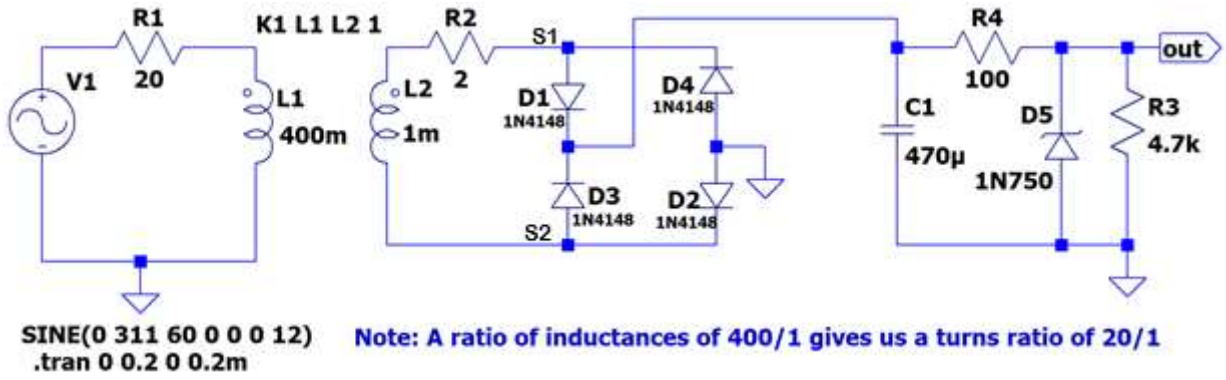


Figure 6.2 Regulated power supply.

ABSOLUTE MAXIMUM RATINGS (T_{amb} = 25 °C, unless otherwise specified)				
PARAMETER	**TEST CONDITION**	**SYMBOL**	**VALUE**	**UNIT**
Repetitive peak reverse voltage		V_{RRM}	100	V
Reverse voltage		V_R	75	V
Peak forward surge current	t_p = 1 µs	I_{FSM}	2	A
Repetitive peak forward current		I_{FRM}	500	mA
Forward continuous current		I_F	300	mA
Average forward current	V_R = 0	$I_{F(AV)}$	150	mA
Power dissipation	l = 4 mm, T_L = 45 °C	P_{tot}	440	mW
	l = 4 mm, $T_L \leq$ 25 °C	P_{tot}	500	mW

Figure 6.3 Absolute Maximum Ratings for the 1N4148 diode.

[6] If you would like to save the circuit under a different name, go to File >> Save As, type the name, and then click OK.

PROCEDURE C: Voltage Comparator

1. Go to File >> New Schematic.
2. First, you have to add all components required to build the circuit (Fig. 6.4) on your schematic, including a circuit ground.
3. Wire all components, and label the **in**, **IC_out** and the **out** nodes.
4. To adjust the voltage source parameters, right click on it, and then fill in the required fields, which are: (DC offset = 0, Amplitude = 1, Freq = 1000, Tdelay = 0, Theta = 0, Phi = 0, Ncycles = 3).
5. Go to Edit >> SPICE analysis, and then fill in the required fields, which are: (Stop time = 3m , Time to start saving data = 0, Maximum Timestep = 3u).
6. Click on the **in, IC_out,** and **out** nodes respectively to display their waveforms.
7. Flip D2 upside down, then change its model to 1N750 (Zener diode).
8. Run the simulation again. Compare the resulting output voltage waveform to that you obtained for procedure B of the op amp experiment.
9. Put the mouse pointer on R_1 (note how pointer changes to a clamp meter shape) then left click on it. The resulting waveform is the op-amp's output current. What is its maximum value? Make sure it is less than 20 mA, which is the maximum output current rating for the majority of general-purpose op amps.
10. Save the circuit schematic before you close the program.

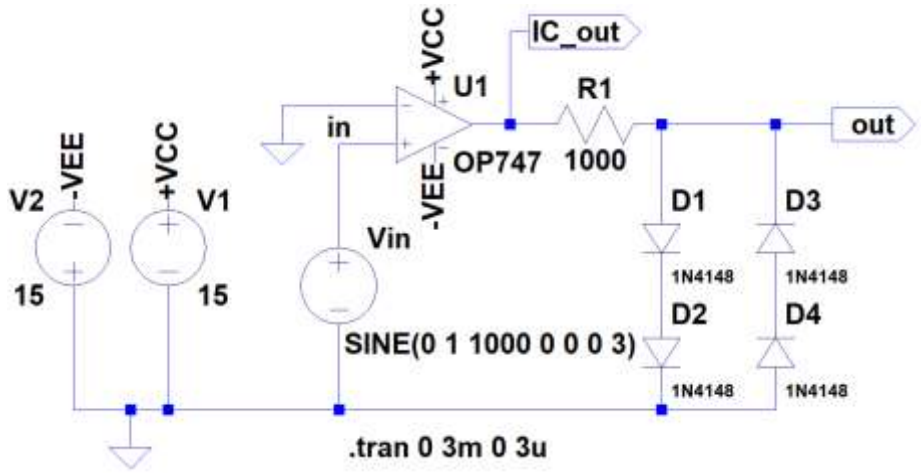


Figure 6.4 A comparator employing a limiter at its output.

POST LAB QUESTIONS (PLQs)

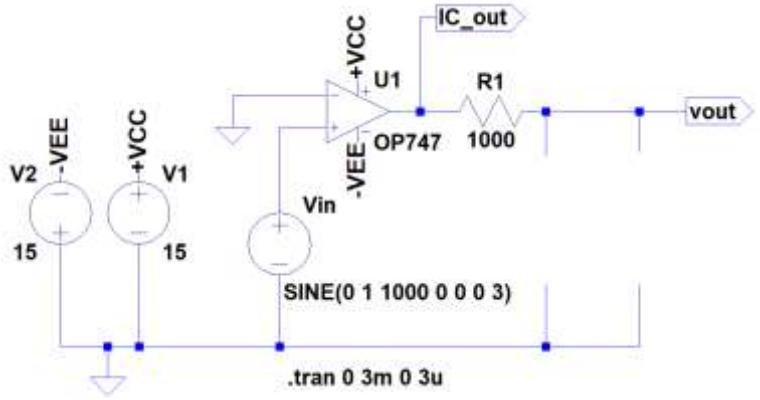


Figure 6.5

Q1) The circuit shown in Fig. 6.5 can be used to illustrate how an op-amp can be used a comparator (Exp. 5) to create a square wave whose limits are designed according to limiting properties of diodes (Exp. 3).

a. Design and sketch a circuit diagram for a voltage limiter whose output is kept **approximately** within the limits of the following range:

$$\mathbf{-1.4}\ \text{V} \le V_O \le \mathbf{5.4}\ \text{V},$$

Use appropriate number of regular silicon diodes, and a single Zener diode operating in the reverse breakdown region. Assume a forward voltage drop of about 0.7 V for the two types of diodes and a nominal breakdown voltage $V_Z = 4.7$ V for the Zener diode. Use two branches of appropriately connected diodes, branch A that conducts current **only** for positive half cycles of the input to give you the upper limiting level and branch B that conducts current **only** for negative half cycles of the input to give you the lower limiting level.

b. Perform an LTspice ***transient*** analyses to verify that your proposed circuits meet specifications. In your simulation, use diode **1N4148** for a regular silicon diode and **1N750** for a Zener diode. To apply a sinusoidal waveform, right click on the voltage source >> advanced, and then select SINE. Fill in the required parameters as follows: DC offset = 0, Amplitude = 1, Freq = 1000, Tdelay = 0, Theta = 0, Phi = 0, Ncycles = 3. To obtain the transient analysis, click on Edit >> SPICE analysis >> Transient, then fill in the required fields: Stop time = 3m, Time to start saving data = 0, Maximum Timestep = 3u.
c. Run the simulation. Click on the input, IC_out, and vout nodes to display their simulated waveforms. Include the bitmaps of your results in your Word document.
d. From the above simulation results, find the upper and lower limiting levels. Watch the following videos if you need further help:
https://lms.ksu.edu.sa/bbcswebdav/users/talmadhi/EE312/LTSPice_TRAN.mp4

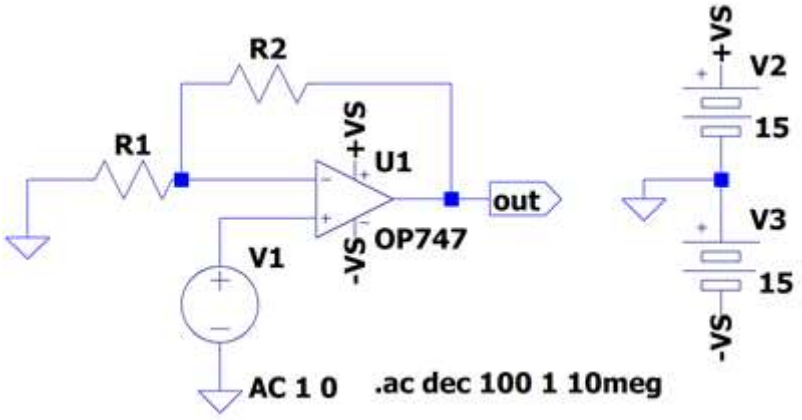


Figure 6.6

Q2) For the the non-inverting amplifier shown in Fig. 6.6 consider R_1 to have a value of **1 kΩ**. Use LTSpice, to perform an ***AC Analysis*** to measure its gain and bandwidth for the following cases.

a. R_2 = **2** × TN kΩ,
b. R_2 = **4** × TN kΩ, where TN stands for your table number in the lab.
c. Comment on how the bandwidth changes if the closed loop voltage gain increases.

For the ***AC analysis*** options, use decade for ***type of sweep***, 100 for ***number of points per decade***, 1 for start frequency and 10meg for stop frequency.

Watch the following video if you need help:
https://lms.ksu.edu.sa/bbcswebdav/users/talmadhi/EE312/LTSpice_AC.mp4

EXPERIMENT

7

Terminal Characteristics of Enhancement N-MOSFET

OBJECTIVES:

- To be get acquainted with the basic principles of operation of a typical enhancement *n*-channel MOS transistor.
- To experimentally obtain the necessary measurements to plot the i_D-v_{DS} characteristics for a sample *n*-channel MOS transistor.
- To learn how to calculate important parameters from the experimentally-obtained data.

MATERIALS:

- Laboratory setup, including a plug-in panel or a breadboard
- One NMOS transistor (e.g., IRF620)
- 1 Resistor (1 kΩ)
- Several wires and bridging plugs

INTRODUCTION

There are two major types of three-terminal semiconductor devices: the MOSFET and the BJT, which we shall study in experiment 8. The basic principle involved in a transistor is using the voltage between two terminals of the transistor to control the current flowing into the third terminal [1].

In an *n*-channel enhancement-type MOSFET, the **gate**-to-**source** voltage, v_{GS}, controls the current flowing through the **drain**, i_D. The value of v_{GS} at which a conducting channel between the drain and the source is induced is called the threshold voltage, V_t. If $v_{GS} < V_t$, the device is said to be **cutoff**. If $v_{GS} > V_t$, however, the device could be either in **triode** or in **saturation**, depending on the value of v_{DS} compared to the gate-to-source overdrive voltage $v_{OV} \equiv v_{GS} - V_t$. In the triode and cutoff modes of operation, the MOS transistor is used mainly as an **electronic switch**. In the saturation mode, and for sufficiently small signals, the MOS transistor acts as a linear voltage-controlled current source, and consequently can be used as an **electronic amplifier**.

In the saturation mode of operation, i_D is related to v_{GS} and v_{DS} by

$$i_D = \frac{k_n'}{2}\frac{W}{L}(v_{GS} - V_t)^2(1 + \lambda v_{DS}) \qquad (7.1)$$

To increase the magnitude of an amplifier's voltage gain employing a MOSFET, the MOSFET has to have a high **transconducatnce**, g_m. At a constant v_{DS} in the saturation region, g_m quantifies how much the drain current changes for a small change in v_{GS}; thus

$$g_m \equiv \frac{\Delta i_D}{\Delta v_{GS}} = \frac{2I_D}{V_{GS} - V_t} \qquad (7.2)$$

The saturation mode is characterized by a weak dependence of i_D on v_{DS} and a much greater dependence on v_{GS}. Caused by a phenomenon known as **channel-length modulation**, the weak dependence of i_D on v_{DS} –**in the saturation region**–at a constant V_{GS} is linear, and can be modeled by a finite resistance[7] r_o between the drain and the source, which is given by

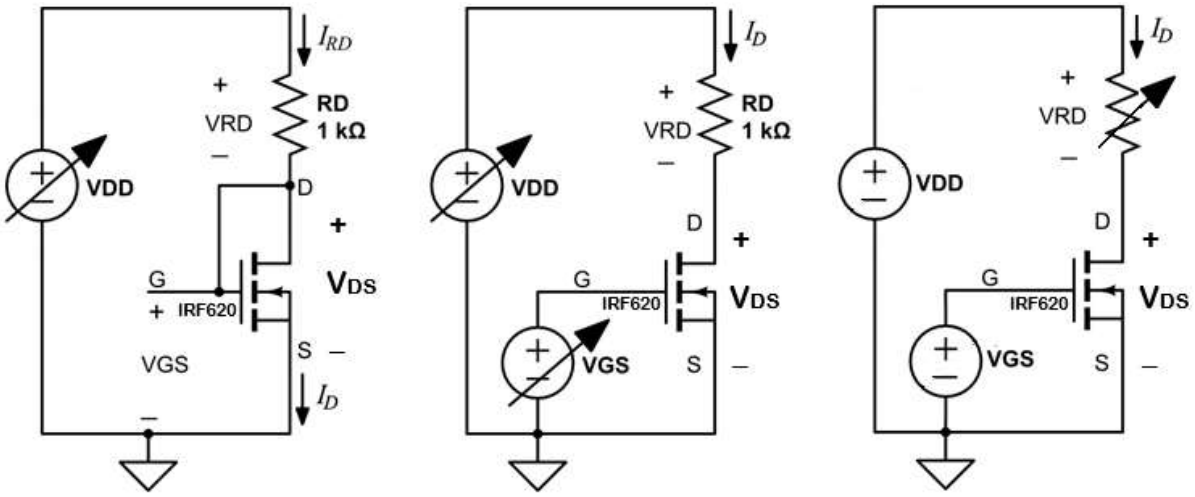
$$r_o \equiv \frac{\Delta v_{DS}}{\Delta i_D} = \frac{1}{\lambda I_D} \qquad (7.3)$$

In the triode mode, for small v_{DS}, the MOSFET effectively acts as a voltage-controlled resistance. Its **closure resistance**[8] r_{DS} is given by

$$r_{DS} \equiv \frac{\Delta v_{DS}}{\Delta i_D} = \frac{1}{k_n' \frac{W}{L} V_{OV}} \qquad [in\ the\ \boldsymbol{triode}\ mode\ \ at\ a given\ V_{GS}\ ] \qquad (7.4)$$

[7] This resistance is important in assessing how effective a MOSFET will be if used in an **amplifier circuit**.
[8] This resistance**, denoted by R_{ON} in datasheets**, is important in **switching applications of the MOSFET.**

PROCEDURE A: Measuring VGS(TH).



(a) (b) (c)

Figure 7.1 Circuit setup for measuring: **(a)** the threshold voltage, **(b)** the i_D-v_{DS} characteristics, and for **(c)** exploring current source behavior**.**

1. Assemble the circuit shown in Fig. 7.1a using an IRF620 as a MOSFET.[9]
2. Switch on your dc supply. Use the CURRENT knob to set the current limit at **0.100** A, and then press the dc switch on.
3. Adjust the output voltage of your dc supply until V_{RD} becomes 0.25V, which translates to a drain current 250 μA.
4. Using a DMM, measure and record the corresponding value of V_{GS}. Note that this value of V_{GS} is usually denoted in datasheets by $\boldsymbol{V_{GS(TH)}}$, and is different than the ***theoretical*** threshold voltage $\boldsymbol{V_t}$ used in equation 7.1.

$$\boldsymbol{V_{GS(TH)}} = \quad\quad$$

5. Switch off the dc power supply.

[9] The IRF620, is a discrete **power MOSFET** whose typical $\boldsymbol{V_{GS(TH)}}$ is around 3 V (see Appendix C), which is relatively large compared to MOSFETs fabricated on an Integrated Circuit (IC). To use this MOSFET for switching a gate-source voltage of around 10 volts is needed as indicated in its datasheet.

PROCEDURE B: Obtaining the Output Terminal Characteristics

1. Remove the connection between the drain and gate and connect the other unit of the dc power supply between the gate and the source terminals of the MOSFET (Fig. 7.1b).
2. Using one unit of the power supply, set and fix V_{GS} at ($\boldsymbol{V_{GS(TH)}}$ **+ 0.15**).
3. Using the other unit of the dc power supply, vary V_{DD} to obtain the V_{DS} values shown in Table 7.1. For each value of V_{DS}, measure and record V_{RD}. *The corresponding values of I_D can be found from Ohm's law.*
4. Increase V_{GS} by **0.15** V. Repeat step 3 while keeping the dc supply V_{GS} fixed.
5. Increase V_{GS} by **another 0.15** V. Repeat step 3 while keeping V_{GS} fixed.
6. Turn off your dc power supply.

PROCEDURE C: The Current Source Behavior

1. Turn the dc supply on. Adjust V_{DD} at 10.0 V and V_{GS} at ($\boldsymbol{V_{GS(TH)}}$ **+ 0.3)**.
2. Start off with a value of 270 Ω for R_D.
3. Using a digital multimeter, measure and record V_{DS} and V_{RD} in Table 7.2. Calculate I_D using Ohm's law.
4. Change R_D to 470 Ω, and then repeat the above step.
5. Change R_D to 1 kΩ, and then repeat the step 3.
6. Switch off your dc supply, disconnect all parts and place them back in their designated places, and tidy up your table.

RESULTS

Table 7.1

V_{GS1} = $\boldsymbol{V_{GS(TH)}}$ + 0.15 V		V_{GS2} = $\boldsymbol{V_{GS(TH)}}$ + 0.3 V		V_{GS3} = $\boldsymbol{V_{GS(TH)}}$ + 0.45 V	
V_{DS} (V)	I_D (mA) = V_{RD} (V)	V_{DS} (V)	I_D (mA) = V_{RD} (V)	V_{DS} (V)	I_D (mA) = V_{RD} (V)
0.00	0.000	0.00	0.000	0.00	0.000
0.05		0.05		0.05	
0.10		0.10		0.10	
0.15		0.15		0.15	
0.30		**0.30**		0.30	
0.45		**0.45**		**0.45**	
0.75		**0.75**		**0.75**	
1.5		**1.5**		**1.5**	
3.00		**3.00**		**3.00**	

Table 7.2 The Effect of varying R_D in the saturation region, while V_{DD} is fixed, on I_D

V_{GS} = $\boldsymbol{V_{GS(TH)}}$ + **0.3** V, V_{DD} = 10 V (fixed)			
R_D	270 Ω	470 Ω	1 kΩ
V_{RD}			
$I_D = \frac{V_{RD}}{R_D}$			
V_{DS}			

POST LAB QUESTIONS (PLQs)

1) Use MATLAB® (see Appendix B) to plot i_D (mA) vs. v_{DS} (V) for each value of v_{GS}.
   i. Label each curve with the corresponding values of v_{GS}.
   ii. Label the three regions of operation on your graph and sketch the boundary between saturation and triode, assuming that $\boldsymbol{V_t \approx V_{GS(TH)}}$.
   iii. Which region represents a voltage-controlled current source behavior?
   iv. Which region represents a voltage-controlled resistance behavior?
2) Using $r_o \equiv \frac{\Delta v_{DS}}{\Delta i_D}$, calculate r_o *for each one of the three curves* given that V_{DS1} = 1.5 V and V_{DS2} = 3 V. Note that r_o is only defined in the **saturation** region.
3) Using $r_{DS} \equiv \frac{\Delta v_{DS}}{\Delta i_D}$, calculate r_{DS} *for each one of the three curves* given that V_{DS1} = 0 V and V_{DS2} = 0.05 V. Note that r_{DS} is only defined in the **triode** region.
4) Based on the above calculations, how do r_o and r_{DS} change with V_{OV}?
5) *Procedure C:*
   i. Why does i_D, which is equal to i_{RD}, stay almost constant despite R_D being increased?
   ii. Calculate the maximum value that can be used for R_D before the MOSFET leaves the saturation region. Assume that $\boldsymbol{V_t \approx V_{GS(TH)}}$.
   iii. In which region the MOSFET operate if R_D is increased beyond the above-calculated value?

EXPERIMENT

8

SPICE Simulation II
MOS Switching Circuits

OBJECTIVES:

- To use a software tool, like LTspice®, to obtain and examine the voltage transfer characteristics (VTC) and the dynamic responses for the:
    - Resistive-load logic inverter, which is basically a common-source circuit.
    - CMOS inverter.
- To explore, with aid of simulation, some factors that affect the dynamic response performance for both circuits.

MATERIALS:

- A Computer with LTspice® installed

INTRODUCTION

Figure 8.1 shows how a logic inverter can be implemented using a voltage-controlled switch. When v_I is low (logic 0), the switch will be open and no current flows in the circuit; therefore, $v_O = V_{DD}$ (logic 1). On the other hand, when v_I becomes high enough to close the switch, current flows in the circuit and v_O will be low (logic 0) due to the closure resistance R_{on} of the switch being significantly lower than R.

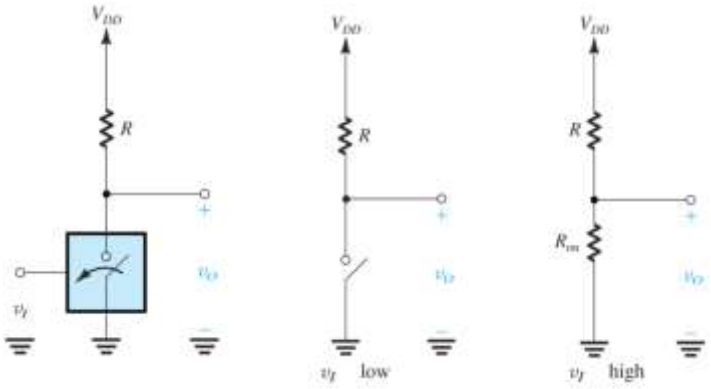


Figure 8.1 Implementing a logic inverter using a resistor and an electronic switch [1].

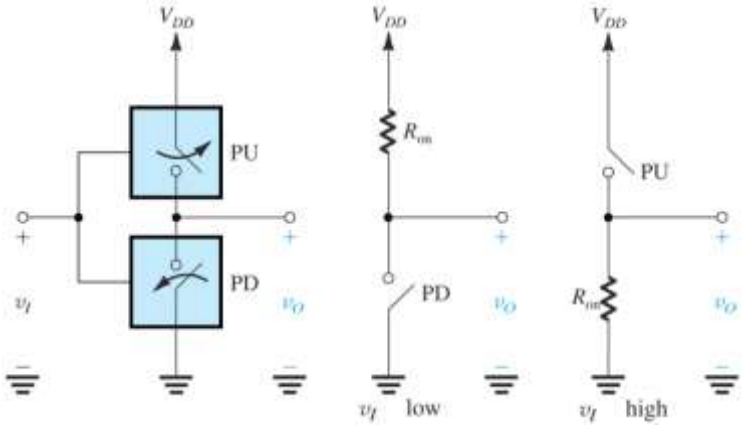
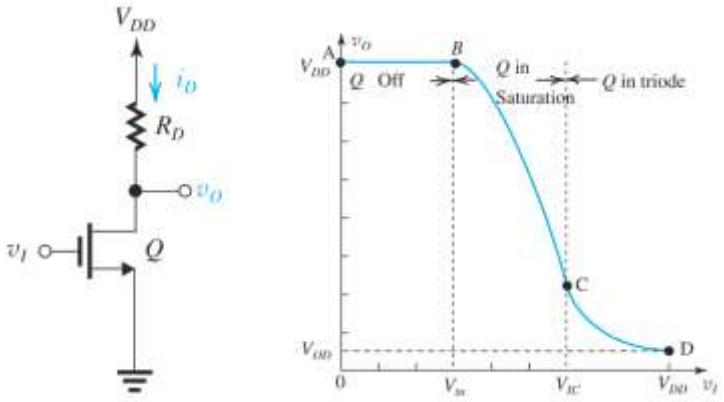


Figure 8.2 Implementing a logic inverter using two complementary electronic switches [1].

Figure 8.2, on the other hand, shows how a logic inverter can be better implemented using two *complementary* voltage-controlled switches. When v_I is low (logic 0), the pull-down (PD) switch will be open and the pull-up (PU) switch closed; therefore, the output node will be pulled up to V_{DD} (logic 1) through a very low resistance R_{on}. On the other hand, when v_I is high (logic 1), the PD switch will be closed and the PU switch open; therefore, the output will be pulled down to ground (logic 0) through R_{on}.

The **Resistive-load inverter** (Fig. 8.3a.), which is basically **a common-source** (CS) circuit, can be used to implement the one-switch logic inverter. In such a circuit, an MOS transistor is employed as voltage-controlled switch. The voltage transfer characteristic of this circuit is shown in Figure 8.3b [1]. The VTC has two extreme regions, where the circuit can be used as a digital logic inverter, and a transition region in the middle where the circuit can be used as an inverting amplifier. For $v_I < V_t$ (segment A to B), the MOS transistor is cut off; hence acting as an **open**

switch. For $v_I \geq V_t$ and less than V_{IC} (segment B to C), v_{GD} is less than V_t ; therefore, the MOSFET is saturated, and the circuit can be used to **amplify** a small signal riding over an appropriate dc offset voltage. For $v_I \geq V_{IB}$ (segment B to C), the MOS transistor operates at $v_{GD} > V_t$, which is a characteristic of the triode region; hence the device is effectively acting as a **closed switch** with a voltage-dependent closure resistance, $\boldsymbol{r_{DS}}$.



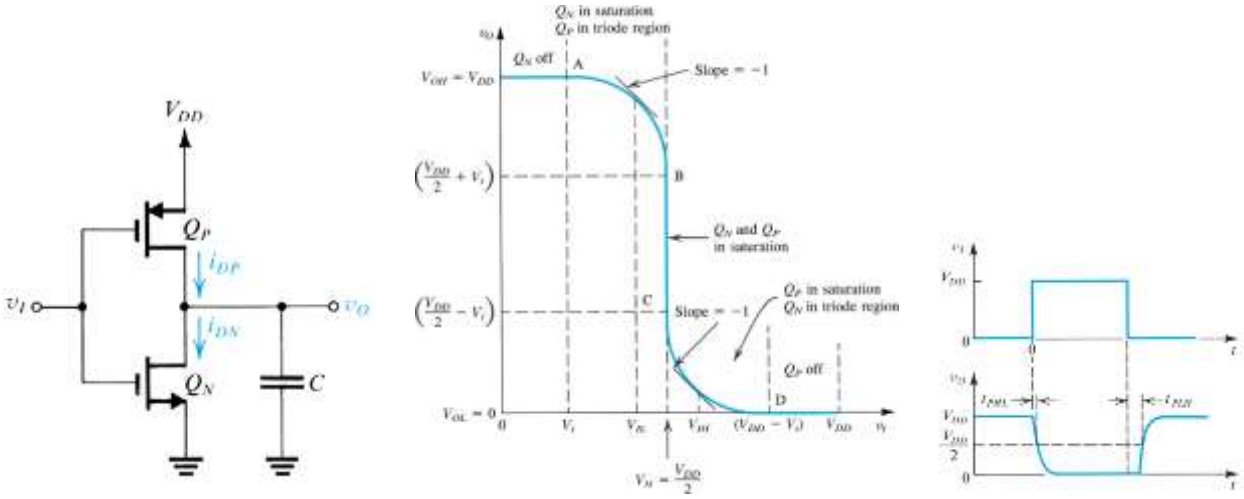
(a) (b)

Figure 8.3 **(a)** The CS circuit and **(b)** its transfer characteristic [1].

On the other hand, the two complementary switches based logic inverter shown in Fig. 8.2 can naturally be implemented by the **CMOS** inverter (Fig. 8.4a). In a CMOS inverter circuit, an NMOS transistor is used as a PD switch, while a PMOS device is used as complementary PU switch. The voltage transfer characteristic of this circuit is shown in Figure 8.4b [1]. The most distinctive feature of this circuit is that the two complementary MOS transistors are not turned on simultaneously– making it possible for the circuit to have very minimized (ideally zero) static power dissipation.

Figure 8.4c shows the dynamic response of the CMOS inverter. Evidently, there are **propagation delays** between the two waveforms due the nonzero switching time of the transistor, which is strongly dependent on its *W/L* ratio. Furthermore, there is capacitance that is inevitably present between the output node and ground due to wiring and device parasitic capacitances. These delays are denoted t_{PHL} (for the output going from Hi to Low) and t_{PLH} (for the output going from Low to Hi). The propagation delay can be approximated by $0.7 \times R_{on(avg)} \times C$, where $R_{on(avg)}$ is the **average switching resistance** of the relevant MOS transistor, which is proportional to (*L/W*) of the transistor.

To have equal propagation delays (i.e., $t_{PHL} = t_{PLH}$), an inverter has to have a matched MOS transistors pair; that is, $W_p = (\mu_n/\mu_p)\, W_n \approx 3W_n$. If two MOS transistors are needed to be connected in series, as in the PD network of a NAND gate or the PU network of a NOR gate, the individual widths have to be doubled.



(a) (b) (c)

Figure 8.4 **(a)** The CMOS inverter, **(b)** its transfer characteristic, and **(c)** its dynamic response. [1].

PROCEDURE A: The Common Source Circuit

1. Download and save the following file containing the SPICE models of a one-µm process:[10] https://lms.ksu.edu.sa/bbcswebdav/users/talmadhi/mosfet1u.lib
   - ❖ You are going to need the path of the file you just downloaded when you add the **.lib** directive.
2. Double click the LTspice icon on the desktop. Go to File >> New Schematic.

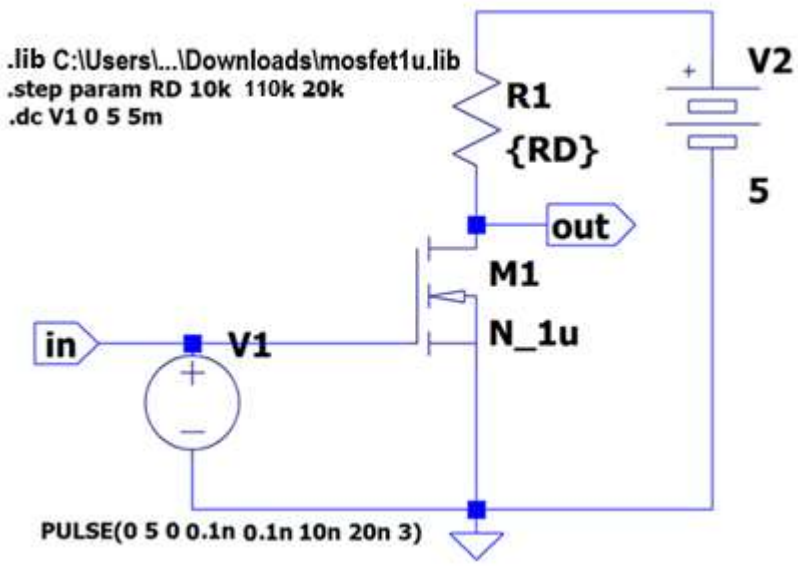


Figure 8.5 Implementing the CS circuit in LTspice.

3. Place all of the components required to build the circuit of Fig. 8.5 on your schematic. To obtain the N_1u MOSFET, add an nmos4 component, then right-click on the NMOS default name and change it to N_1u.
4. Wire all components, and label the **in** and **out** nodes.
5. To add the **.lib** and **.step** commands, go to Edit >> .op SPICE directive'S', or just press the 'S' key, then type the .lib command as shown in Fig. 8.5, then press Ctrl+M and type the .step command.
6. Right click on V1, then select *Advanced*, and then select PULSE. Fill in the required fields, which are: (Vinitail = 0, Von = 5, Tdelay = 0, Trise = 0.1n, Tfall = 0.1n, Ton = 10n, Tperiod = 20n, Ncycles = 3).

[10] To examine the contents of this file, visit https://lms.ksu.edu.sa/bbcswebdav/users/talmadhi/EE312/mosfet1u.txt.

7. To add a DC sweep command (**.dc**),[11] go to Edit >> SPICE analysis, and then fill in the required fields, which are: (Name of 1st source to sweep = V1, Type of sweep = Linear, Start value = 0, Stop value = 5, Increment = 5m), then click OK.
8. To run your simulation, go to Simulate >> run, or just click on the run icon.
9. Place the mouse pointer on the out node on the schematic, and then make a left click to see the VTC curves of the circuit.
    - How does the value of R_D in the transition region affect the magnitude of the slope, and hence the voltage gain of the circuit?
    - How does the value of R_D affect output low level V_{OL}?
10. To do a **transient analysis**, go to Edit >> SPICE analysis, click on the Transient tab, and then fill in the required fields, which are: (Stop time = 60n[12] , Time to start saving data = 0, Maximum Timestep = 0.01n).
11. Click on the run icon.
12. Place the mouse pointer on the **out** node on the schematic, and then make a left click by the mouse to see the output voltage waveforms.
13. Place the mouse pointer on the **in** node on the schematic, and then make a left click to see the input voltage waveform.
    - Which logic gate does the circuit implement?
14. Use the mouse to zoom in (click and drag to draw a rectangular box) around the transition area, where the input falls and the output waveforms rise)
    - Which value of R_D corresponds to maximum propagation delay?
    - Summarize the effect of R_D on the circuit performance if used as an amplifier and if used as a logic inverter (NOT gate).
15. Save your circuit by pressing Ctrl+S.

[11] In order to obtain the voltage transfer characteristic (VTC) of the circuit.
[12] This can be found by multiplying Tperiod by Ncycles.

PROCEDURE B: The CMOS Logic Inverter

1. Go to File >> New Schematic.

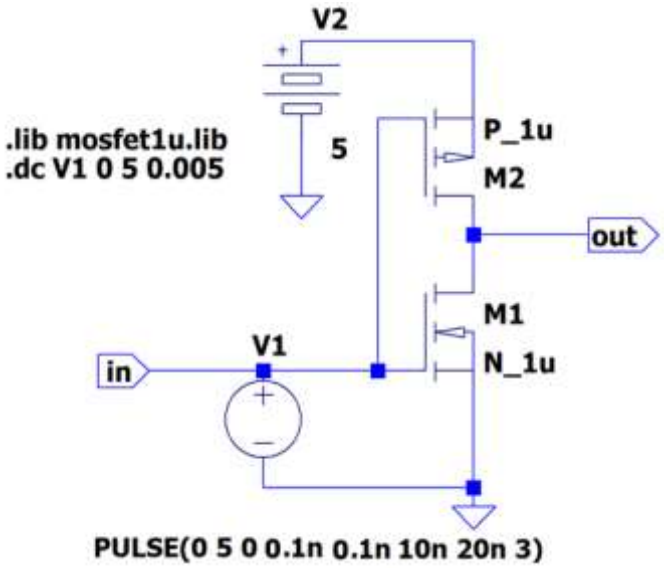


Figure 8.6 Implementing the CMOS inverter (NOT gate) in LTspice.

2. Place all of the components required to build the circuit of Fig. 8.6 on your schematic. To obtain the N_1u MOSFET, add an nmos4 component, then right-click on the NMOS default name and change it to N_1u. To obtain the P_1u MOSFET, add pmos4 component, then right-click on the PMOS default name and change it to P_1u.
3. Wire all components, add a ground (GND), and label the **in** and **out** nodes.
4. Adjust V1 as you have done in procedure A.
5. Add the **.lib** command as you have done in procedure A.
6. Add the **.dc** command as you have done in procedure A.
7. Click on the run icon.
8. Place the mouse pointer on the out node on the schematic, and then make a left click to see the VTC curve of the circuit.

We will now do a transient analysis to see the dynamic response of the circuit.

9. Go to Edit >> SPICE analysis, click on the Transient tab, and then fill in the required fields, which are:
   (Stop time = 60n, Time to start saving data = 0, Maximum Timestep = 0.01n).
10. Click on the run icon.
11. Place the mouse pointer on the **out** node on the schematic, and then make a left click by the mouse to see the output voltage waveforms.
12. Place the mouse pointer on the **in** node on the schematic, and then make a left click to see the input voltage waveform.
    - *Which logic gate does the circuit implement?*
13. Use the mouse to zoom in (click and drag to draw a rectangular box) around the transition area, where the input falls and the output waveforms rise)
    - *Find the propagation delay when the output changes from low to high, i.e., t_{PLH}.*
14. Save your circuit by pressing Ctrl+S.

POST LAB QUESTIONS (PLQs)

1) *Procedure A:* Summarize the effect of increasing R_D on voltage gain if the circuit is used as an amplifier, and on the low-to high propagation delay if it used as a logic inverter (NOT gate).
2) Mention three advantages of the CMOS inverter (Fig 8.6) over the inverter shown in Fig 8.5.
3) Do an LTspice transient analysis of the circuit shown in Fig. 8.7. Measure the high-to-low propagation delay $\boldsymbol{t_{PHL}}$ between *V(in)* and *V(out)*, and $\boldsymbol{t_{PHL}}$ between *V(in)* and *V(inv1_out)*.
4) Measure the low-to-high propagation delay $\boldsymbol{t_{PLH}}$ between *V(in)* and *V(out)*, and $\boldsymbol{t_{PLH}}$ between *V(in)* and *V(inv1_out)*.
5) What is your observation regarding the propagation delay of the output voltage waveform of the third inverter *V(out)* compared to that of the first inverter *V(inv1_out)*?

You need to submit screen captures of your schematics and input/output waveforms for Q3 and Q4.

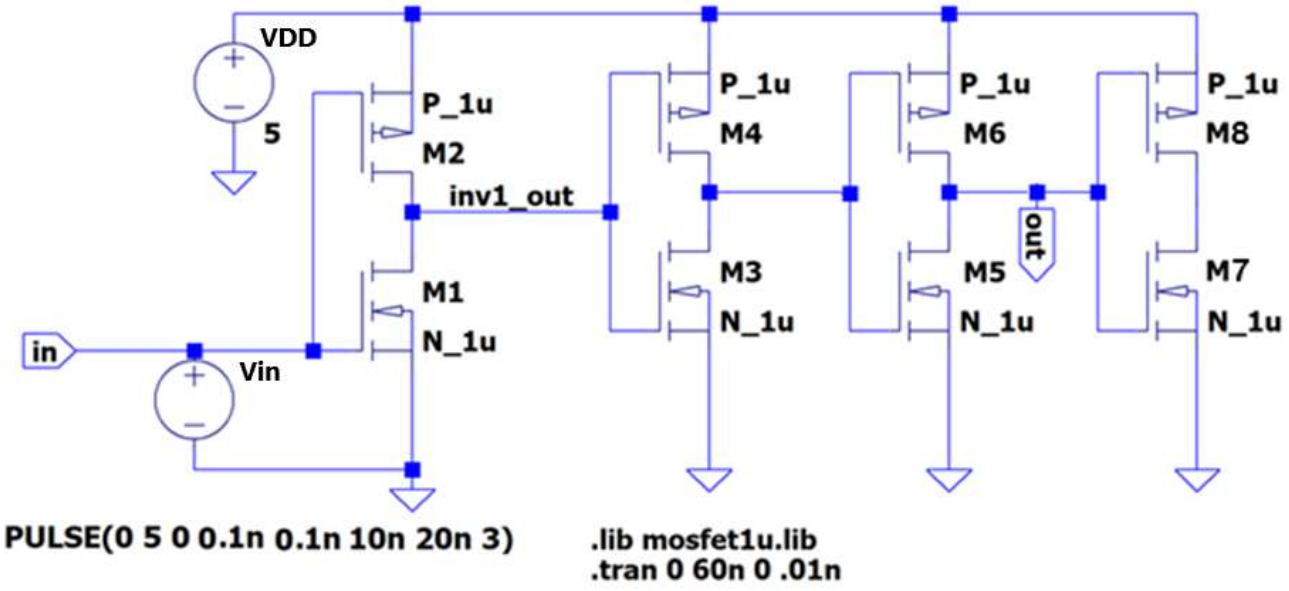


Figure 8.7 A circuit to explore the effect of cascading on propagation delay.

EXPERIMENT

9

The BJT CE Circuit with Emphasis on Switching

OBJECTIVES:

- To experimentally obtain and carefully examine the voltage transfer characteristic (VTC) of a simple common-emitter (grounded emitter) circuit.
- To discuss and understand the implications of the VTC of that circuit on using it as either a common-emitter amplifier or as a resistive-load inverter.
- To design a simple electronic switch based on a BJT.

MATERIALS:

- Laboratory setup, including a plug-in panel or a proto-board
- One *npn* BJT transistor (e.g., BD137)
- Resistors (27 kΩ, 2.7 kΩ)
- Several wires and bridging plugs

INTRODUCTION

The BJT is the second three-terminal device we encounter in this course. Despite the fierce competition with the MOSFET, the BJT is more popular in discrete-circuit design. Moreover, because of its reliable performance in sever environmental conditions, it is *the* dominant device in automotive applications [1]. Radio frequency (RF) circuits, and very high-speed digital circuits are two other fields where BJTs excel. Manufacturers generally classify their BJTs into three broad categories: general-purpose/small signal devices, power devices, and RF (radio frequency/microwave) devices [3].

There are two types of BJTs: the *npn* transistor and its dual, the *pnp* transistor. In this experiment we are going to concentrate on the *npn* type, with the BD137 as an example. One example of a BD137 datasheet is provided in Appendix C for easy reference, but one can always go online to search for datasheets for any particular device.

The three terminals of the BJT are connected to three semiconductor layers: the **emitter** (E), the **base** (B), and the **collector** (C). As such, it consists of two *pn* junctions, the emitter-base junction (**EBJ**) and the collector-base junction (**CBJ**). Based on the bias condition of these two junctions, three distinct modes of operation are obtained, as shown in Table 9.1. The **cutoff mode** and the **saturation mode** are employed in switching applications (e.g., logic circuits). The **active mode** is the one used if the transistor is to operate as a linear amplifier.

Since a BJT has two pn junctions, it can be tested by a DMM set to its diode-test mode. Both junction should read about 0.6 V when forward-biased and an OL when reverse-biased. If a DMM is connected between the collector and emitter, it should read OL.

Figure 9.1 shows the voltage polarities and current flow directions for an *npn* transistor biased to operate in the active mode. The base of a BJT does a similar role to the gate of a MOSFET, except that it does need a dc current in order for the BJT to function properly. Consequently, the emitter current I_E is not equal to the collector current I_C, as the difference between them equals the base current I_B according to Kirchhoff's current law. In the active mode of operation, I_C is directly proportional to I_E via the **common-base current gain** α, and directly proportional to I_B via the **common-emitter current gain** β, which is usually denoted $\mathbf{h_{FE}}$ in datasheets.

When a BJT is used to **switch** high-current loads **on and off**, the load is connected in series with the collector (Fig. 9.3), and consequently the collector current is known in advance. The ratio between $I_{CSat} = I_{load}$ and I_B is taken to be **less than 20** to insure that the BJT has minimum V_{CE} drop when the load is energized. This ratio is known as $\boldsymbol{\beta_{forced}}$, and as a good rule of thumb for **switching applications design**, we take it to be around 10.

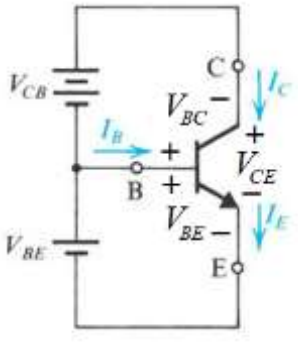


Figure 9.1 Voltage polarities and current flow for an *npn* transistor.

Table 9.1 Modes of Operation and the Corresponding Bias Conditions for a npn BJT.

Mode	***EBJ*** bias condition	***CBJ*** bias condition
Cutoff	$v_{BE} < V_{BE\,\text{on}} \Rightarrow$ reverse/slightly-forward	$v_{BC} < V_{BC\,\text{on}} \Rightarrow$ reverse/slightly-forward
Active	$v_{BE} \geq V_{BE\,\text{on}} \Rightarrow$ forward	$v_{BC} < V_{BC\,\text{on}} \Rightarrow$ reverse/slightly-forward
Saturation	$v_{BE} \geq V_{BE\,\text{on}} \Rightarrow$ forward	$v_{BC} \geq V_{BC\,\text{on}} \Rightarrow$ forward
$V_{BE\,\text{on}} \approx 0.5$ and $V_{BC\,\text{on}} \approx 0.4$ **(approximate values)**		
General Relationships: $I_E = I_B + I_C$ $V_{BC} = V_{BE} - V_{CE}$		
In the **active** mode: $v_{CE} \geq 0.3$ V $\beta \equiv I_C / I_B$ $\alpha \equiv I_C / I_E = \beta / (\beta+1)$	In the **saturation** mode: $v_{CE} = V_{CE\,\text{sat}} \leq 0.2$ V $\beta_{forced} \equiv I_{C\text{sat}} / I_B < \beta_{min}$ **For design purposes:** $V_{CE\,\text{sat}} \approx 0.2$ V; $10 \leq \boldsymbol{\beta_{forced}} \leq 20$.	

Real semiconductor devices do have limitations and maximum ratings that should be observed by circuit designers. Refer to the datasheet of the BD137 transistor given on Appendix C. Note that the maximum allowable V_{CE} is 60 V, and the maximum allowable power dissipation P_D at an ambient temperature T_A of 25 °C is 1.25 W. That value should be derated for higher ambient temperatures using the following simple equation:

$$P_D|_{higher\ temp} = P_D|_{at\ 25\,°C} - DF \times (higher\ temp - 25) \qquad (9.1)$$

Where the power derating factor *DF* at T_A = 25 °C is 10 mV/°C as given in the datasheet.

PROCEDURE A: The Transfer Characteristic of the CE Circuit

a. Assemble the circuit shown in Fig. 9.2.

b. Using one dc supply, fix V_{CC} at 5 V, and using another dc supply, vary V_I to obtain V_O values that are shown in Table 9.2. For each value of V_O, measure V_{BE} and V_I and record their values in the table.

c. Do the required calculations in Table 9.2 using the following equations

$$I_B = (V_I - V_{BE})/R_B \quad (9.1)$$

$$I_C = (V_{CC} - V_O)/R_C \quad (9.2)$$

$$I_E = I_C + I_B \quad (9.3)$$

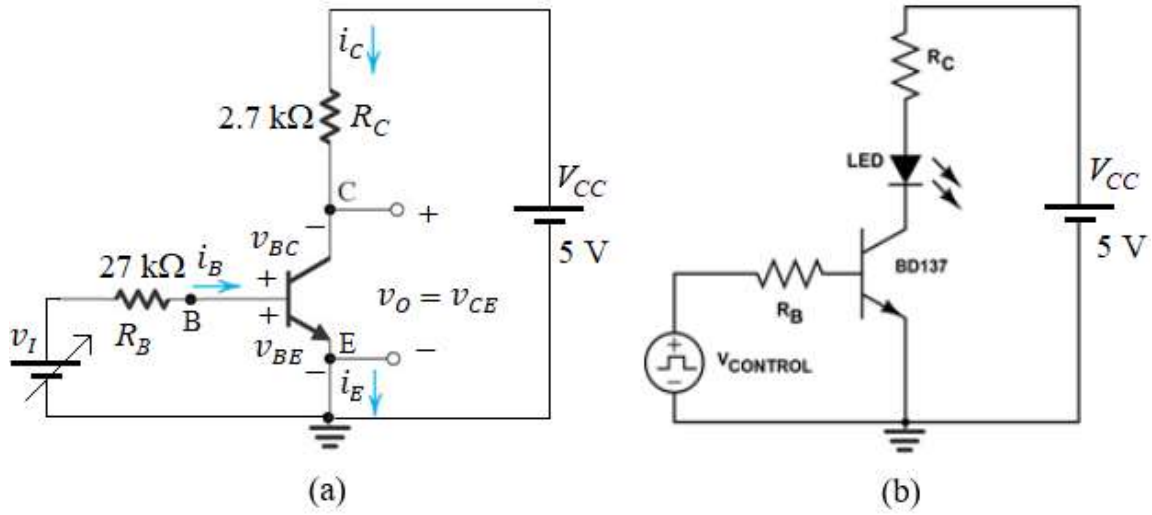


Figure 9.2 **(a)** The Common Emitter Circuit. **(b)** Using the CE configuration for switching.

PROCEDURE B: Switching Application of the CE Circuit

Suppose an LED needs about 10 mA to glow brightly, with a forward voltage drop around 2 volts. You need a transistor to switch the LED on and off because you do not want the controlling signal source to carry the load

current (Fig. 9.2b). The current of the LED in this case is equal to I_{Csat}. What current does the controlling signal source carry in this case?

1. Calculate the required value of R_C using the following equation:

$$R_C = \frac{V_{CC} - V_{LED} - 0.2}{I_{Csat}} \qquad (9.3)$$

2. Assuming $\beta_{forced} = 10$, calculate the required I_B:

$$I_B = \frac{I_{Csat}}{\beta_{forced}} \qquad (9.4)$$

3. Calculate the required R_B using the following equation:

$$R_B = \frac{V_{Contol_Hi} - 0.7}{I_B} \qquad (9.5)$$

4. Implement and test the circuit in the lab using a **square wave** for $V_{Control}$, with ***Hi_Level*** = 5 V and ***Lo_Level*** = 0 V, and frequency of 0.2 Hz. For R_C and R_B, you can use two resistance decade boxes, but for a final design you should find the nearest standard values to the calculated ones.
5. Test the circuit if the frequency of controlling signal source is increased to 2 Hz and 60 Hz as well.
    - Why is the switching not noticeable in the case of 60 Hz?

RESULTS

Table 9.2 Measured Data (Rounded Up to 3 Decimal Places)

Vary V_I to obtain	To be **Measured**		To be **Calculated**				
$V_{CE} = V_O$ (V)	V_{BE} [V]	V_I [V]	$V_{BC} = V_{BE} - V_{CE}$ [V]	Mode of operation	I_B [mA]	I_C [mA]	I_E [mA]
5.000	0.000	0.000	–5.000	Cutoff	0.000	0.000	0.000
4.900							
4.500							
3.500							
3.400							
2.000							
1.000							
0.500							
0.200							
0.100							

POST LAB QUESTIONS (PLQs)

1) Complete the calculations required in Table 9.2.
2) Plot v_O vs. v_I and label it with the three modes of operation of the BJT.
3) Which mode of the above three modes of operation can be used for voltage amplification based on the magnitude of slope? Why?
4) From Table 9.2, calculate the voltage gains $G_V = \Delta v_O / \Delta v_I$ and $A_V = \Delta v_O / \Delta v_{BE}$ when v_O changes from 3.5 to 3.4 V.
5) Why is $|G_V|$ (the voltage gain relative to v_I) usually less than $|A_V|$ (the voltage gain relative to v_{BE})?
6) If the LED shown in the circuit of Fig 9.2b is a blue LED, which has a nominal forward voltage of 3.4 V at a current of 30 mA, calculate the values of the required R_C and R_B. Find nearest standard values for these resistors from Appendix F. Assume β_{forced} = 10.

EXPERIMENT

10

The Common Emitter Amplifier

OBJECTIVES:

- To assemble a common emitter amplifier circuit, perform dc and small signal measurements, and calculate some important parameters.
- To experimentally see the distortion resulting from not maintaining the input signal small enough to guarantee linear model validity.
- To learn a practical technique for designing a dc biasing scheme that will produce a stable Q point despite potential variations in temperature and/or β.

MATERIALS:

- Laboratory setup, including a plug-in panel or a proto-board
- One *npn* BJT transistor (e.g., BD137)
- Resistors (4.7 kΩ, 5.6 kΩ, 27 kΩ, 47 kΩ)
- Electrolytic capacitors (2.2 μF (two), 470 μF)

INTRODUCTION

The Common emitter (CE) amplifier is the most famous of the three BJT amplifier configurations. The other two being the common base (used to achieve higher bandwidth) and the common collector (used as a voltage buffer). The discrete CE amplifier receives the input signal at the base (via a coupling capacitor) and delivers the output from the collector.

Figure 10.1 shows only the dc circuit of such an amplifier. The dc bias scheme used here is known as the voltage divider bias with emitter degeneration resistance R_E. R_E serves to stabilize the dc bias point against variations in β. Moreover, including R_E increases the amplifier's input impedance, and extends its linear range. However, the presence of R_E reduces the transconductance g_m by a factor of $(1 + g_m R_E)$, hence reducing the voltage gain by the same factor.

Shown in Fig. 10.2 is the complete CE amplifier circuit. Capacitors C_1 and C_2, known as **coupling** capacitors, are used to act as a short circuit at signal frequencies of interest while blocking dc–hence, the other common name **dc-blocking** capacitors. A **bypass** capacitor, C_E, is used in order to bypass R_E at signal frequencies; therefore, nearly eliminate its undesirable effect on A_v.

The CE amplifier is characterized by a 180° phase shift; it's an indication of the inverse relationship (negative slope) between its input and output as you may recall from experiment 9.

With **efficiently bypassed R_E**, the theoretical voltage gains of the CE amplifier are given by the following expressions [1]:

$$G_v \equiv \frac{v_o}{v_{sig}} = -g_m(R_C \parallel r_o \parallel R_L)\left\{\frac{R_1 \parallel R_2 \parallel (\beta+1)r_e}{R_{sig} + [R_1 \parallel R_2 \parallel (\beta+1)r_e]}\right\} \qquad (10.1)$$

$$A_v \equiv \frac{v_o}{v_i} = -g_m(R_C \parallel r_o \parallel R_L) \approx -g_m R_C \quad if\ R_C \ll (r_o \parallel R_L) \qquad (10.2)$$

Where, $g_m = \frac{I_C}{V_T}$ and $r_e = \frac{V_T}{I_E} = \alpha\frac{V_T}{I_C} = \frac{\alpha}{g_m}$ $\qquad (10.3, 10.4)$

$$\Rightarrow A_v \approx -g_m R_C = -\alpha\frac{R_C}{r_e} = -\frac{I_C R_C}{V_T} \quad [No\ Load!] \qquad (10.5)$$

PROCEDURE A: dc Measurements

1. Using a DMM, measure each resistor and record its value in Table 10.1.
2. Construct the circuit shown in Fig. 10.1.
3. Using a DMM set to measure dc voltage, measure and record the values of V_{BE}, V_{BC}, V_{CE}, V_{R1}, V_{R2}, V_{RC} and V_{RE} on Fig. 10.3.
4. Using the *measured* values of resistors and the above voltage measurements, calculate I_C, I_E, I_1 and I_2 by Ohm's law, and then using Kirchhoff's current law, calculate I_B.

Table 10.1

R_C	R_E	R_1	R_2
I_C	I_E	I_{R1}	I_{R2}
I_B		$\beta \equiv I_C / I_B =$	$\alpha \equiv \beta/(\beta+1) =$
$Av = -\frac{I_C R_C}{V_T} =$		$Av = \frac{Vo(pp)}{Vi(pp)} =$	
Verify that (**$V_{BE} \geq 0.5$) and ($V_{BC} < 0.4$)** (for active mode operation):			

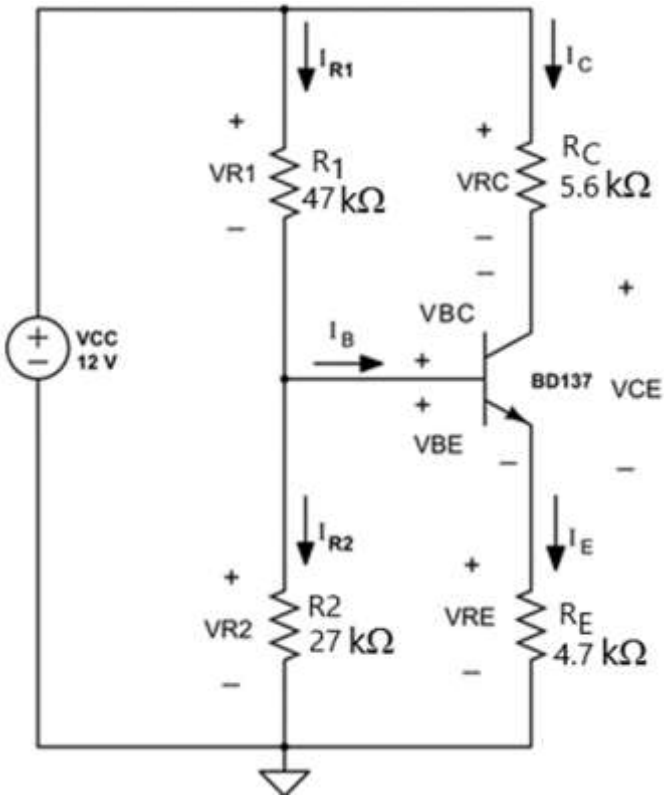


Figure 10.1 CE amplifier using R_E and a single supply.

PROCEDURE B: Small Signal Measurements

1. Assemble the circuit shown in Fig 10.2.
2. Turn on your scope, wait for few seconds, press the **Default** key, and then the **Autoset** key. Display $v_i(t)$ and $v_O(t)$ on **CH1** and **CH2** respectively.
3. Using a signal generator, apply to your circuit a 20-$mV_{p\text{-}p}$, 1-kHz sinusoidal signal.
4. Add **Pk-Pk** automatic measurements to **CH1** and **CH2** of your scope as you have learned previously.
5. Display and record the waveforms of $v_i(t)$ and $v_O(t)$ on Fig. 10.4. What is the phase difference between these two signals?
6. Calculate the actual voltage gain: $A_v = V_{o\,(pp)} / V_{i\,(pp)}$ and the idealized theoretical **no-load** voltage gain A_v given by equation (10.5). Record the two gains in Table 10.1.
7. Increase $V_{i\,(p\text{-}p)}$ to 200 mV. Record the waveforms of $v_i(t)$ and $v_O(t)$ on Fig. 10.5.

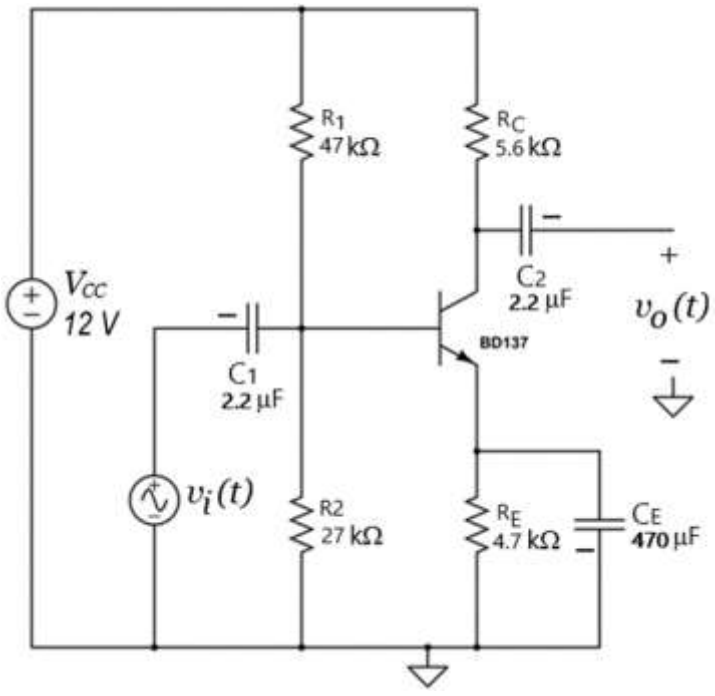


Figure 10.2 Complete CE amplifier circuit.

RESULTS

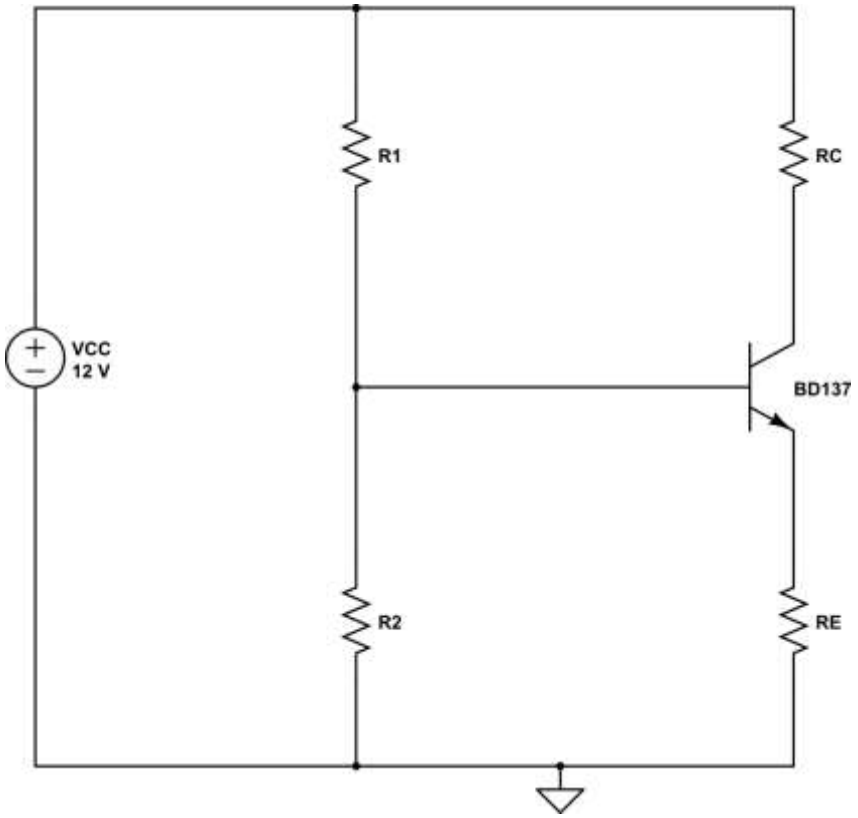


Fig. 10.3 Label this figure with the measured DC voltages.

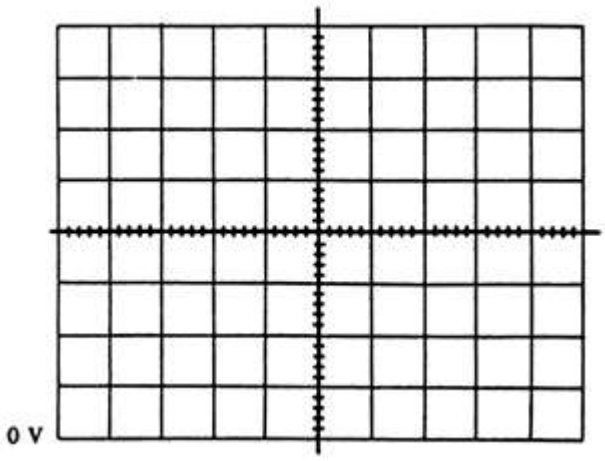


Figure 10.4

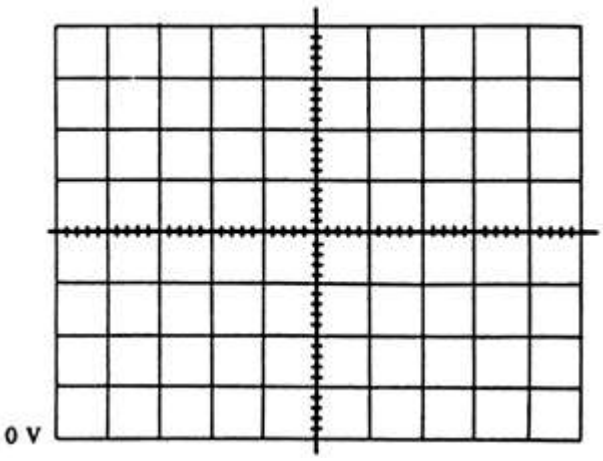


Figure 10.5

POST LAB QUESTIONS (PLQs)

1. *Procedure A*: Complete all required calculations in Table 10.1.
2. *Procedure A*: Verify that:
   i. $V_{R1} + V_{R2} \approx V_{CC}$
   ii. $V_{RC} + V_{CE} + V_{RE} \approx V_{CC}$
   iii. $V_{CE} = V_{BE} - V_{BC}$
3. *Procedure B*: Describe the effect of increasing the input signal amplitude on the shape of the output voltage waveform. Is the BJT working linearly for such large input amplitudes?
4. Using the bias scheme design tips given on appendix H, and assuming $V_{BE(on)}$ = 0.7 V in the circuit of Fig. 10.6a, specify E96 standard resistors[13] to bias the BJT at I_E = TN mA, where TN is your table number in the lab.
5. Verify your design in part (a) using **DC op pnt** LTspice analysis. In your simulation, change the default NPN BJT to **2N2222**.
6. Add a signal source and three capacitors to your schematic as shown in Fig. 10.6b. Run a **Transient** analysis and measure the output voltage peak-to-peak value, and then calculate the voltage gain. You need to include the circuit's schematic and the simulation results of your design in your report.

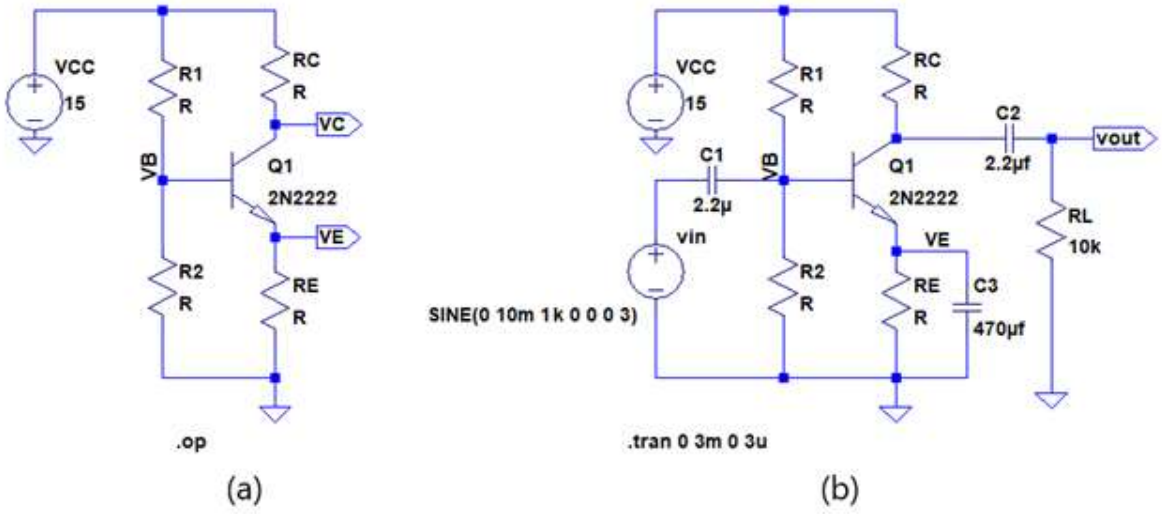


Figure 10.6 Common emitter amplifier; (a) dc circuit, (b) complete amplifier.

[13]See Appendix F. A value like 374 means that 374 Ω, 3.74 kΩ, 37.4 kΩ, 374 kΩ and 3.74 MΩ are standards as well.

Appendix A

References

[1] A. S. Sedra and K. C.Smith, *Microelectronic Circuits*, New York: Oxford University Press, 2010.

[2] R. Boylestad and L. Nashelesky, *Electronic Devices and Circuit Theory, 7th ed.*, Upper Saddle River, NJ: Prentice-Hall Inc., 1992.

[3] T. Floyd, *Electronic Devices, 7th ed*., Upper Saddle River, NJ: Pearson Education Inc., 2005.

[4] S. Franco, *Analog Circuit Design*, New York: McGraw-Hill Education, 2015.

[5] D. Neamen, *Microelectronics, Circuit Analysis and Design, 3rd ed.*, New York: McGraw-Hill, 2007.

[6] S. Franco, *Design with Operational Amplifiers and Analog Integrated Circuits*, New York: McGraw-Hill Education, 2002.

[7] P. Horowitz and W. Hill, *The Art of Electronics, 3rd ed.*, New York: Cambridge University Press, 2015.

[8] V. Gaudet and K. C. Smith, *Laboratory Explorations to Accompany Microelectronic Circuits, 7th ed.*, New York: Oxford University Press, 2015.

[9] R. Boylestad and L. Nashelesky, *Laboratory Manual for Electronic Devices and Circuit Theory, 7th ed.*, Upper Saddle River, NJ: Prentice-Hall Inc., 1999.

[10] P. Schers and S. Monk, *Practical Electronics for Inventors, 3rd ed.*, New York: McGraw-Hill Education, 2013.

[11] S. Monk, Hacking electronics, New York: McGraw-Hill, 2013.

[12] F. Mims III, *Electronic sensor circuits & Projects*, Niles, IL: Master Publishing Inc., 2007.

[13] F. Mims III, *Engineer's Notebook*, Eagle Rock, Virginia: LLH Master Publishing, 1992.

Appendix B

MATLAB® Codes

(Experiment 2)

```
% lines starting with "%" are COMMENTS; no need to enter them in Matlab command window

% Enter the commands in BLACK font ONE at a time

clf

% store voltage readings for the rectifier diode below

VD=[                                ];

% store current readings for the rectifier diode below

ID=[                                ];

% store voltage readings for Zener diode below

VDz=[                                     ];

% store current readings for Zener diode below

IDz=[                                     ];

plot(VD,ID,'-dk',VDz,IDz,'-*k')

axis([-6 1 -25 25])

grid on

legend('Rectifier Diode','Zener Diode')

title({'The i-v characteristics for a an ordinary Si diode and a
Zener diode';'put your name and no. here'})

% Do not forget to type you student name and number inside the above title( ) command!

xlabel('VD (V)')

ylabel('ID (mA)')
```

MATLAB® Code

(Experiment 7)

```
% lines starting with "%" are COMMENTS; no need to enter them in Matlab command window

% plotting ID-VDS characteristics for different values of VGS for E-NMOS

% Enter the commands in BLACK font ONE at a time

clf

% store your readings for VDS below

VDS=[                                   ];

% store current readings for VGS1 below

ID1=[                                      ];

% store current readings for VGS2 below

ID2=[                                      ];

% store current readings for VGS3 below

ID3=[                                      ];

figure(1)

plot(VDS,ID1,'-ob',VDS,ID2,'-dk',VDS,ID3,'-*r')

grid on

legend('VGS1 = VGS(TH) + 0.15','VGS2 = VGS(TH)+ 0.3','VGS3 =
VGS(TH)+ 0.45')

% Do not forget to type you student name and number inside the title( ) command below!

title({'The ID-VDS characteristics for the IRF620 NMOS, VGS(TH)=
';'put your name and no. here'})

xlabel('VDS (V)')

ylabel('ID (mA)')
```

Appendix C

Datasheets

FAIRCHILD

November 2014

1N4001 - 1N4007
General-Purpose Rectifiers

Features

- Low Forward Voltage Drop
- High Surge Current Capability



DO-41
COLOR BAND DENOTES CATHODE

Ordering Information

Part Number	Top Mark	Package	Packing Method
1N4001	1N4001	DO-204AL (DO-41)	Tape and Reel
1N4002	1N4002	DO-204AL (DO-41)	Tape and Reel
1N4003	1N4003	DO-204AL (DO-41)	Tape and Reel
1N4004	1N4004	DO-204AL (DO-41)	Tape and Reel
1N4005	1N4005	DO-204AL (DO-41)	Tape and Reel
1N4006	1N4006	DO-204AL (DO-41)	Tape and Reel
1N4007	1N4007	DO-204AL (DO-41)	Tape and Reel

Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only. Values are at T_A = 25°C unless otherwise noted.

Symbol	Parameter	Value							Unit
		1N 4001	1N 4002	1N 4003	1N 4004	1N 4005	1N 4006	1N 4007	
V_{RRM}	Peak Repetitive Reverse Voltage	50	100	200	400	600	800	1000	V
$I_{F(AV)}$	Average Rectified Forward Current .375 " Lead Length at T_A = 75°C	1.0							A
I_{FSM}	Non-Repetitive Peak Forward Surge Current 8.3 ms Single Half-Sine-Wave	30							A
I^2t	Rating for Fusing (t < 8.3 ms)	3.7							A^2sec
T_{STG}	Storage Temperature Range	-55 to +175							°C
T_J	Operating Junction Temperature	-55 to +175							°C

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Thermal Characteristics

Values are at T_A = 25°C unless otherwise noted.

Symbol	Parameter	Value	Unit
P_D	Power Dissipation	3.0	W
$R_{\theta JA}$	Thermal Resistance, Junction-to-Ambient	50	°C/W

Electrical Characteristics

Values are at T_A = 25°C unless otherwise noted.

Symbol	Parameter	Conditions	Value	Unit
V_F	Forward Voltage	I_F = 1.0 A	1.1	V
I_{rr}	Maximum Full Load Reverse Current, Full Cycle	T_A = 75°C	30	µA
I_R	Reverse Current at Rated V_R	T_A = 25°C	5.0	µA
		T_A = 100°C	50	
C_T	Total Capacitance	V_R = 4.0 V, f = 1.0 MHz	15	pF

Typical Performance Characteristics

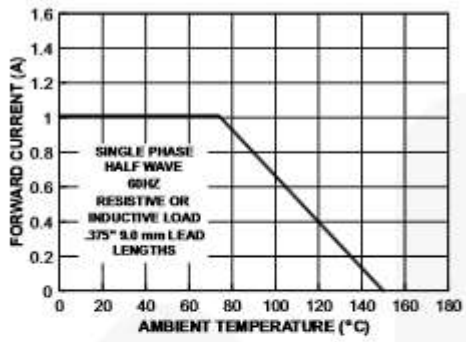


Figure 1. Forward Current Derating Curve

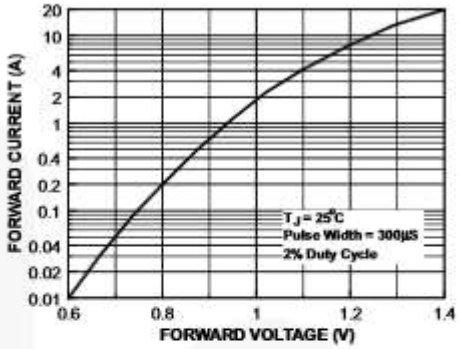


Figure 2. Forward Characteristics

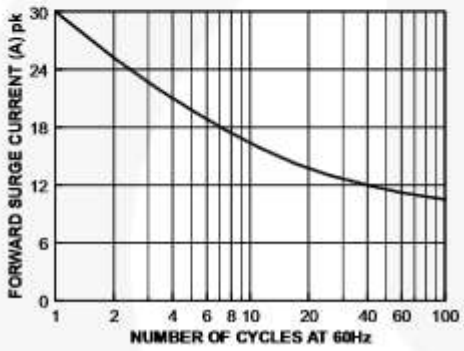


Figure 3. Non-Repetitive Surge Current

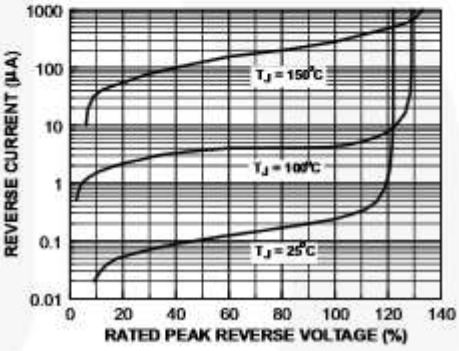


Figure 4. Reverse Characteristics

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FAIRCHILD SEMICONDUCTOR®

IRF620

Data Sheet **January 2002**

5.0A, 200V, 0.800 Ohm, N-Channel Power MOSFET

This N-Channel enhancement mode silicon gate power field effect transistor is an advanced power MOSFET designed, tested, and guaranteed to withstand a specified level of energy in the breakdown avalanche mode of operation. All of these power MOSFETs are designed for applications such as switching regulators, switching convertors, motor drivers, relay drivers, and drivers for high power bipolar switching transistors requiring high speed and low gate drive power. These types can be operated directly from integrated circuits.

Formerly developmental type TA9600.

Ordering Information

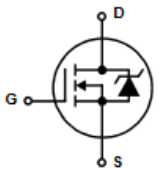
PART NUMBER	PACKAGE	BRAND
IRF620	TO-220AB	IRF620

NOTE: When ordering, use the entire part number.

Features

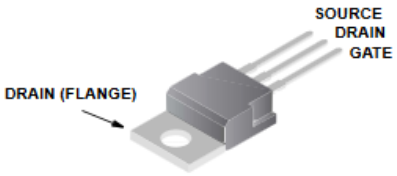
- 5.0A, 200V
- $r_{DS(ON)} = 0.800\Omega$
- Single Pulse Avalanche Energy Rated
- SOA is Power Dissipation Limited
- Nanosecond Switching Speeds
- Linear Transfer Characteristics
- High Input Impedance
- Related Literature
  - TB334 "Guidelines for Soldering Surface Mount Components to PC Boards"

Symbol



Packaging

JEDEC TO-220AB



IRF620

Absolute Maximum Ratings $T_C = 25^oC$, Unless Otherwise Specified

		IRF620	UNITS
Drain to Source Voltage (Note 1)	V_{DS}	200	V
Drain to Gate Voltage (R_{GS} = 20kΩ) (Note 1)	V_{DGR}	200	V
Continuous Drain Current	I_D	5.0	A
$T_C = 100^oC$	I_D	3.0	A
Pulsed Drain Current (Note 3)	I_{DM}	20	A
Gate to Source Voltage	V_{GS}	±20	V
Maximum Power Dissipation	P_D	40	W
Linear Derating Factor		0.32	W/°C
Single Pulse Avalanche Energy Rating (Note 4)	E_{AS}	85	mJ
Operating and Storage Temperature	T_J, T_{STG}	-55 to 150	°C
Maximum Temperature for Soldering			
Leads at 0.063in (1.6mm) from Case for 10s	T_L	300	°C
Package Body for 10s, See TB334	T_{pkg}	260	°C

CAUTION: Stresses above those listed in "Absolute Maximum Ratings" may cause permanent damage to the device. This is a stress only rating and operation of the device at these or any other conditions above those indicated in the operational sections of this specification is not implied.

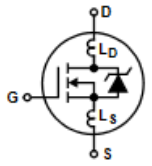
NOTE:

1. $T_J = 25^oC$ to 125^oC

Electrical Specifications $T_C = 25^oC$, Unless Otherwise Specified

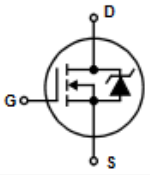
PARAMETER	SYMBOL	TEST CONDITIONS	MIN	TYP	MAX	UNITS
Drain to Source Breakdown Voltage	BV_{DSS}	V_{GS} = 0V, I_D = 250µA, (Figure 10)	200	-	-	V
Gate Threshold Voltage	$V_{GS(TH)}$	V_{DS} = V_{GS}, I_D = 250µA	2.0	-	4.0	V
Zero Gate Voltage Drain Current	I_{DSS}	V_{DS} = Rated BV_{DSS}, V_{GS} = 0V	-	-	25	µA
		V_{DS} = 0.8 x Rated BV_{DSS}, V_{GS} = 0V, $T_J = 125^oC$	-	-	250	µA
On-State Drain Current (Note 2)	$I_{D(ON)}$	$V_{DS} > I_{D(ON)} \times r_{DS(ON)MAX}$, V_{GS} = 10V	5.0	-	-	A
Gate to Source Leakage Current	I_{GSS}	V_{GS} = ±20V	-	-	±100	nA
Drain to Source On Resistance (Note 2)	$r_{DS(ON)}$	V_{GS} = 10V, I_D = 2.5A, (Figures 8, 9)	-	0.8	1.2	Ω
Forward Transconductance (Note 2)	g_{fs}	$V_{DS} > I_{D(ON)} \times r_{DS(ON)MAX}$, I_D = 2.5A (Figure 12)	1.3	2.5	-	S
Turn-On Delay Time	$t_{d(ON)}$	V_{DD} = 100V, $I_D \approx$ 5.0A, R_G = 9.1Ω, R_L = 20Ω, MOSFET Switching Times are Essentially Independent of Operating Temperature	-	20	40	ns
Rise Time	t_r		-	30	60	ns
Turn-Off Delay Time	$t_{d(OFF)}$		-	50	100	ns
Fall Time	t_f		-	30	60	ns
Total Gate Charge (Gate to Source + Gate to Drain)	$Q_{g(TOT)}$	V_{GS} = 10V, I_D = 5.0A, V_{DS} = 0.8 x Rated BV_{DSS}, $I_{G(REF)}$ = 1.5mA, (Figure 14) Gate Charge is Essentially Independent of Operating Temperature	-	11	15	nC
Gate to Source Charge	Q_{gs}		-	5.0	-	nC
Gate to Drain "Miller" Charge	Q_{gd}		-	6.0	-	nC
Input Capacitance	C_{ISS}	V_{DS} = 25V, V_{GS} = 0V, f = 1MHz, (Figure 11)	-	450	-	pF
Output Capacitance	C_{OSS}		-	150	-	pF
Reverse Transfer Capacitance	C_{RSS}		-	40	-	pF
Internal Drain Inductance	L_D	Measured from the Contact Screw on Tab to Center of Die	-	3.5	-	nH
		Measured from the Drain Lead, 6mm (0.25in) from Package to Center of Die	-	4.5	-	nH
Internal Source Inductance	L_S	Measured from the Source Lead, 6mm (0.25in) from Header to Source Bonding Pad	-	7.5	-	nH
Thermal Resistance Junction to Case	$R_{\theta JC}$		-	-	3.12	°C/W
Thermal Resistance Junction to Ambient	$R_{\theta JA}$	Free Air Operation	-	-	62.5	°C/W

Modified MOSFET Symbol Showing the Internal Devices Inductances



Source to Drain Diode Specifications

PARAMETER	SYMBOL	TEST CONDITIONS	MIN	TYP	MAX	UNITS
Continuous Source to Drain Current	I_{SD}	Modified MOSFET Symbol Showing the Integral Reverse P-N Junction Rectifier	-	-	5.0	A
Pulse Source to Drain Current (Note 3)	I_{SDM}		-	-	20	A
Source to Drain Diode Voltage (Note 2)	V_{SD}	$T_J = 25^oC$, I_{SD} = 5.0A, V_{GS} = 0V, (Figure 13)	-	-	1.8	V
Reverse Recovery Time	t_{rr}	$T_J = 150^oC$, I_{SD} = 5.0A, dI_{SD}/dt = 100A/µs	-	350	-	ns
Reverse Recovery Charge	Q_{RR}	$T_J = 150^oC$, I_{SD} = 5.0A, dI_{SD}/dt = 100A/µs	-	2.3	-	µC



Typical Performance Curves Unless Otherwise Specified (Continued)

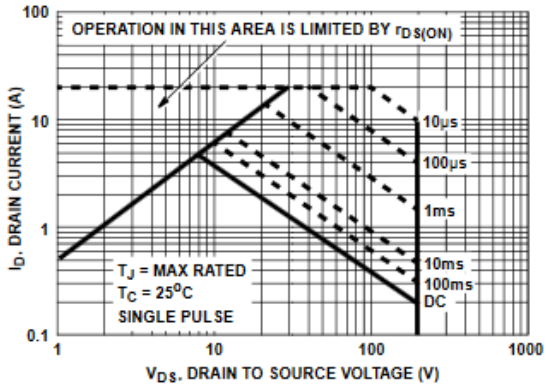


FIGURE 4. FORWARD BIAS SAFE OPERATING AREA

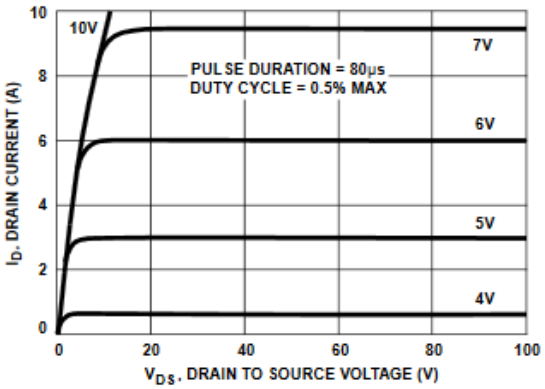


FIGURE 5. OUTPUT CHARACTERISTICS

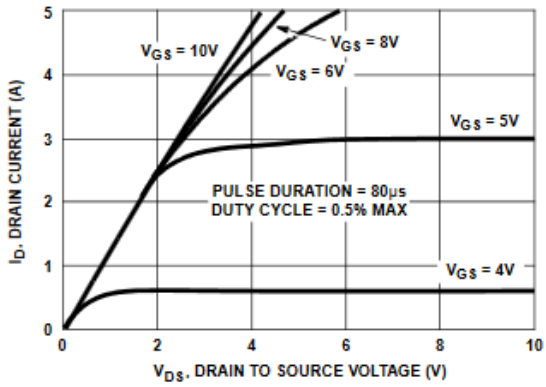


FIGURE 6. SATURATION CHARACTERISTICS

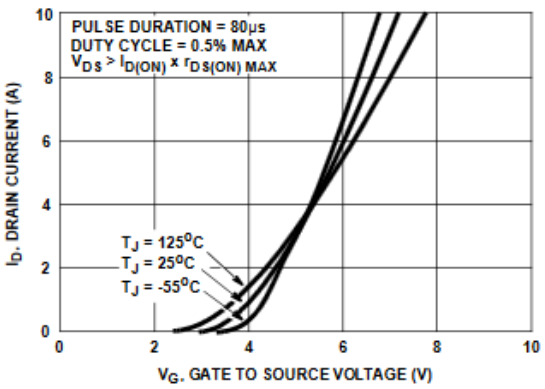
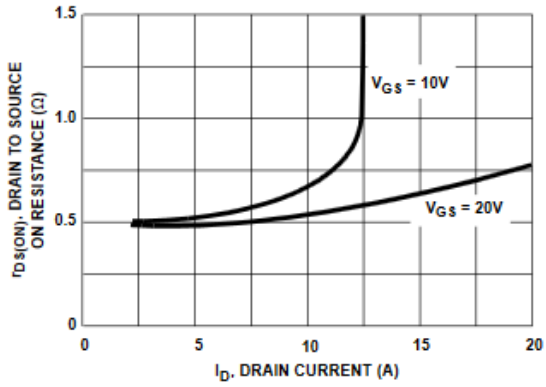


FIGURE 7. TRANSFER CHARACTERISTICS



NOTE: Heating effect of 2.0µs pulse is minimal.

FIGURE 8. DRAIN TO SOURCE ON RESISTANCE vs GATE VOLTAGE AND DRAIN CURRENT

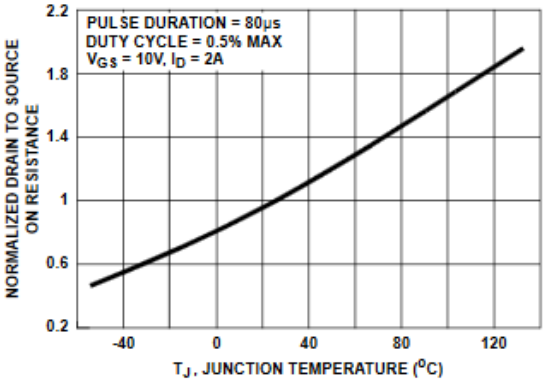


FIGURE 9. NORMALIZED DRAIN TO SOURCE ON RESISTANCE vs JUNCTION TEMPERATURE

BD135G, BD137G, BD139G

Plastic Medium-Power Silicon NPN Transistors

This series of plastic, medium−power silicon NPN transistors are designed for use as audio amplifiers and drivers utilizing complementary or quasi complementary circuits.

Features

- High DC Current Gain
- BD 135, 137, 139 are complementary with BD 136, 138, 140
- These Devices are Pb−Free, Halogen Free/BFR Free and are RoHS Compliant*

MAXIMUM RATINGS

Rating	Symbol	Value	Unit
Collector−Emitter Voltage BD135G BD137G BD139G	V_{CEO}	 45 60 80	Vdc
Collector−Base Voltage BD135G BD137G BD139G	V_{CBO}	 45 60 100	Vdc
Emitter−Base Voltage	V_{EBO}	5.0	Vdc
Collector Current	I_C	1.5	Adc
Base Current	I_B	0.5	Adc
Total Device Dissipation @ T_A = 25°C Derate above 25°C	P_D	 1.25 10	Watts mW/°C
Total Device Dissipation @ T_C = 25°C Derate above 25°C	P_D	 12.5 100	Watts mW/°C
Operating and Storage Junction Temperature Range	T_J, T_{stg}	−55 to +150	°C

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

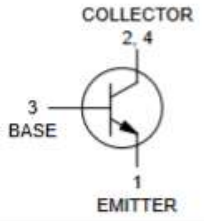
THERMAL CHARACTERISTICS

Characteristic	Symbol	Max	Unit
Thermal Resistance, Junction−to−Case	$R_{\theta JC}$	10	°C/W
Thermal Resistance, Junction−to−Ambient	$R_{\theta JA}$	100	°C/W

ON Semiconductor®

http://onsemi.com

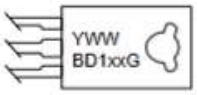
1.5 A POWER TRANSISTORS NPN SILICON 45, 60, 80 V, 12.5 W





TO−225
CASE 77−09
STYLE 1

MARKING DIAGRAM



Y = Year
WW = Work Week
BD1xx = Device Code
xx = 35, 37, 39
G = Pb−Free Package

BD135G, BD137G, BD139G

ELECTRICAL CHARACTERISTICS (T_C = 25°C unless otherwise noted)

Characteristic	Symbol	Min	Max	Unit
Collector−Emitter Sustaining Voltage* (I_C = 0.03 Adc, I_B = 0) BD135G BD137G BD139G	BV_{CEO}*	 45 60 80	 – – –	Vdc
Collector Cutoff Current (V_{CB} = 30 Vdc, I_E = 0) (V_{CB} = 30 Vdc, I_E = 0, T_C = 125°C)	I_{CBO}	 – –	 0.1 10	μAdc
Emitter Cutoff Current (V_{BE} = 5.0 Vdc, I_C = 0)	I_{EBO}	–	10	μAdc
DC Current Gain (I_C = 0.005 A, V_{CE} = 2 V) (I_C = 0.15 A, V_{CE} = 2 V) (I_C = 0.5 A V_{CE} = 2 V)	h_{FE}*	 25 40 25	 – 250 –	–
Collector−Emitter Saturation Voltage* (I_C = 0.5 Adc, I_B = 0.05 Adc)	$V_{CE(sat)}$*	–	0.5	Vdc
Base−Emitter On Voltage* (I_C = 0.5 Adc, V_{CE} = 2.0 Vdc)	$V_{BE(on)}$*	–	1	Vdc

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

*Pulse Test: Pulse Width ≤ 300 μs, Duty Cycle ≤ 2.0%.

TYPICAL CHARACTERISTICS

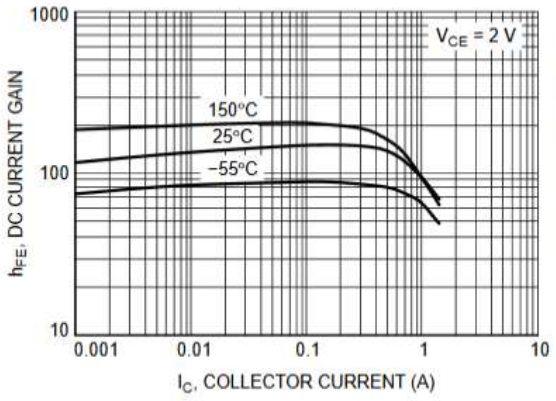


Figure 1. DC Current Gain

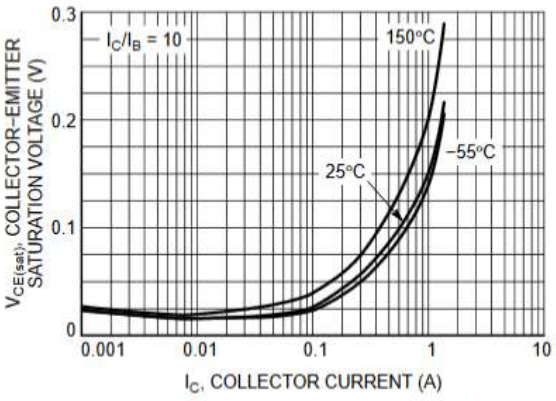


Figure 2. Collector−Emitter Saturation Voltage

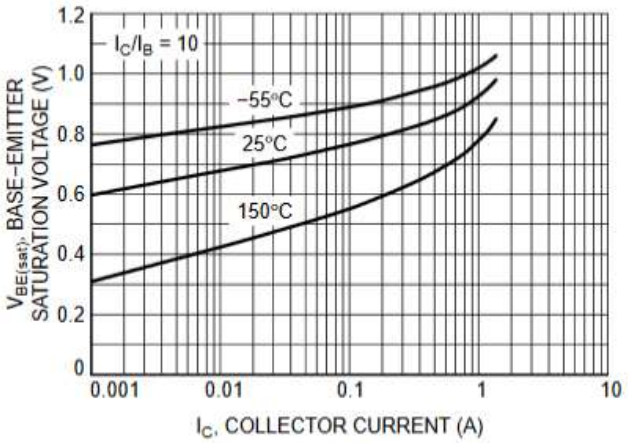


Figure 3. Base−Emitter Saturation Voltage

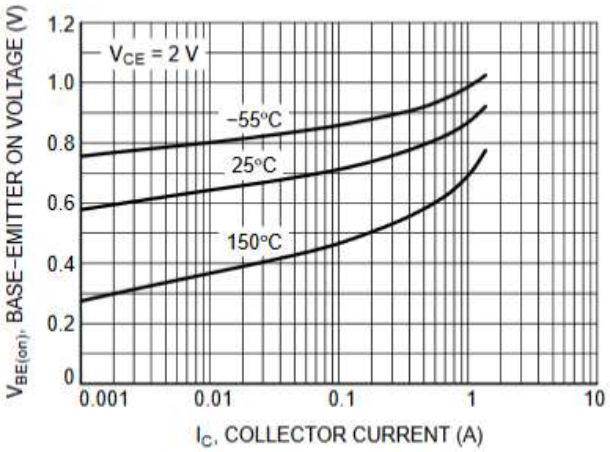


Figure 4. Base−Emitter On Voltage

Appendix D

Type	Part Number	Ratings	Type	Part Number	Ratings
Rectifier Diode	1N1183	50V 35A	Rectifier Diode	1N4001	50V 1A
Rectifier Diode	1N1184	100V 35A	Rectifier Diode	1N4002	100V 1A
Rectifier Diode	1N1186	200V 35A	Rectifier Diode	1N4003	200V 1A
Rectifier Diode	1N1187	300V 35A	Rectifier Diode	1N4004	400V 1A
Rectifier Diode	1N1199	50V 35A	Rectifier Diode	1S310	50V 0.5A
Rectifier Diode	1N3910	100V 30A	Rectifier Diode	1S311	100V 0.5A
Rectifier Diode	1N3911	200V 30A	Rectifier Diode	ERA3804	400V 0 .5A
Rectifier Diode	1N3491	50V 25A	Switching Diode	1N4500	80V 500mA
Rectifier Diode	1N3492	100V 25A	Switching Diode	BA220	10V 400mA
Rectifier Diode	1N3493	200V 25A	Switching Diode	BAS45	125V 225mA
Rectifier Diode	1N3494	300V 25A	Switching Diode	1N916	100V 200mA
Rectifier Diode	1N3900	100V 20A	Switching Diode	1N3600	75V 200mA
Rectifier Diode	1N3901	200V 20A	Switching Diode	1N4148	75V 200mA
Rectifier Diode	MUR1610CT	100V 16A	Switching Diode	1N4450	40V 200mA
Rectifier Diode	MUR1620CT	200V 16A	Switching Diode	1N4152	40V 200mA
Rectifier Diode	1N1200	100V 12A	Switching Diode	1N4154	35V 200mA
Rectifier Diode	1N1202	200V 12A	Switching Diode	1N4534	75V 150mA
Rectifier Diode	1N1204	400V 12A	Switching Diode	1N4531	100V 125mA
Rectifier Diode	MUR805	50V 8A	Switching Diode	1N4532	75V 125mA
Rectifier Diode	MUR810	100V 8A	Switching Diode	1N3064	75V 75mA
Rectifier Diode	MBR1535CT	35V 7.5A	Varactor Diode	1N5443	30V 10pf
Rectifier Diode	1N3879	50V 6A	Varactor Diode	1N5449	30V 27pf
Rectifier Diode	1N3880	100V 6A	Varactor Diode	1N5453	30V 56pf
Rectifier Diode	1N3881	200V 6A	Varactor Diode	1N5456	30V 100pf
Rectifier Diode	1N3882	300V 6A	Varactor Diode	BB112	12V 470p
Rectifier Diode	MUR405	50V 4A	Schottky Diode	10KQ100	100V 10A
Rectifier Diode	MUR410	100V 4A	Schottky Diode	10KQ30	30V 10A
Rectifier Diode	MUR420	200V 4A	Schottky Diode	5KQ60	60V 5A
Rectifier Diode	1N4719	50V 3A	Schottky Diode	5KQ90	90V 5A
Rectifier Diode	1N4720	100V 3A	Schottky Diode	1N5821	30V 3A
Rectifier Diode	1N4721	200V 3A	Schottky Diode	1N5822	40V 3A
Rectifier Diode	1N5400	50V 3A	Schottky Diode	ERB83006	60V 2A
Rectifier Diode	1N5401	100V 3A	Schottky Diode	ERB84009	90V 2A
Rectifier Diode	1N5402	200V 3A	Schottky Diode	ERA83006	60V 1A
Rectifier Diode	BY296	100V 2A	Schottky Diode	ERA85009	90V 1A
Rectifier Diode	BY297	200V 2A	Diode Bridges	MDA2502	200V 25A, Full Wave
Rectifier Diode	BYV27-50	50V 2A	Diode Bridges	MDA2504	400V 25A, Full Wave
Rectifier Diode	BYV27-100	100V 2A	Diode Bridges	KBPC801	80V 8A, Full Wave
Rectifier Diode	1N5391	50V 1.5A	Diode Bridges	KBPC804	250V 8A, Full Wave
Rectifier Diode	1N5392	100V 1.5A	Diode Bridges	KBU4D	200V 4A, Full Wave
Rectifier Diode	1R5BZ41	100V 1.5A	Diode Bridges	KBU4G	400V 4A, Full Wave
Rectifier Diode	1N5393	200V 1.5A	Diode Bridges	KBU4J	600V 4A, Full Wave

Appendix E

ZENER DIODE TABLE								
Volt	0.4 Watt		0.5 Watt		1 Watt		5 Watt	
2.4			1N5221	1N4617		UZ87=UZ88		UZ51=52=53
2.5			1N5222			UZ81=UZ82		UZ57=58=59
2.7			1N5223	1N4618				
2.8			1N5224					
3.0			1N5225	1N4619				
3.3	1N746		1N5226	1N4620	1N4728		1N5333	
3.6	1N747		1N5227	1N4621	1N4729		1N5334	
3.9	1N748		1N5228	1N4622	1N4730		1N5335	
4.3	1N749		1N5229	1N4623	1N4731		1N5336	
4.7	1N750		1N5230	1N4624	1N4732		1N5337	
5.1	1N751		1N5231	1N4625	1N4733		1N5338	
5.6	1N752		1N5232	1N4626	1N4734		1N5339	
6.0			1N5233	1N469			1N5340	
6.2	1N753		1N5234	1N4627	1N4735		1N5341	
6.8	1N754	1N957	1N5235	1N4628	1N4736	UZ8806	1N5342	UZ5806
7.5	1N755	1N958	1N5236	1N4629	1N4737	UZ8807	1N5343	UZ5807
8.2	1N756	1N959	1N5237	1N4630	1N4738	UZ8808	1N5344	UZ5808
8.7			1N5238	1N4695			1N5345	
9.1	1N757	1N960	1N5239	1N4631	1N4739	UZ8809	1N5346	UZ5809
10.0	1N758	1N961	1N5240	1N4632	1N4740	UZ8810	1N5347	UZ5810
11.0		1N962	1N5241	1N4633	1N4741		1N5348	
12.0	1N759	1N963	1N5242	1N4634	1N4742	UZ8812	1N5349	UZ5812
13.0	1N717	1N964	1N5243	1N4635	1N4743	UZ8813	1N5350	UZ5813
14.0			1N5244				1N5351	UZ5814
15.0	1N718	1N965	1N5245	1N4636	1N4744	UZ8815	1N5352	UZ5815
16.0	1N719	1N966	1N5246	1N4637	1N4745	UZ8816	1N5353	UZ5816
17.0			1N5247				1N5354	
18.0	1N720	1N967	1N5248	1N4638	1N4746	UZ8818	1N5355	UZ5818
19.0			1N5249				1N5356	
20.0	1N721	1N968	1N5250	1N4639	1N4747	UZ8820	1N5357	UZ5820
22.0	1N722	1N969	1N5251	1N4640	1N4748	UZ8822	1N5358	UZ5822
24.0	1N723	1N970	1N5252	1N4641	1N4749	UZ8824	1N5359	UZ5824
25.0			1N5253				1N5360	
27.0	1N724	1N971	1N5254	1N4642	1N4750	UZ8827	1N5361	UZ5827
28.0			1N5255				1N5362	
30.0	1N725	1N972	1N5256	1N4643	1N4751	UZ8830	1N5363	UZ5830
33.0	1N726	1N973	1N5257	1N4644	1N4752	UZ8833	1N5364	UZ5833
36.0	1N727	1N974	1N5258	1N4645	1N4753	UZ8836	1N5365	UZ5836
39.0	1N728	1N975	1N5259	1N4646	1N4754	UZ8840	1N5366	UZ5840
43.0	1N729	1N976	1N5260	1N4647	1N4755		1N5367	
47.0	1N730	1N977	1N5261	1N4648	1N4756	UZ8845	1N5368	
51.0	1N731	1N978	1N5262		1N4757	UZ8850	1N5369	UZ5850
56.0	1N732	1N979	1N5263		1N4758	UZ8856	1N5370	UZ5856
60.0			1N5264				1N5371	UZ5860
62.0	1N733	1N980	1N5265		1N4759	UZ8860	1N5372	
68.0	1N734	1N981	1N5266		1N4760	UZ8870	1N5373	
75.0	1N735	1N982	1N5267		1N4761	UZ8875	1N5374	UZ5875
82.0	1N736	1N983	1N5268		1N4762	UZ8880	1N5375	UZ5880
87.0			1N5269				1N5376	
91.0	1N737	1N984	1N5270		1N4763	UZ8890	1N5377	UZ5890
100.0	1N738	1N985	1N5271		1N4764	UZ8110	1N5378	UZ5310
110.0	1N739	1N986	1N5272			UZ8111	1N5379	UZ5311
120.0	1N740	1N987	1N5273			UZ8112	1N5380	UZ5312
130.0	1N741	1N988	1N5274			UZ8113	1N5381	UZ5313
140.0			1N5275			UZ8114	1N5382	UZ5314
150.0	1N742	1N989	1N5276			UZ8115	1N5383	UZ5315
160.0	1N743	1N990	1N5277			UZ8116	1N5384	UZ5316
170.0			1N5278			UZ8117	1N5385	UZ5317
180.0	1N744	1N991	1N5279			UZ8118	1N5386	UZ5318
190.0			1N5280			UZ8119	1N5387	UZ5319
200.0	1N745	1N992	1N5281			UZ8120	1N5388	UZ5320

Appendix F

Standard Components Values

Resistors

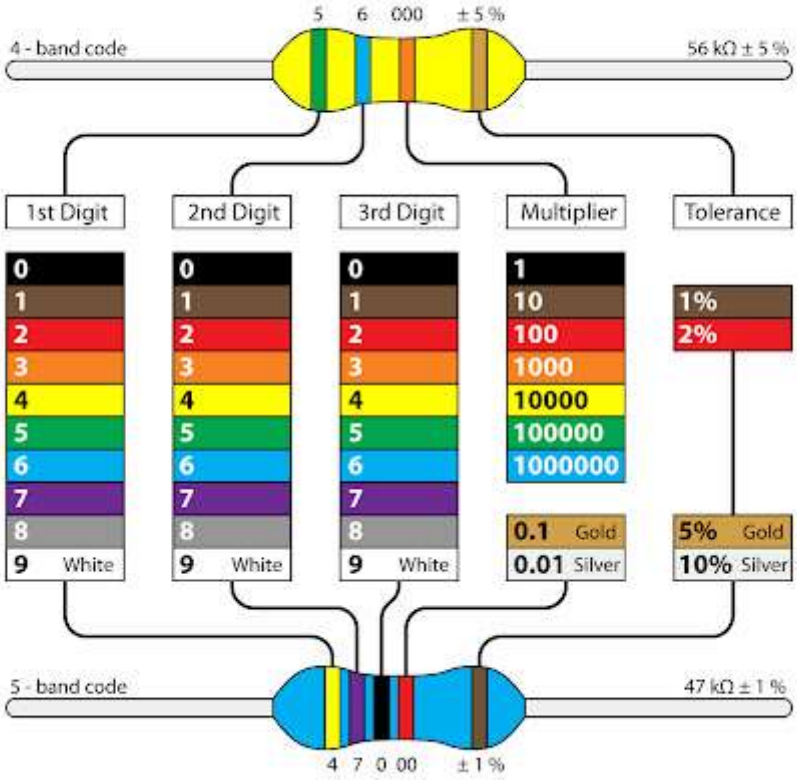
Standard Series Values in a Decade for Resistances and Capacitances

E3 TO E192

ACCORDING TO IEC 60063

E192	E96	E48	E192	E96	E48	E192	E96	E48	E192	E96	E48		E24	E12	E6	E3
100	100	100	178	178	178	316	316	316	562	562	562		10	10	10	10
101			180			320			569				11			
102	102		182	182		324	324		576	576			12	12		
104			184			328			583				13			
105	105	105	187	187	187	332	332	332	590	590	590		15	15	15	
106			189			336			597				16			
107	107		191	191		340	340		604	604			18	18		
109			193			344			612				20			
110	110	110	196	196	196	348	348	348	619	619	619		22	22	22	22
111			198			352			626				24			
113	113		200	200		357	357		634	634			27	27		
114			203			361			642				30			
115	115	115	205	205	205	365	365	365	649	649	649		33	33	33	
117			208			370			657				36			
118	118		210	210		374	374		665	665			39	39		
120			213			379			673				43			
121	121	121	215	215	215	383	383	383	681	681	681		47	47	47	47
123			218			388			690				51			
124	124		221	221		392	392		698	698			56	56		
126			223			397			706				62			
127	127	127	226	226	226	402	402	402	715	715	715		68	68	68	
129			229			407			723				75			
130	130		232	232		412	412		732	732			82	82		
132			234			417			741				91			
133	133	133	237	237	237	422	422	422	750	750	750					
135			240			427			759							
137	137		243	243		432	432		768	768						
138			246			437			777							
140	140	140	249	249	249	442	442	442	787	787	787					
142			252			448			796							
143	143		255	255		453	453		806	806						
145			258			459			816							
147	147	147	261	261	261	464	464	464	825	825	825					
149			264			470			835							
150	150		267	267		475	475		845	845						
152			271			481			856							
154	154	154	274	274	274	487	487	487	866	866	866					
156			277			493			876							
158	158		280	280		499	499		887	887						
160			284			505			898							
162	162	162	287	287	287	511	511	511	909	909	909					
164			291			517			920							
165	165		294	294		523	523		931	931						
167			298			530			942							
169	169	169	301	301	301	536	536	536	953	953	953					
172			305			542			965							
174	174		309	309		549	549		976	976						
176			312			556			988							

Resistor colour code



Appendix G

In this lab, two types of boards will be used to assemble our circuits: the plug-in panel (Fig. G.1) and the breadboard, an example of which is shown in Fig. G.2.

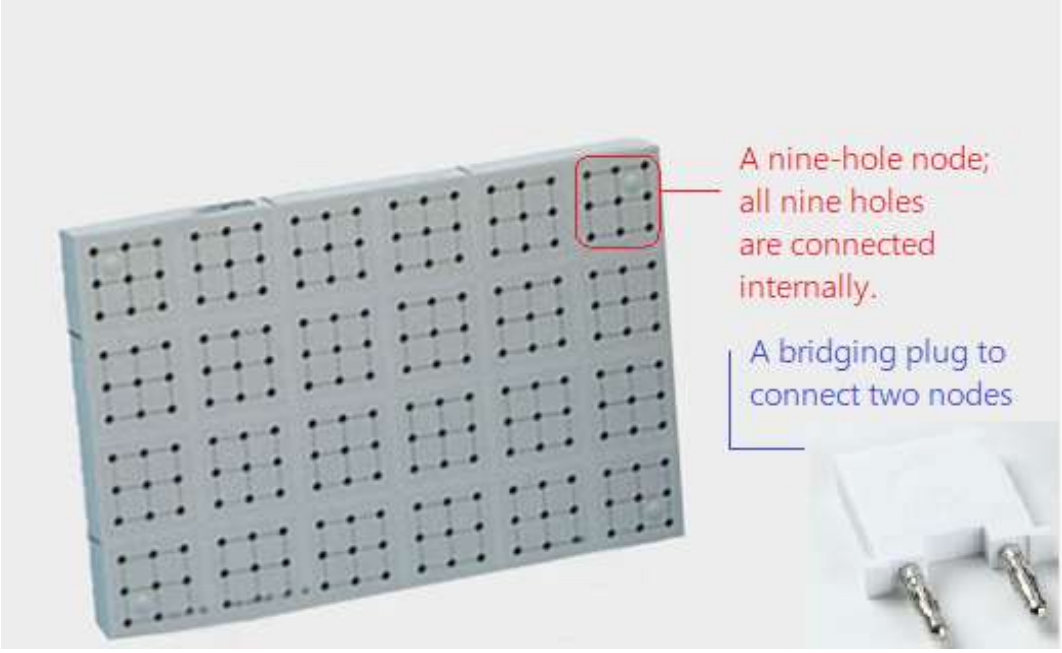


Figure G.1 Plug-in panel used to assemble simple circuits in this lab.

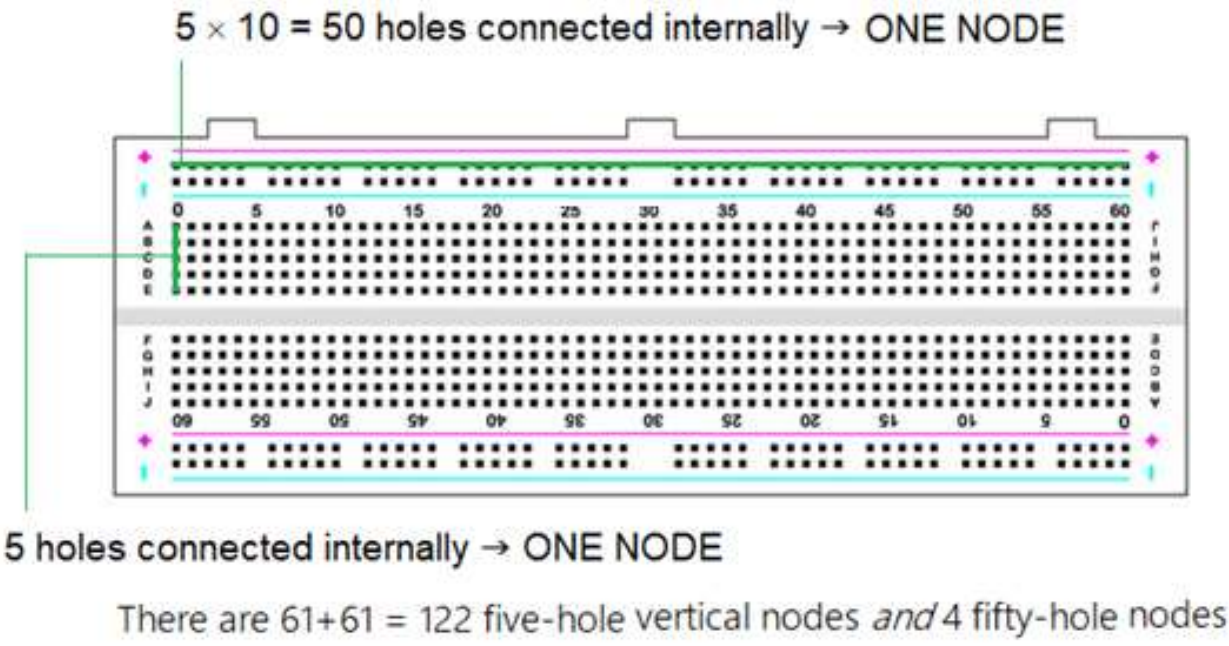


Figure G.2 Breadboard, a type of board used for mini projects and prototype designs.

Digital storage oscilloscopes (DSO) are modern types of scopes with a wide variety of useful features compared to the old CRO analog ones. Figure G.3 shows the front view of the GDS-1104B oscilloscope made by GwINSTEK®, which is available in the electronics lab at the College of Engineering, King Saud University. It can display four waveforms simultaneously on its four channels. Moreover, it can perform many useful automatic V/I and time measurements. Furthermore, it has powerful *Math* and *Advanced Math* capabilities.

The vertical sensitivity (measured in VOLTS/DIV) of each channel can be set independently based on the amplitude of the displayed waveform. The display of the scope is divided into eight major vertical divisions and ten major horizontal divisions. Note how a major division is divided into **five** subdivisions.

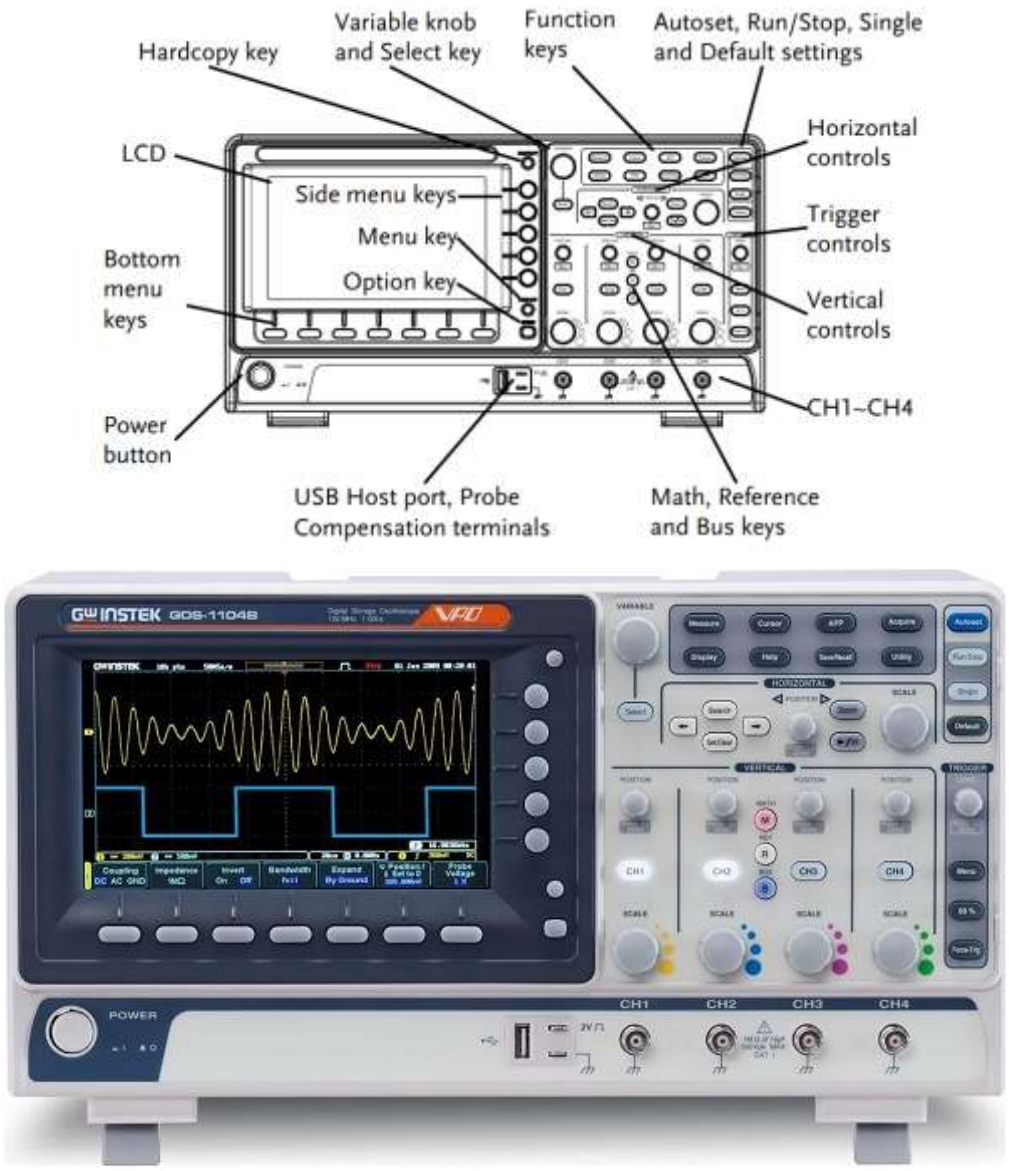


Figure G.3 A front view of the GwINSTEK® GDS-1104B oscilloscope.

A waveform or function generator is an essential piece of equipment in any electronics lab. It can be used to obtain several waveform types to test our circuits. It has the capabilities of controlling the frequency, amplitude, and the offset of its output waveform. Figure G.4 shows the front view of the PeakTeck® 4124 waveform generator which is currently available in the electronics lab at the College of Engineering, KSU.



Figure G.4 A front view of the PeakTeck® 4124 waveform generator.

A digital multi-meter (DMM) is a test equipment that is used to measure multiple electrical quantities. It offers testing some electronic devices, like diodes, as well. They can be hand-held (battery operated) or bench type (Fig. G.7).



FIGURE G.7 Example of a bench digital multimeter from *MCP*.

Coaxial cables (Fig. G.6) are used to connect the output of function generators/oscilloscopes to circuit boards. One side is usually terminated with a BNC connector and the other can be of banana-, alligator-, or mini-grabber-type.

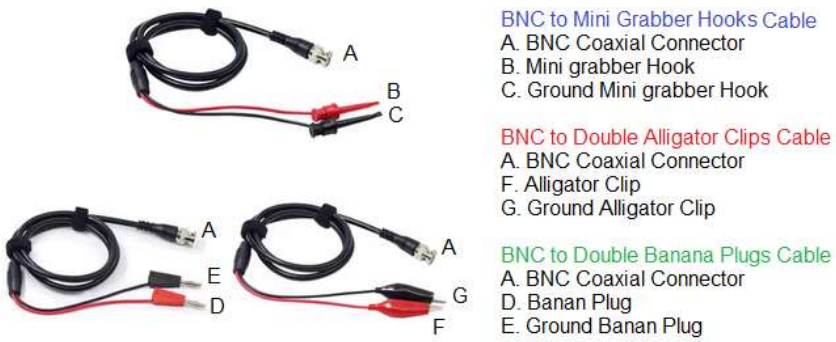


Figure G.6 Common types of coaxial-cable probes and leads.

Appendix H

DESIGN TIPS

H.1 *Voltage Reference*

One diode or more, depending on the desired dc output voltage, can be used to implement a **voltage reference**[14] (Fig. 1.7).

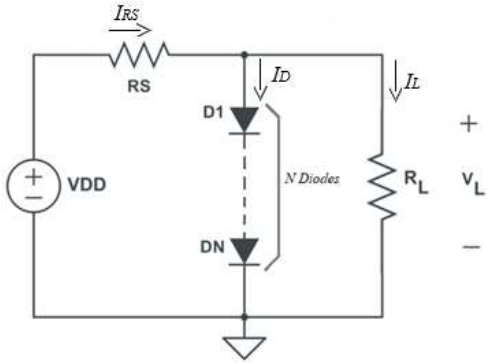


FIGURE H.1 Diode-based voltage reference

Suppose that you have a dc supply whose output voltage value is not exactly constant, i.e., $V_{DD(min)} \leq V_{DD} \leq V_{DD(max)}$. You are required to design a voltage reference circuit to feed a variable load, which is specified to operate at a given *constant* V_L. To maintain a constant V_L, the current in the resistor R_S must stay roughly constant. When the load demands more current, it can get it from the current going through the diodes.

- Assume a given diode current I_D greater then 1mA, say 10 mA.
- Look for a diode (see Appendix D) with *average forward current rating* which is about 50% greater than the sum of the load current and the diode current.
- Now, for that type of diode, find out how much forward voltage drop, V_D, it has when it passes a current[15] of I_D.
- The number of the required diodes[16] N can be found by:

$$N = \frac{V_L}{V_D} \qquad \text{(H.1)}$$

- Calculate R_S using:

$$R_s \approx \frac{V_{DD(\min)} - V_L}{I_{L(\max)} + I_D} \qquad \text{(H.2)}$$

- Use Appendix F to find the nearest E96 standard resistor to the above calculated value.
- To use a practical-size resistor, make sure that the maximum power dissipation in R_S does not exceed 1 W

[14] A voltage reference is a circuit that maintains its output dc voltage approximately constant despite reasonable changes in the load current or the input voltage

[15] You can find out that either in the diode datasheet available online, or you can measure it using a simple series circuit and a multimeter.

[16] If N > 5, it is better to use a Zener diode operating in the reverse breakdown region.

H.2 *Simple Unregulated Power Supply*

For a given dc load with a maximum current $I_{L(MAX)}$ operating at a given desirable dc voltage V_O, use the following checklist to complete a full-wave rectifier design. Note that the ***maximum*** voltage across a *single* secondary winding of the transformer is denoted V_{SM}. The acceptable *peak to peak* ripple in the output voltage (if no regulator is used) is denoted $V_{r(pp)}$.

1. Determine *C* using the following equation:

$$C = \frac{I_{L(MAX)}}{120 \times V_{r(pp)}} \qquad \text{(H.3)}$$

if not specified, assume $V_{r(pp)} = 0.05 \times V_O$

2. Select the nearest standard value of *C* from the table given in Appendix F.
3. Determine the diode average forward current using the following equation:

$$i_{Dav} \cong I_{L(MAX)} \left(1 + \pi \sqrt{\frac{V_{SM}}{2V_{r(pp)}}}\right) \qquad \text{(H.4)}$$

Determine $\mathbf{I_{F(avg)}}$ of the diode. A good practice is to have $\mathbf{I_{F(avg)}} = 1.5\ i_{Dav}$

4. Calculate the peak inverse voltage (*PIV*) that a diode in the circuit will have to withstand without breaking down. For the circuit shown in Fig. 4.2,

$$PIV = V_{SM} - 0.7 - (-V_{SM}) \cong 2V_{SM} \qquad \text{(H.5)}$$

5. Determine the $\mathbf{V_{RRM}}$ rating of the diode based on the calculated *PIV*. $\mathbf{V_{RRM}}$ should be greater than the expected *PIV* by enough margin. A good practice is to have $\mathbf{V_{RRM}} \cong 4V_S$ (rms)
6. Use Appendix D to look for a rectifier diode that has $\mathbf{I_{F(avg)}}$ and $\mathbf{V_{RRM}}$ ratings equal or slightly greater than those calculated in tips 3 and 5.

H.3 *Simple Voltage Regulator Design*

A voltage regulator is necessary for a dc power supply when ac input and/or the load is variable (See Fig. H.2). The input to the regulator $v_{UNREG}\ (t)$ has an average value and a ripple component.

1. Look for a Zener diode whose ***nominal*** breakdown voltage V_Z is approximately equal to the specified $V_O + 0.6$ (see for example Appendix E).
2. From the datasheet, find out the corresponding the *maximum current* I_{ZM}. If not given, calculate it from the maximum power rating. The maximum load current $I_{L(max)}/(\beta + 1)$ must be less than 75% I_{ZM}.

3. To insure operation deep into the breakdown region, $I_{Z(min)}$ that flows in the Zener diode when the current of the load is maximum, should be greater than the Zener knee current I_{ZK} by a couple of milliamperes.

4. Determine the value of the required R_S as follows,

$$R_S \approx \frac{V_{unreg(\min)} - V_Z}{I_{Z(min)} + [\frac{I_{L(max)}}{\beta+1}]} \qquad \text{(H.6)}$$

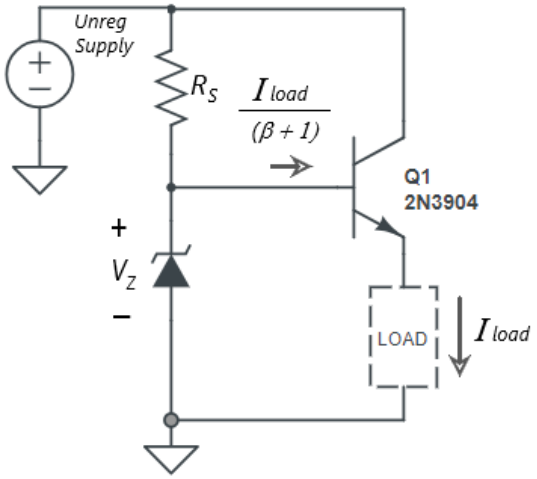


FIGURE H.2 Voltage regulator using a Zener diode and a voltage follower.

H.4 *Discrete Common Source Amplifier Biasing Scheme*

If the common-source configuration is to be utilized in an amplifier circuit, a proper bias scheme to ensure a stable operating point in the saturation mode should be used. A classical discrete-circuit bias arrangement for the MOSFET is shown in Fig. 10.6a. It is the most commonly used scheme if a single power supply is available to the designer. It utilizes a voltage divider between two resistors, R_1 and R_2, to establish an appropriate voltage at the gate to put the transistor in the saturation mode.

It is highly desirable to select the values of the bias resistors in such a way to make the operating point Q (I_{DQ}, V_{DSQ}) not strongly dependent on V_t and $k_n'W/L$, which vary with temperature and across samples of a given transistor type.

Starting off with a known value of $k_n'W/L$ and V_t, and for a given I_D (based on power budget) and a given $I_{R1} = I_{R2}$ (preferably in the μA range), a proposed sizing scheme of the resistors goes like this:

- Start with $V_{RD} = V_{DS} = V_S = (1/3) \times V_{DD} \Rightarrow$ calculate R_S from Ohm's law.
- Select $R_D = R_S$ (*note*: $I_S = I_D$).
- Calculate $V_{OV} = \sqrt{\frac{2 \times I_D}{k_n' \times \frac{W}{L}}}$
- Calculate $V_{GS} = V_t + V_{OV}$
- Calculate $V_G = V_{GS} + V_S$
- Using the specified value of I_{R2}, calculate R_2 from Ohm's law, noting that the voltage dropped across R_2 is the same as V_G.
- Using the specified value of $I_{R1} = I_{R2}$, calculate R_1 from Ohm's law, noting that the voltage dropped across R_1 is $(V_{DD} - V_G)$.
- Use Appendix F to find the nearest standard resistors to those you have calculated.

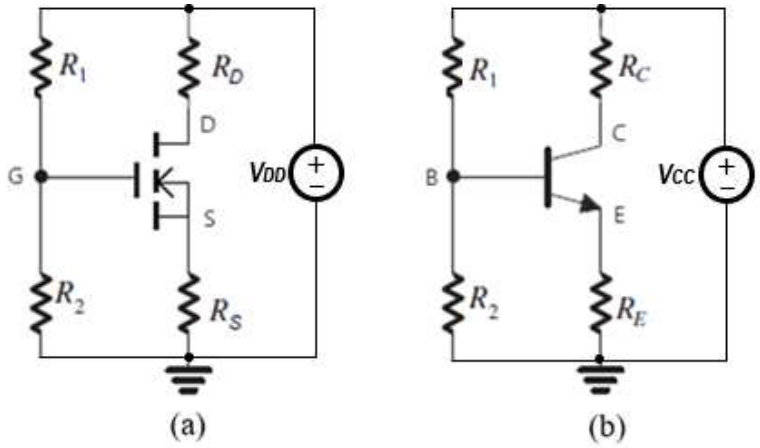


Figure H.3 Discrete amplifiers bias arrangements for **(a)** CS amplifier, and **(b)** CE amplifier.

H.5 *Discrete Common Emitter Amplifier Biasing Scheme*

If the common-emitter configuration is to be utilized in an amplifier circuit, a proper bias scheme to ensure a stable operating point in the active mode should be used. A classical discrete-circuit bias arrangement for the BJT is shown in Fig. 10.6b. It is the most commonly used scheme if a single power supply is available to the designer. It utilizes a voltage divider between two resistors, R_1 and R_2, to establish an appropriate voltage at the base to put the transistor in the active mode.

It is highly desirable to select the values of the bias resistors in such a way to make the operating point Q (I_{CQ}, V_{CEQ}) not strongly dependent on V_{BE} and β, which vary with temperature and across samples of a given transistor manufacturer.

Starting off with a $V_{BE(on)}$ of 0.7 V and a known value of β, and for a given I_E (based on power budget), a proposed sizing scheme of the resistors goes like this:

- Start with $V_{RC} = V_{CE} = V_E = (1/3) \times V_{CC} \Rightarrow$ calculate R_E from Ohm's law.
- Select $R_C = R_E$ (*note*: for large β, $I_C \approx I_E$).
- Calculate $V_B = V_{BE(on)} + V_E$
- Select a value for I_{R2} such that $\mathbf{0.1 \times} \boldsymbol{I_E} \leq I_{R2} \leq \mathbf{0.3 \times} \boldsymbol{I_E}$.
- Using the above value of I_{R2}, calculate R_2 from Ohm's law, noting that the voltage dropped across R_2 is the same as V_B.
- Since $I_{R1} \approx I_{R2}$, calculate R_1 from Ohm's law, noting that the voltage dropped across R_1 is $(V_{DD} - V_B)$.
- Use Appendix F to find the nearest standard resistors to those you have calculated.